

S5K3P9SX04

1/3.1" 16MP CMOS Image Sensor for RGB Bayer Pattern

Revision 0.09

April 2018

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Data Sheet

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Chip Handling Guide

Precaution against Electrostatic Discharge

When using semiconductor devices, ensure that the environment is protected against static electricity:

1. Wear antistatic clothes and use earth band.
2. All objects that are in direct contact with devices must be made up of materials that do not produce static electricity.
3. Ensure that the equipment and work table are earthed.
4. Use ionizer to remove electron charge.

Contamination

Do not use semiconductor products in an environment exposed to dust or dirt adhesion.

Temperature/Humidity

Semiconductor devices are sensitive to:

- Environment
- Temperature
- Humidity

High temperature or humidity deteriorates the characteristics of semiconductor devices. Therefore, do not store or use semiconductor devices in such conditions.

Mechanical Shock

Do not to apply excessive mechanical shock or force on semiconductor devices.

Chemical

Do not expose semiconductor devices to chemicals because exposure to chemicals leads to reactions that deteriorate the characteristics of the devices.

Light Protection

In non- Epoxy Molding Compound (EMC) package, do not expose semiconductor IC to bright light. Exposure to bright light causes malfunctioning of the devices. However, a few special products that utilize light or with security functions are exempted from this guide.

Radioactive, Cosmic and X-ray

Radioactive substances, cosmic ray, or X-ray may influence semiconductor devices. These substances or rays may cause a soft error during a device operation. Therefore, ensure to shield the semiconductor devices under environment that may be exposed to radioactive substances, cosmic ray, or X-ray.

EMS (Electromagnetic Susceptibility)

Strong electromagnetic wave or magnetic field may affect the characteristic of semiconductor devices during the operation under insufficient PCB circuit design for Electromagnetic Susceptibility (EMS).

Revision History

Revision No.	Date	Description	Author(s)
0.01	June 06, 2017	Initial draft (out of version_3)	Shachaf Amir
0.02	June 25, 2017	- Changing typical digital voltage to 1.05v - Updating Typical Functional Operation Modes Table - Updating Operating Conditions Table	Shachaf Amir
0.03	July 14, 2017	- Updating physical specifications diagrams and application circuit diagram - PDAF pattern update to TBD	Shachaf Amir
0.04	July 20, 2017	- Remove all PDAF chapter - Added capacitors values in Application circuit diagram	Shachaf Amir
0.05	Aug 10, 2017	Typo fix	Oren Reinherz
0.06	Aug 20, 2017	- Mode table fix - Update CRA Image Heights	Oren Reinherz & Shachaf Amir
0.07	Sep 25, 2017	Updating Gyro synchronizer chapter	Shachaf Amir
0.08	Mar 20, 2018	- Update section 2.1 - top view and pad for dummy - Update section 2.2 - application circuit for dummy - Update section 6.2 - temperature - Update section 6.3 - supply current	Daniel KJ Lee
0.09	Apr 16, 2018	Fixing Figure 25 Dual Sync Pad number	SH Kim

Table of Contents

1 PRODUCT OVERVIEW	11
1.1 Introduction	11
1.2 Features	12
2 PAD CONFIGURATION	13
2.1 Pad Description	14
2.2 Application Circuit	18
2.3 Pixel Array Information	19
3 POWER SEQUENCE.....	20
3.1 Operating Modes.....	20
3.2 Power-Up Sequence	21
3.3 Power-Down Sequence	23
3.4 Software Standby Mode Sequence	24
4 CONTROL INTERFACE	25
4.1 SPI Control Interface.....	26
4.1.1 SPI Timing Definitions	27
4.1.2 SPI Sequences	29
4.1.3 Master SPI Control Interface	30
4.2 Camera Control Interface (CCI).....	31
5 FUNCTIONAL FEATURES.....	37
5.1 Block Diagram	37
5.2 Pixel Array Addresses	39
5.3 Horizontal Mirror and Vertical Flip.....	40
5.4 Averaged Readout (Binned Readout)	42
5.5 Frame Rate Control	43
5.5.1 Integration Time Control (Electronic Shutter Control)	43
5.5.2 Functional Operation Modes	44
5.6 PLL and Clock Generator	45
5.6.1 Clock Relationships	46
5.6.2 Master Clock Waveform Diagram.....	48
5.7 Gain Control	49
5.7.1 Analog Gain Control	49
5.7.2 Digital Gain Control.....	49
5.8 Embedded Line	50
5.9 NVM OTP Memory.....	50
5.9.1 OTP Read/Write Procedure.....	50
5.10 Wide Dynamic Range (WDR) Support	51
5.11 Phase Detection Auto Focus (PD AF) Support.....	51
5.12 Dual camera input Sync.....	52
5.13 Gyro synchronizer	53
5.14 Test Pattern.....	54

5.15 Output Data Interface.....54

6 ELECTRICAL CHARACTERISTICS.....59

6.1 Absolute Maximum Rating59

6.2 Operating Conditions60

6.3 DC Characteristics61

6.4 AC Characteristics62

7 LENS SPECIFICATION (CRA : CHIEF RAY ANGLE SPEC.).....63

7.1 Target CRA63

7.2 Pixel-Shading characteristics.....64

7.3 Spectral Response.....65

7.4 IR-cut filter selection66

8 REGISTER DESCRIPTION67

8.1 Configuration Register67

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List of Figures

Figure Number	Title	Page Number
Figure 1	Chip Dimension Top View.....	13
Figure 2	Module Application Circuit Top View	18
Figure 3	Pixel Array Information.....	19
Figure 4	System State Diagram	20
Figure 5	Power-Up Sequence	22
Figure 6	Power-Down Sequence	23
Figure 7	SPI Control Interface.....	26
Figure 8	SPI Write Timing Diagram Mode 11	27
Figure 9	SPI Read Timing Diagram Mode 11	27
Figure 10	SPI Write/Read Mode 00	28
Figure 11	SPI WRITE Sequence (1 Cycle)	29
Figure 12	SPI Read.....	29
Figure 13	CCI Timing	31
Figure 14	CCI Write Timing Example.....	33
Figure 15	CCI Read Timing Example	34
Figure 16	CCI Read Multiple Timing Example	35
Figure 17	Function Block Diagram	37
Figure 18	Window of Interest on Pixel Array.....	39
Figure 21	Horizontal Mirror and Vertical Flip @ Normal/Binning Output mode	41
Figure 23	Virtual Frame Format Example	43
Figure 24	Clock System Block Diagram.....	45
Figure 25	Clock Relationships	46
Figure 26	PLL Frequency Synthesis	47
Figure 27	Master Clock Waveform Diagram	48
Figure 34	Dual Camera system	52
Figure 35	Dual camera sync initialization sequence	53
Figure 36	Gyro synchronizer Schematic	53
Figure 37	CSI-2 and CCI Transmitter and Receiver Interface	56
Figure 38	Four Lane Transmitter and Four Lane Receiver Example	56
Figure 39	Low Level Protocol Packet Overview.....	57
Figure 40	Long Packet Structure	57
Figure 41	RAW10 Transmission	58
Figure 42	RAW10 Frame Format.....	58
Figure 43	Target CRA	63
Figure 44	Shading Characteristics of Green	64
Figure 45	Shading Characteristics of Color ratio	64
Figure 46	Spectral Response.....	65
Figure 47	IR Response of Blue channel	66

List of Tables

Table Number	Title	Page Number
Table 1	Pad Description	14
Table 2	Operating Mode Summary	21
Table 3	Operating Mode Registers	21
Table 4	Power-Up Sequence Timing Constraints	22
Table 5	Power-Down Sequence Timing Constraints	24
Table 6	Abort Frame Timing Transition Options upon Entering SW Reset	24
Table 7	SPI/CCI PAD Sharing	25
Table 8	SPI Write/Read Timing Specification	28
Table 9	I2C Standard Mode Timing Specifications	31
Table 10	I2C Fast Mode Timing Specifications	32
Table 11	I2C Fast Mode plus (FM+) Timing Specifications	32
Table 12	I2C ID Address (XCE Pad)	36
Table 13	Typical Functional Operation Modes	44
Table 14	PLL Component Output Frequency	47
Table 15	MCLK timing spec	48
Table 16	Analog Gain Examples	49
Table 17	Absolute Maximum Rating	59
Table 18	Operating Conditions	60
Table 19	DC Characteristics	61
Table 20	AC Characteristics	62
Table 21	Image Height (Field)	63
Table 22	Spectral Response	65
Table 23	Configuration Register	67

List of Conventions

Register RW Access Type Conventions

Type	Definition	Description
R	Read Only	The application has permission to read the Register field. Writes to read-only fields have no effect.
RW	Read/Write	The application has permission to read and writes in the Register field. Written value effects on the next frame.
RW/R	Read/Write	The application has permission to read and writes in the Register field. Written value effects only on exit from stand-by.
RW/C	Read/Write	The application has permission to read and writes in the Register field. Changing value typically causes configuration change (either in abort timing or preserve timing modes).
RW/SR	Read/Write	The application has permission to read and writes in the Register field. Changing value may cause entering stand-by / software reset.

Register Value Conventions

Expression	Description
x	Undefined bit
X	Undefined multiple bits
?	Undefined, but depends on the device or pin status
Device dependent	The value depends on the device
Pin value	The value depends on the pin status

Reset Value Conventions

Expression	Description
0	Clears the register field
1	Sets the register field
x	Don't care condition

Warning: Some bits of control registers are driven by hardware or write operation only. As a result the indicated reset value and the read value after reset might be different.

1

Product Overview

1.1 Introduction

The S5K3P9SX04 is a highly integrated 16M pixel camera chip that includes a CMOS image sensor (CIS), image correction functionality and serial transmission using 4-lane MIPI. It is designed for fast yet low power operation, delivering full resolution capture at 30 frames per second (fps) and full field of view (4:3), 4M video at 60fps.

The S5K3P9SX04 has RGB Bayer Pattern. It is fabricated by the SAMSUNG CMOS image sensor process to realize a high-efficiency and low-power photo sensor. The sensor consists of 4640x3488 effective pixels that meet with the 1/3-inch optical format.

The CIS has on-chip 10-bit ADC arrays to digitize the pixel output and on-chip Correlated Double Sampling (CDS) to drastically reduce Fixed Pattern Noise (FPN). It incorporates on-chip camera functions such as defect correction, exposure setting, white balance setting and image data compression.

The S5K3P9SX04 CIS is programmable through a CCI or SPI serial interface and includes on-chip one-time programmable (OTP) non-volatile memory (NVM).

The S5K3P9SX04 is suitable for a low-power camera module with a 2.8V/1.8V/1.05V power supply.

1.2 Features

- 16Mp sensor with 1/3"optics
- Unit Pixel size : 1.0μm
- Effective resolution : 4640 (H) × 3488 (V)
- Active resolution : 4656 (H) × 3504 (V)
- Color Filter : RGB Bayer Pattern
- Shutter Type : Electronic rolling shutter and global reset
- Max. Normal Frame Rate : 30fps @ Full
- Max. Video Frame Rate : 180fps @ VGA (crop)
- Data rate : 2,150Mbps/lane
- ADC Accuracy : 10bits
- Dual sensor synchronize
- Interfaces:
 - Fine interface frequency control using additional dedicated PLL for EMI avoidance and integration flexibility.
 - MIPI CSI2: four lanes (2.15Gbps per lane)
 - Output formats: RAW8 (using DPCM/PCM compression), RAW10
- Control interface:
 - SPI interface: Three-wire serial communication circuit up to 40MHz.
 - CCI (Camera Control Interface) High speed I2C-compatible - Two-wire serial communication circuit up to 1MHz.
- 64 Kbit on-chip OTP memory Mapped BPC and chip ID.
- Analog gain x16
- Vertical and horizontal flip mode
- Continuous frame capture mode
- 2/2 – average-sub-sampling readout
- Pixel skip readout function
- Mapped Bad pixel correction
- Built-in test pattern generation
- Supports EIS function
- Supply voltage : 2.8V for analog, 1.8V or 2.8V for I/O, and 1.05V for digital core supply
- Operating temperature : -20°C to +70°C

2 Pad Configuration

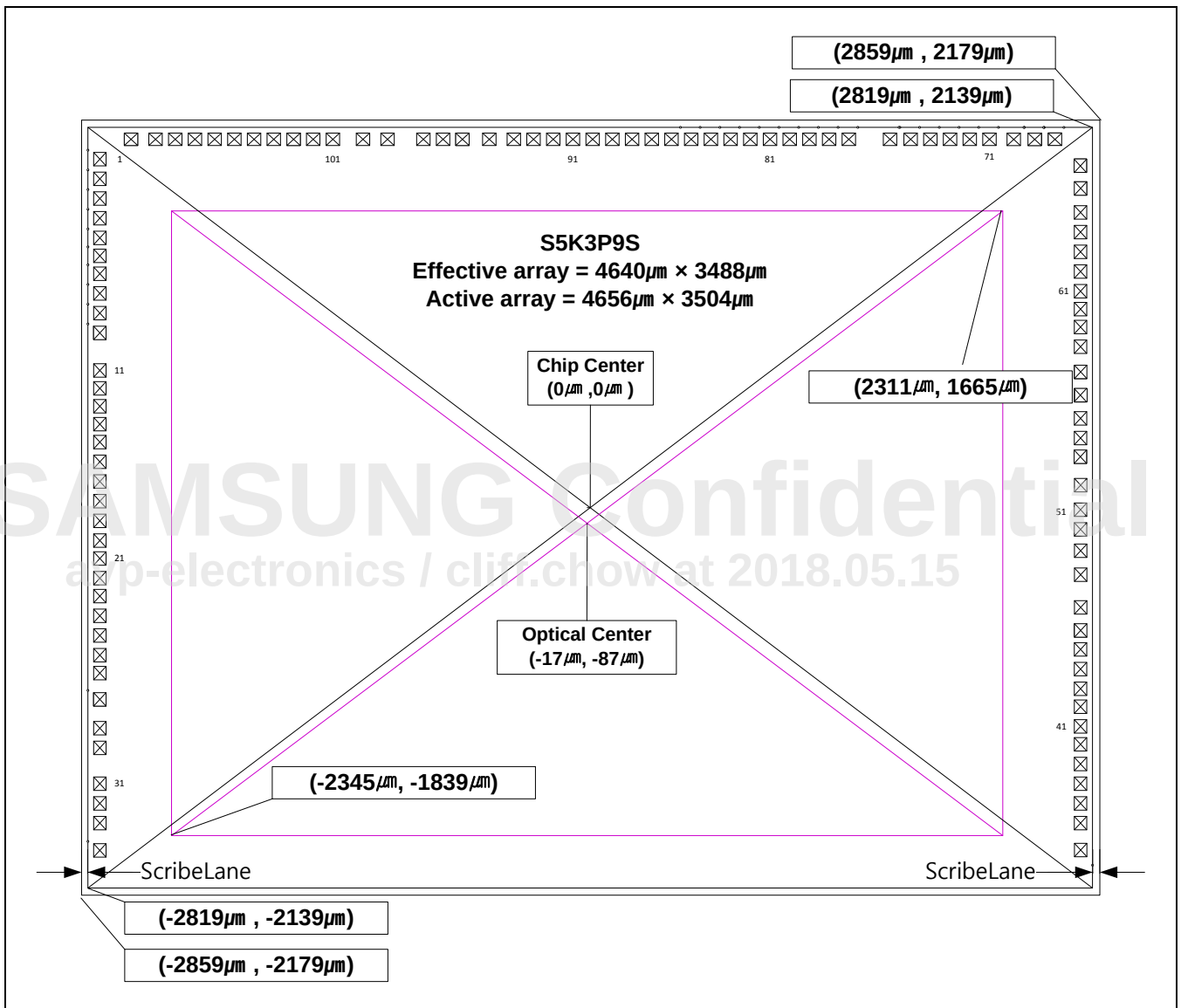


Figure 1 Chip Dimension Top View

2.1 Pad Description

Table 1 Pad Description

Pad No.	Pad Name	Type	A/D	Description
1	VDDIO_0	P	D	I/O Power
2	VSSIO_0	G	D	I/O Ground
3	M_SDO	IOD	D	Master SPI Data-Out
4	M_SDI	IOD	D	Master SPI Data-In
5	GPIO_3	IOZ	D	General In/Out 3
6	GPIO_2	IOZ	D	General In/Out 2
7	GPIO_1	IOZ	D	General In/Out 1
8	VDDD_0	P	D	Digital Power
9	VSSD_0	G	D	Digital Ground
10	NC	G		No connection
11	NC	IOZ		No connection
12	VSYNC	IOZ	D	Vertical sync output for parallel and MIPI interface
13	VSYNC_IN	IOZ	D	Vertical sync input to enable dual sensor function
14	HSYNC	IOZ	D	Horizontal sync output for parallel and MIPI interface (optional)
15	SDI	IOD	D	SPI mode: Serial communication data input
16	SCK	IOD	D	CCI mode: CCI data, connect external 2.2Kohm resistor
16	SDO	IOZ	D	SPI mode: Serial Data output (SPI) / CCI mode: leave open
17	SCK	IOD	D	SPI mode: Serial communication clock input
19	VDDD_1	P	D	CCI mode: CCI input clock, connect external 2.2Kohm resistor
18	XCE	IOD	D	SPI mode: Serial communication enable / CCI mode: CCI slave address selection (0: 20h/21h 1: 5Ah/5Bh)
19	RSTN	I	D	Master Reset, Active low
20	VDDD_1	P	D	Digital Power
21	VSSD_1	G	D	Digital Ground
22	TST	I	D	Test: EDS test configuration pin. Tie low in normal operation
23	VSSIO_1	G	D	I/O Ground
24	VDDIO_1	P	D	I/O Power
25	I2C_SPI_N_SEL	I	D	SPI/CCI Select (1: CCI, 0: SPI)
26	VDDD_2	P	D	Digital Power

Pad No.	Pad Name	Type	A/D	Description
27	VSSD_2	G	D	Digital Ground
28	NC	G		No connection
29	VSSD_ATOP_0	G	D	Digital Ground
30	VDDD_ATOP_0	P	D	Core power 1.05v
31	VSSA_ATOP_0	G	A	Analog Ground
32	VDDA_ATOP_0	P	A	Analog Power
33	VSSA_ATOP_1	G	A	Analog Ground
34	VPIX_0	P	A	Analog Power
35	VPIX_1	P	A	Analog Power
36	VDDA_ATOP_1	P	A	Analog Power
37	VSSA_ATOP_2	G	A	Analog Ground
38	VDDA_ATOP_2	P	A	Analog Power
39	VSSA_ATOP_3	G	A	Analog Ground
40	VDDD_ATOP_1	P	D	Core power 1.05v
41	VDDD_ATOP_2	P	D	Core power 1.05v
42	VSSD_ATOP_1	G	D	Digital Ground
43	NC	G		No connection
44	VDDA_ATOP_3	P	A	Analog Power
45	VSSA_ATOP_4	G	A	Analog Ground
46	NC	G		No connection
47	VNTG	AIO	A	Connect to 4.7uF cap
48	VTG	AIO	A	Connect to 0.1uF cap
49	VRGSL	AIO	A	Connect to 0.47uF cap
50	VDDA_ATOP_4	P	A	Analog Power
51	VSSA_ATOP_5	G	A	Analog Ground
52	NC	AIO		No connection
53	NC	G		No connection
54	VSSD_3	G	D	Digital Ground
55	VDDD_3	P	D	Digital Power
56	NC	G		No connection
57	NC	G		No connection
58	VDDD_4	P	D	Digital Power
59	VSSD_4	G	D	Digital Ground
60	NC	IOZ		No connection
61	VSSD_5	G	D	Digital Ground
62	VDDD_5	P	D	Digital Power

Pad No.	Pad Name	Type	A/D	Description
63	NC	IOZ		No connection
64	VSSIO_2	G	D	I/O Ground
65	VDDIO_2	P	D	I/O Power
66	NC	G		No connection
67	NC	G		No connection
68	NC	G		No connection
69	NC	G		No connection
70	NC	AO		No connection
71	VDDA_CID	P	A	OTP Analog Power
72	VSSA_CID	G	A	OTP Analog Ground
73	NC	AO		No connection
74	NC	AO		No connection
75	NC	AO		No connection
76	NC	AIO		No connection
77	VSSD_6	G	D	Digital Ground
78	VDDD_6	P	D	Digital Power
79	VDDD_M_0	P	D	Digital Power
80	VSSD_M_0	G	D	MIPI Digital Ground
81	M_DPDATA3	AIO	D	Serial Data Output 3 - positive
82	M_DNDATA3	AIO	D	Serial Data Output 3 - negative
83	M_DPDATA2	AIO	D	Serial Data Output 2 - positive
84	M_DNDATA2	AIO	D	Serial Data Output 2 - negative
85	VSSD_7	G	D	Digital Ground
86	M_DPCLK	AIO	D	Serial Clock Output - positive
87	M_DNCLK	AIO	D	Serial Clock Output - negative
88	VDDD_7	P	D	Digital Power
89	M_DPDATA1	AIO	D	Serial Data Output 1 - positive
90	M_DNDATA1	AIO	D	Serial Data Output 1 - negative
91	M_DPDATA0	AIO	D	Serial Data Output 0 - positive
92	M_DNDATA0	AIO	D	Serial Data Output 0 - negative
93	VSSD_M_1	G	D	MIPI Digital Ground
94	VDDD_M_1	P	D	Digital Power
95	VDDA_M_0	P	A	Analog Power
96	NC	G		No connection
97	VDDA_C	P	A	Analog Power
98	VSSA_C	G	A	Analog Ground

Pad No.	Pad Name	Type	A/D	Description
99	VSSD_P	G	D	Digital Ground
100	VDDA_P	P	A	Analog Power
101	VDDD_P	P	D	PLL Digital Power
102	NC	AIO		No connection
103	NC	G		No connection
104	VDDD_8	P	D	Digital Power
105	VSSD_8	G	D	Digital Ground
106	MCLK	I	D	External Input Clock
107	VSSIO_3	G	D	I/O Ground
108	VDDIO_3	P	D	I/O Power
109	M_SCK	IOD	D	Master SPI clock
110	M_CSN	IOD	D	Master SPI CS
111	NC	G		No connection

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Figure 2 illustrates the top view of a module application circuit.

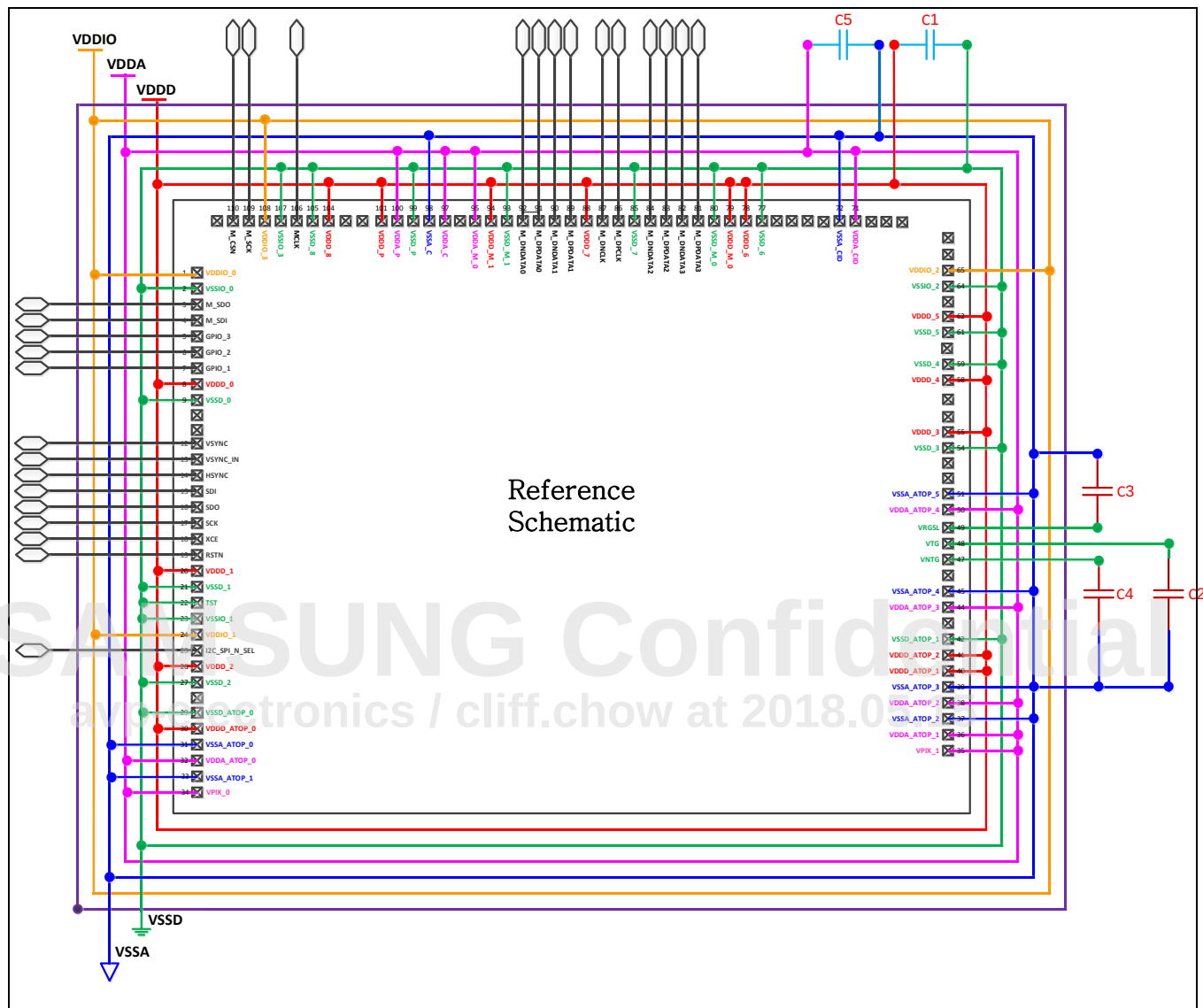


Figure 2 **Module Application Circuit Top View**

2.3 Pixel Array Information

Figure 3 illustrates the top view of pixel array information.

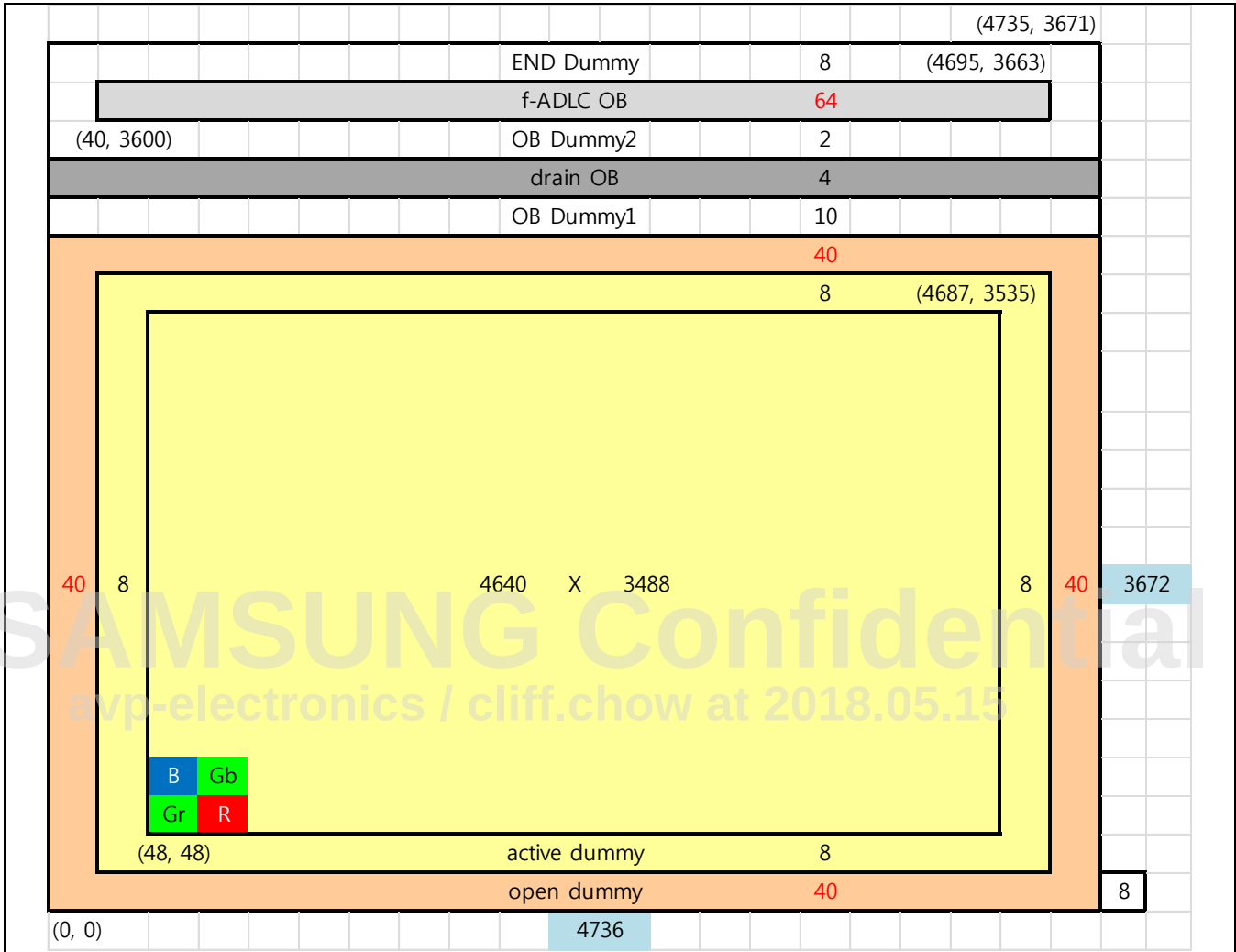


Figure 3 Pixel Array Information

3 Power Sequence

3.1 Operating Modes

The Sensor module has four operating modes ([Table 2](#)). Moving from one mode to another is achieved by issuing the appropriate mode command via the CCI serial control interface, the RSTN (XSHUTDOWN) signal changing state and the power supplies. By default, S5K3P9SX04 powers up with the CSI-2 serial data interface enabled. [Figure 4](#) defines the valid mode changes for the sensor module.

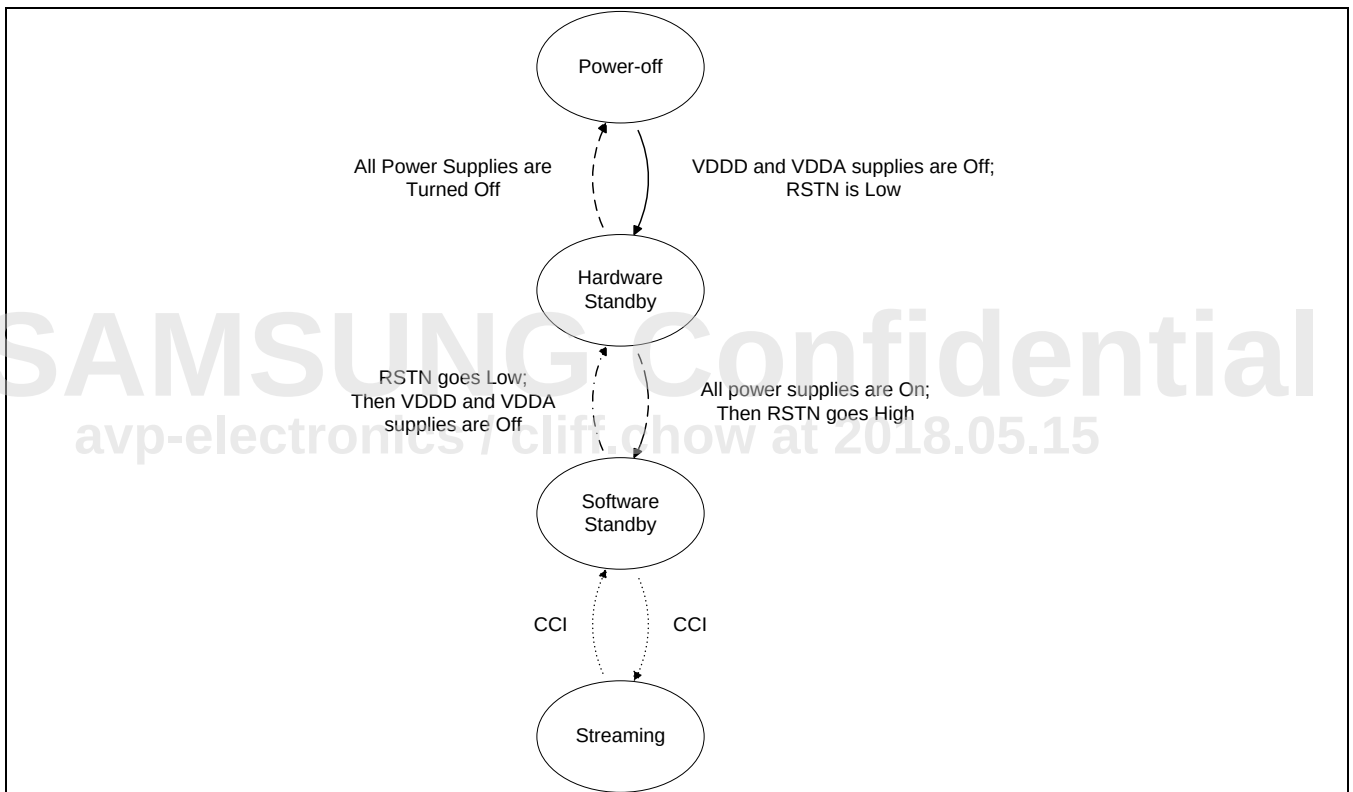


Figure 4 System State Diagram

Table 2 Operating Mode Summary

Power State	Description
Power-off	Power supplies are turned off.
Hardware standby	No communication with the sensor is possible.
Software standby	CCI communication with sensor is possible.
Streaming	The sensor module is fully powered and is streaming image data on the CSI-2 bus.

[Table 3](#) describes the registers that control the operating mode of the camera module

Table 3 Operating Mode Registers

Start	Reset	Name	Type	Width	Description
0x0100	0x00	smiaRegs_rw_general_setup_mode_select	RW/SR	[0:0]	0 = Software Standby 1 = Streaming (Active Video)

3.2 Power-Up Sequence

The digital and analog supply voltages can be powered up in any order, e.g., VDDD then VDDA or VDDA then VDDD.

On power up, RSTN (XSHUTDOWN) should be low when the power supplies are brought up, then the sensor module will go into hardware standby mode. As long as RSTN is low and VDDD is down, the sensor module stays in hardware standby mode.

The assertion of RSTN ensures that the CCI register values are initialized correctly to their default values.

When RSTN will go 'high', all PADs will exit from FAIL-SAFE mode, and switch to Normal operating mode.

The MCLK clock can either be initially low and then enabled during software standby mode or MCLK can be a free running clock.

NOTE:

1. At hardware standby mode, external VDDD and VDDA should be OFF.
2. External clock (MCLK) is recommended to be off in Software Standby to reduce power consumption.

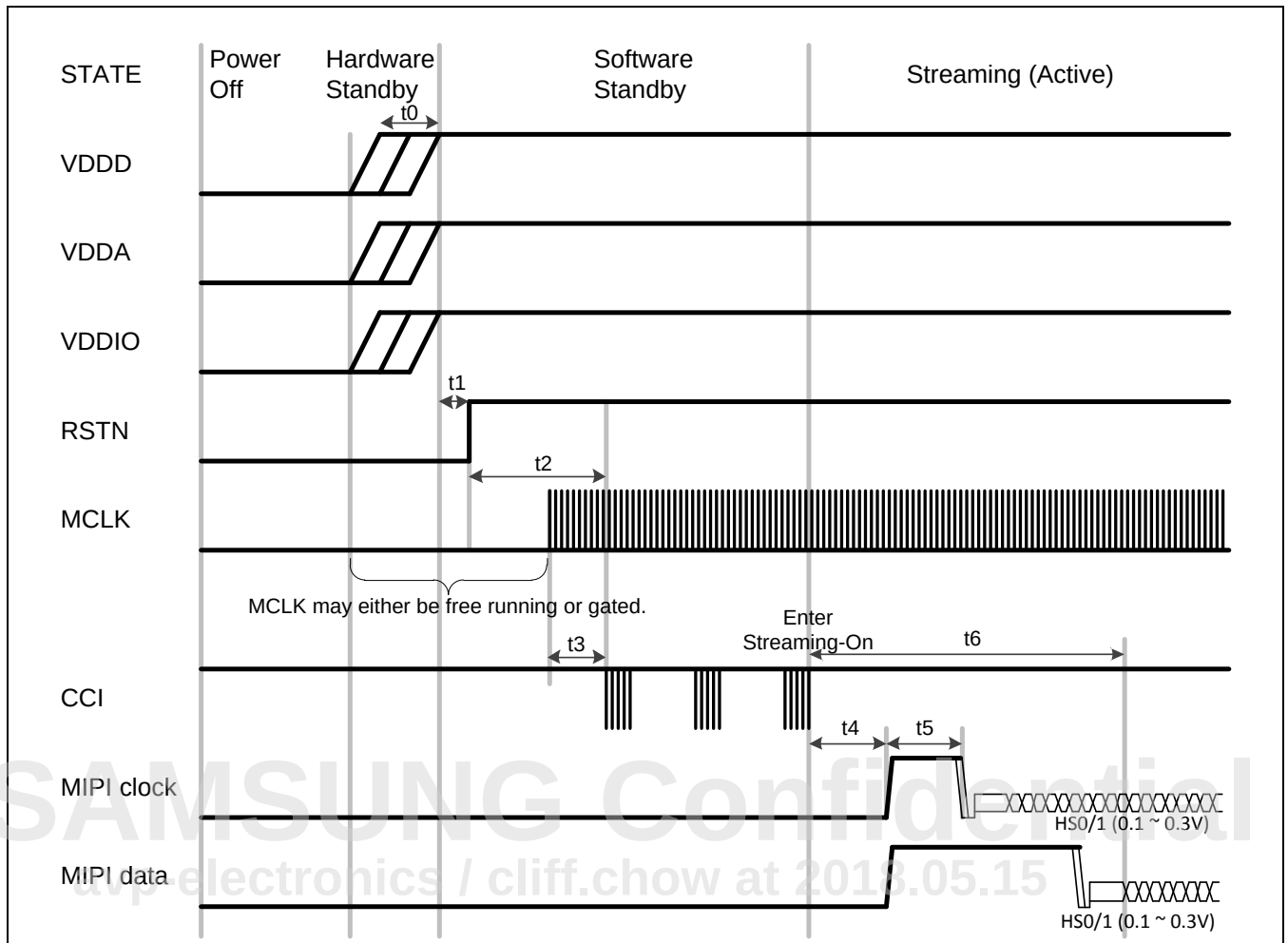


Figure 5 Power-Up Sequence

Table 4 Power-Up Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
VDDA/VDDD/VDDIO rising time	t_0	VDDA/VDDD/VDDIO may rise in any order. The rising separation can vary from 0ns to indefinite.		ns
VDDD rising to RSTN (XSHUTDOWN) rising	t_1	0.0	—	ns
RSTN (XSHUTDOWN) rising to first CCI transaction	t_2	10	—	μ s
Minimum No. of MCLK (EXTCLK) cycles prior to the first CCI transaction	t_3	23,000	—	MCLK cycles
PLL startup/lock time	t_4	—	1	ms
DPHY initialization period (TINIT)	t_5	0.1	—	ms
Entering streaming mode - The first frame start sequence	t_6	+ the delay according to the coarse integration time value		-

3.3 Power-Down Sequence

The digital and analog supply voltages can be powered down in any order, e.g., VDDD then VDDA or VDDA then VDDD.

Similar to the power-up sequence, the MCLK (EXTCLK): input clock may be either gated or continuous.

If the CCI command to exit streaming is received while a frame of valid active data is being output, then the sensor module must wait for the frame end code before entering software standby mode. Frame end code may come either after all frame pixels were transmitted or during the frame when next line transmission is completed - based on a configuration register (Refer to S5K3P9SX04 application notes for details)

If the CCI command to exit streaming mode is received during the inter frame time, then the sensor module must enter software standby mode immediately.

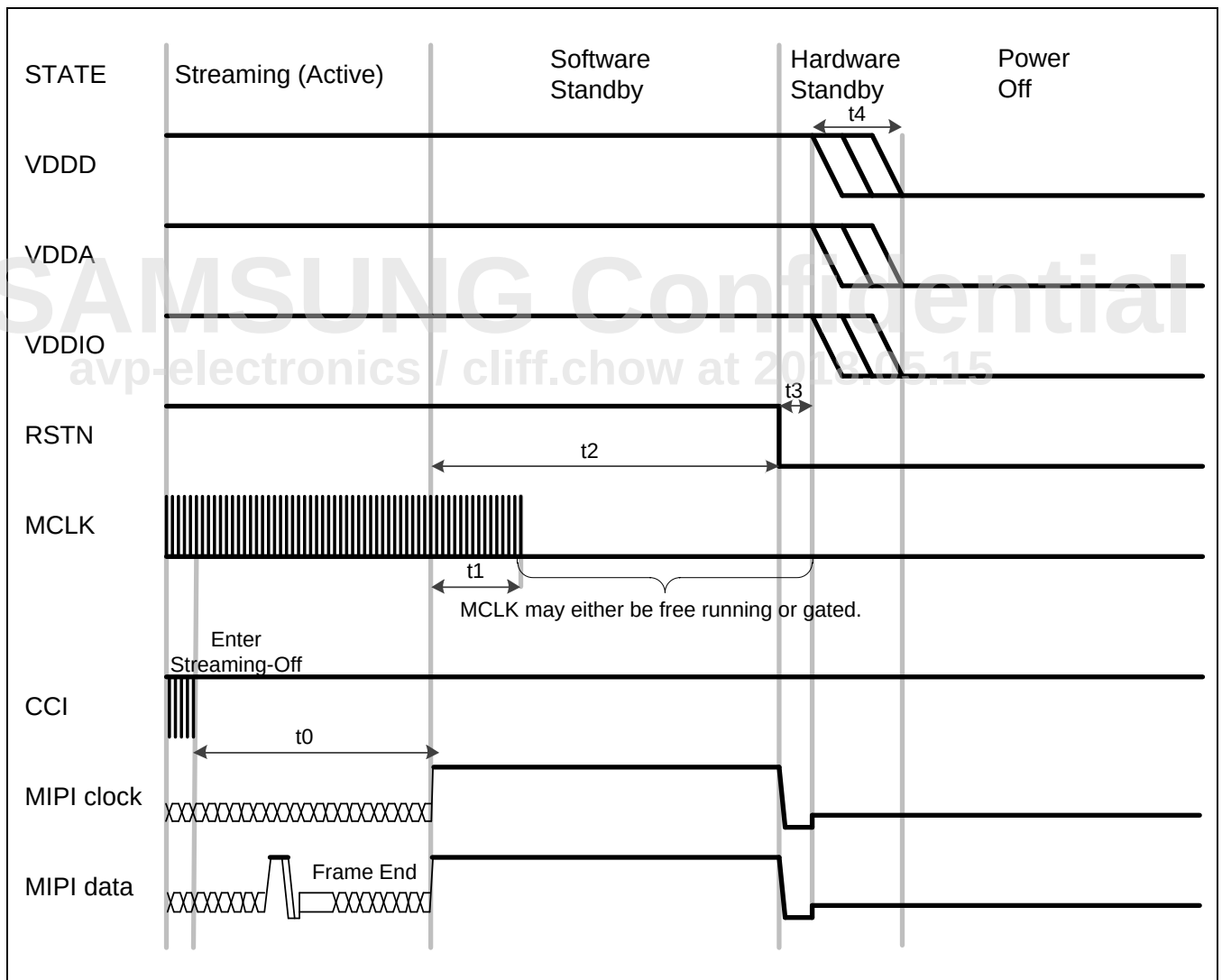


Figure 6 Power-Down Sequence

Table 5 Power-Down Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
Enter Software Standby CCI command-Device in Software Standby mode	t0	If outputting a frame of MIPI, the data waits for the MIPI frame end code before entering software standby; otherwise enter software standby mode immediately.		–
Minimum number of MCLK (EXTCLK) cycles after the last CCI transaction or MIPI frame end code.	T1	512	–	MCLK cycles
Last CCI Transaction or MIPI frame end code - RSTN (XSHUTDOWN) falling	t2	512	–	
RSTN (XSHUTDOWN) falling to VDDD or VDDA falling	t3	0.0	–	ns
VDDA/VDDD/VDDIO falling time	t4	VDDA/VDDD/VDDIO may fall in any order. The falling separation can vary from 0 ns to indefinite		

NOTE: For minimal power at hardware standby mode external VDDD and VDDA should be OFF.

3.4 Software Standby Mode Sequence

Entering software standby mode is the same as entering hardware standby mode but without RSTN (XSHUTDOWN) assertion to low and without applying external VDDD power down.

There are several options to define how the last frame before entering the Software Reset is handled, according to the following parameter.

Table 6 Abort Frame Timing Transition Options upon Entering SW Reset

Timing Transition Type	Setting
abort_timing_on_sw_stby	0 = Abort timing Immediately, even during read out.
	1 = Abort timing after read out ends
	2 = Abort timing on end of frame.

4

Control Interface

S5K3P9SX04 control is done using register writes.

S5K3P9SX04 can be controlled using the Camera Control Interface (CCI).

Table 7 SPI/CCI PAD Sharing

SPI	CCI
SDI	SDA
SCK	SCL
XCE	I2C_ID
SDO	–
I2C_SPI_N_SEL	

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4.1 SPI Control Interface

The S5K3P9SX04 may use the SPI interface for control registers communication.

SPI interface is described in the diagram below.

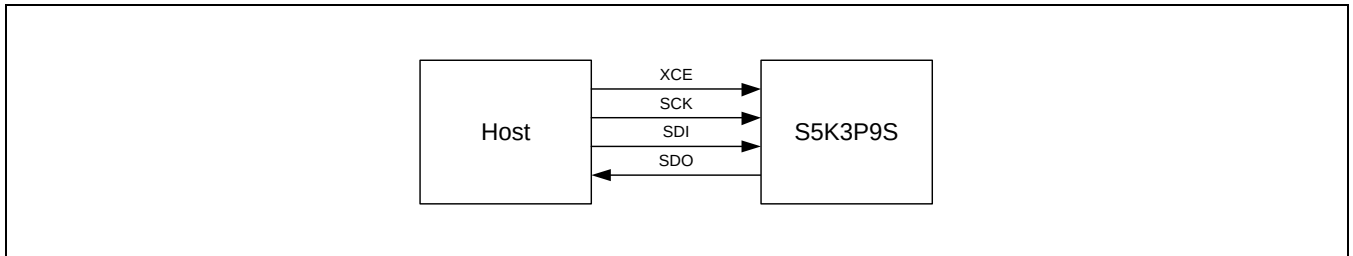


Figure 7 SPI Control Interface

The following table describes SPI signals:

Pin Name	Description
XCE	Serial communication enable
SCK	Serial communication clock input
SDI	Serial data input
SDO	Serial data output

If it is not necessary to read the control registers and chip ID, the SDO pad can be left unconnected.

The SPI control interface implemented in S5K3P9SX04 will be presented in detail in the next section.

4.1.1 SPI Timing Definitions

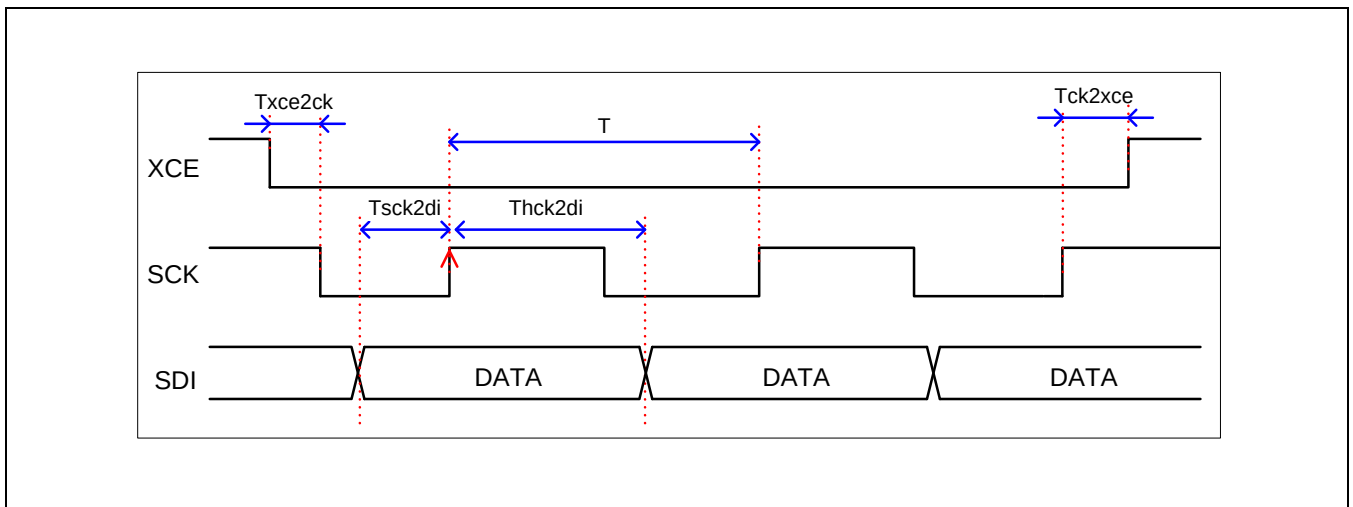


Figure 8 SPI Write Timing Diagram Mode 11

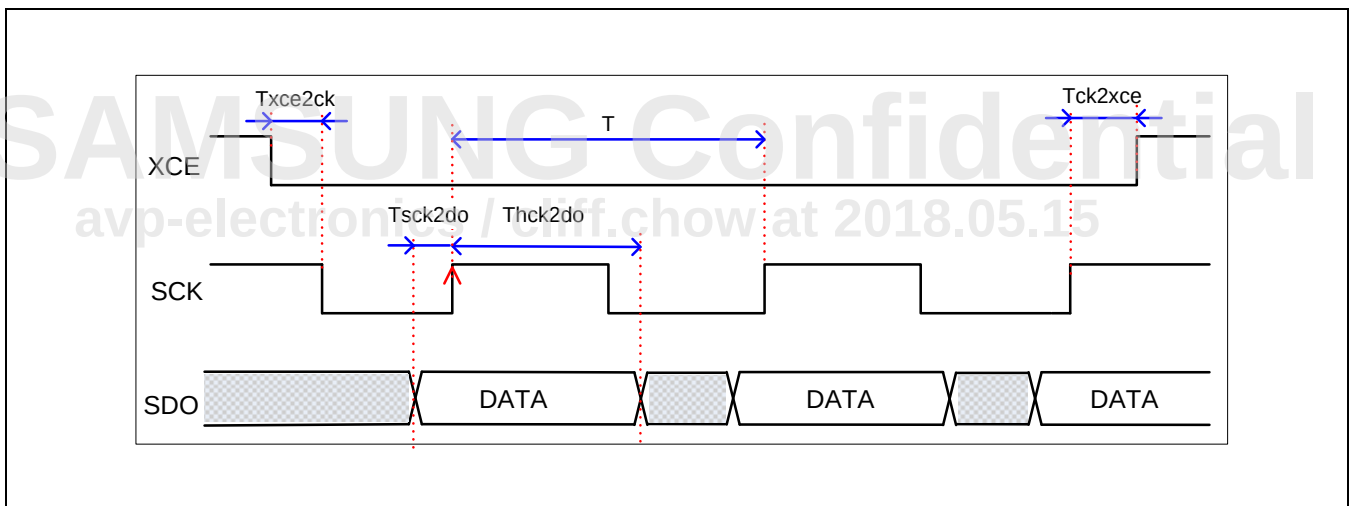


Figure 9 SPI Read Timing Diagram Mode 11

Table 8 SPI Write/Read Timing Specification

Description	Symbol	Min.	Typ.	Max.	Unit
MCLK (external clock) frequency	Fext	12	–	100	MHz
SPI clock frequency	F (1)	–	–	Fext/3	
	F (2)	–	–	20	
SPI clock period	T	1/F	–	–	nSec
SPI clock duty cycle	Tdc	45	–	55	%
XCE negedge to SCK edge	Txce2ck	T	–	–	nSec
SCK posedge to XCE edge	Tck2xce	2T	–	–	
SCK to SDI setup time	Tsck2di	7	–	–	
SCK to SDI hold time	Thck2di	5	–	–	
SCK to SDO setup time	Tsck2do	T/2-20	–	–	
SCK to SDO hold time	Thck2do	T/2	–	–	

NOTE:

1. External clock is used.
2. PLL stable. Internal system clock set to Max. 60MHz

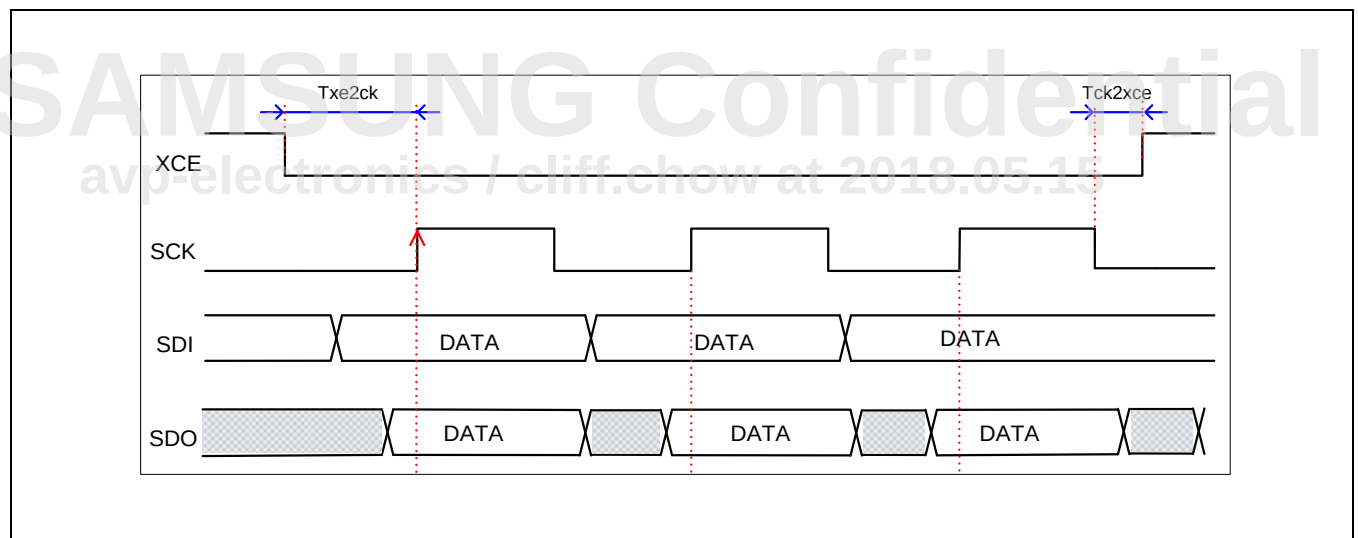


Figure 10 SPI Write/Read Mode 00

4.1.2 SPI Sequences

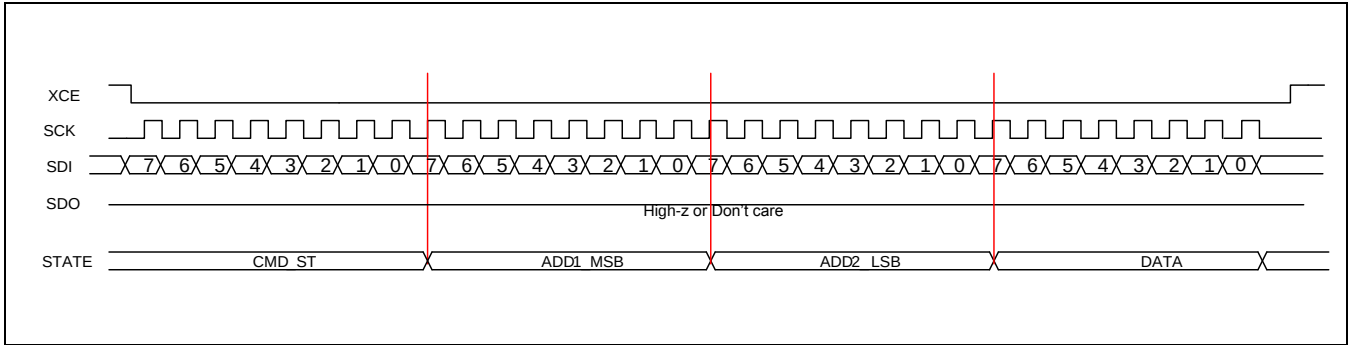


Figure 11 SPI WRITE Sequence (1 Cycle)

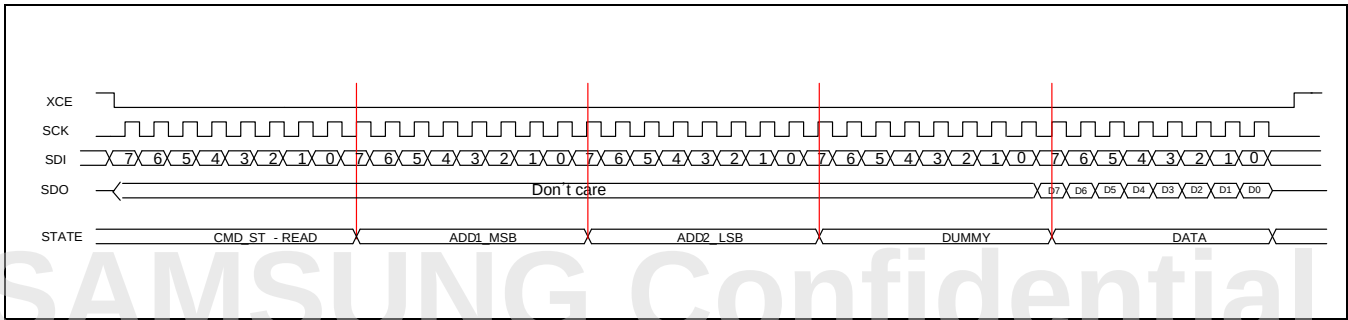


Figure 12 SPI Read

4.1.3 Master SPI Control Interface

In addition to SPI slave interface, the S5K3P9SX04 has Master SPI interface control as well. Its designated function is to receive gyro data and sync it to ISP output for EIS function use.

MSPI may connect to other types of slaves as well.

MSPI interface is described in sections above.

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4.2 Camera Control Interface (CCI)

S5K3P9SX04 supports the Camera Control Interface (CCI), which is an I2C fast-mode compatible interface for controlling the transmitter. S5K3P9SX04 always acts as a slave in the CCI bus. CCI is capable of handling several slaves in the bus, but multi-master mode is not supported. Typically, only the receiver and transmitter are connected to the CCI bus. This makes a pure SW implementation possible.

Typically the CCI is separate from the system I2C bus, but I2C-compatibility ensures that it is also possible to connect the transmitter to the system I2C bus. CCI is a subset of the I2C protocol, including the minimum combination of obligatory features for the I2C slave device specified in the I2C specification. Therefore, transmitters complying with the CCI specification can also be connected to the system I2C bus. However, it is important to ensure that the I2C masters do not try to utilize these I2C features, which are not supported in transmitters complying with the CCI specification. Each transmitter conformed to the CCI specification may have additional features implemented to support I2C.

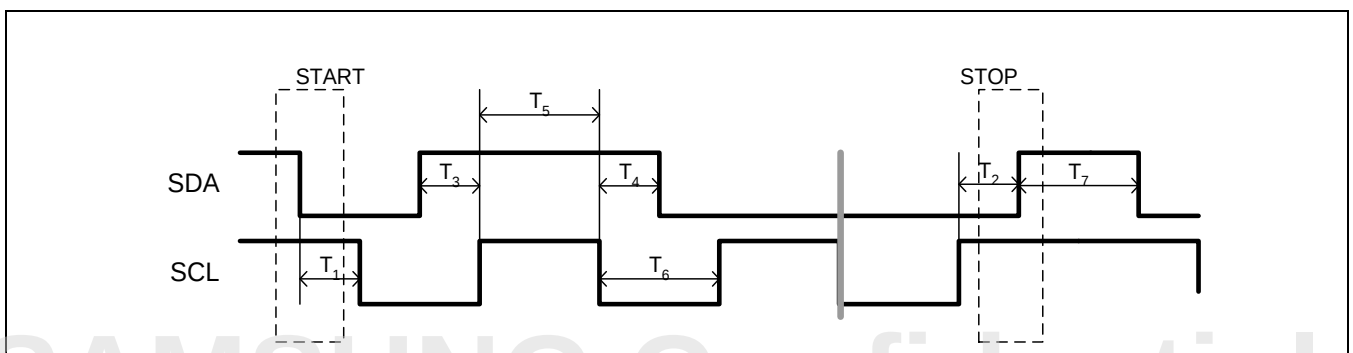


Figure 13 CCI Timing

Table 9 I2C Standard Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	—	0	100	kHz
Hold time for START condition	T_1	4.0	—	μ s
Setup time for STOP condition	T_2	4.0	—	
Data setup time	T_3	250	—	ns
Data hold time	T_4	0	3.45	μ s
High period of the SCL clock	T_5	4.0	—	
Low period of the SCL clock	T_6	4.7	—	
Bus free time between STOP and START conditions	T_7	4.7	—	
Rise time for both SDA and SCL signals	—	—	1000	ns
Fall time for both SDA and SCL signals	—	—	300	
Capacitive load for each bus line	CB	—	400	pF

Table 10 I2C Fast Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	—	0	400	kHz
Hold time for start condition	T ₁	0.6	—	μs
Setup time for stop condition	T ₂	0.6	—	
Data setup time, external clock (MCLK) above 12.8MHz	T ₃	0.1	—	μs
Data setup time, external clock (MCLK) below 12.8MHz		0.6	—	
Data hold time	T ₄	0	0.9	μs
High period of the SCL clock	T ₅	0.6	—	
Low period of the SCL clock	T ₆	1.3	—	
Bus free time between stop and start conditions	T ₇	1.3	—	
Rise time for both SDA and SCL signals	—	—	300	ns
Fall time for both SDA and SCL signals	—	—	300	
Capacitive load for each bus line	CB	—	400	pF

Table 11 I2C Fast Mode plus (FM+) Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	—	0	1	MHz
Hold time for start condition	T ₁	0.26	—	μs
Setup time for stop condition	T ₂	0.26	—	
Data setup time, external clock (MCLK) above 24MHz	T ₃	0.05	—	μs
Data setup time, external clock (MCLK) between 12.8-24MHz		0.1	—	
Data setup time, external clock (MCLK) below 12.8MHz		0.6	—	
Data hold time	T ₄	0	—	μs
High period of the SCL clock	T ₅	0.26	—	
Low period of the SCL clock	T ₆	0.5	—	
Bus free time between stop and start conditions	T ₇	0.5	—	
Rise time for both SDA and SCL signals	—	—	120	ns
Fall time for both SDA and SCL signals	—	—	120	

Capacitive load for each bus line	CB	—	550	pF
-----------------------------------	----	---	-----	----

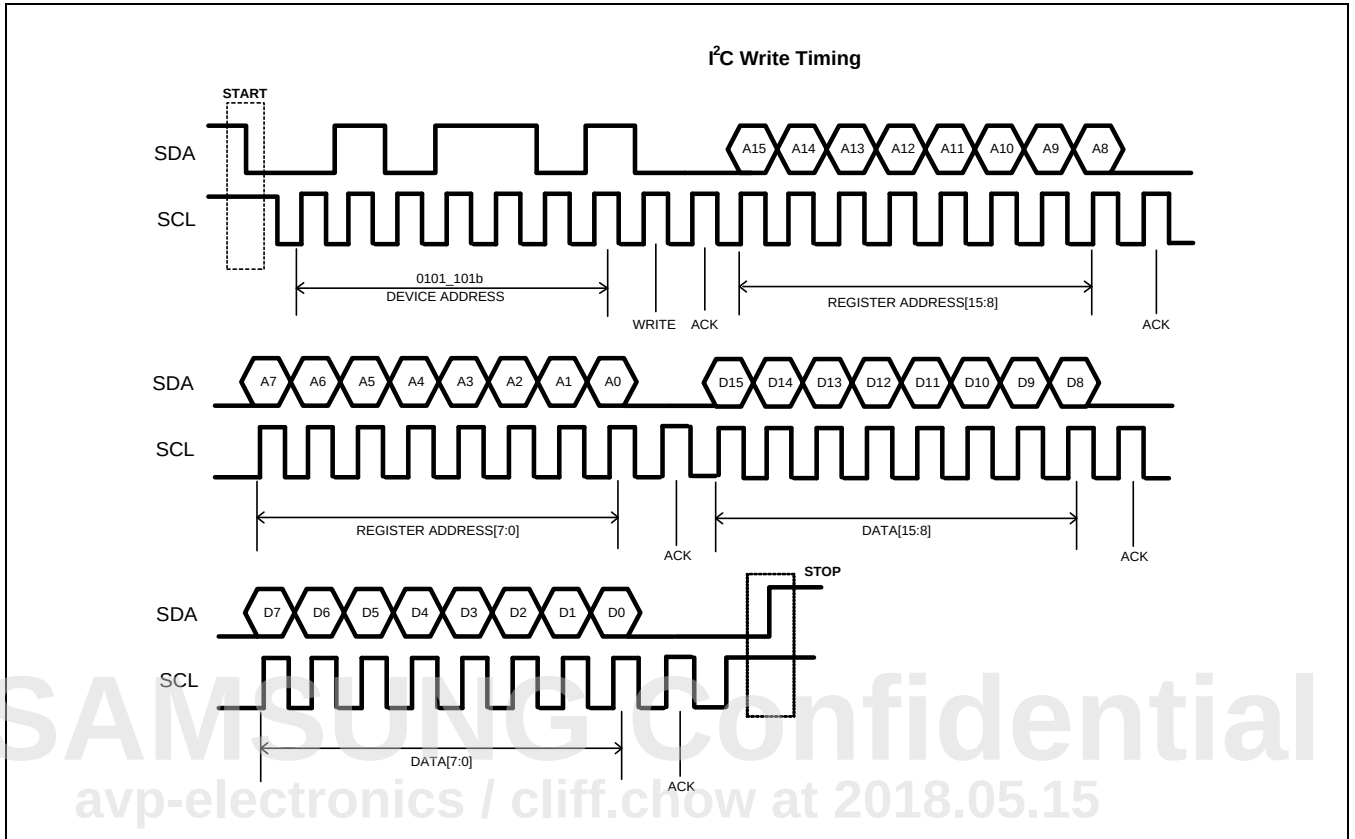


Figure 14 CCI Write Timing Example

NOTE: The device address can be changed by pin configuration of I2C_ID, as described in the pad description.

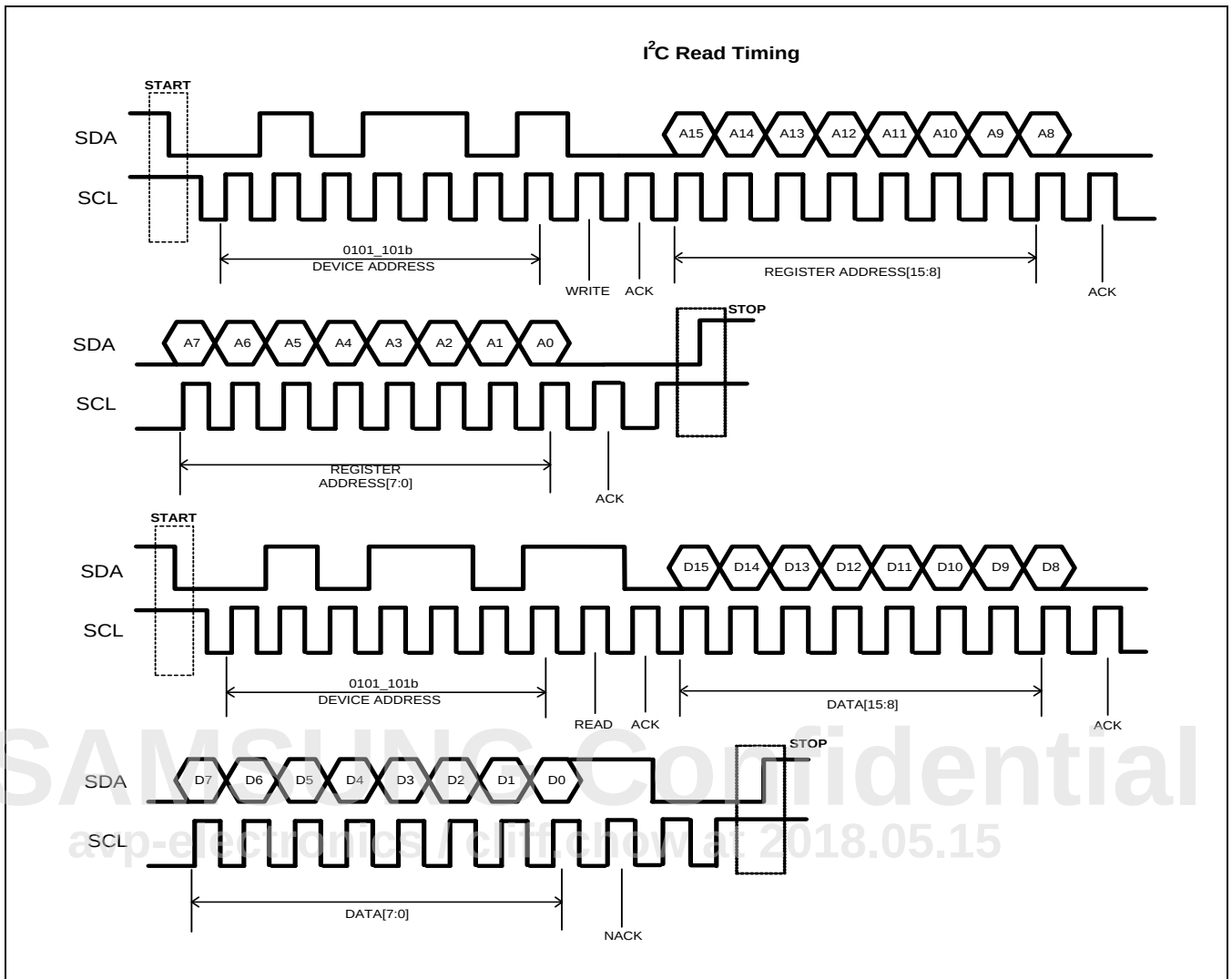


Figure 15 CCI Read Timing Example

NOTE: The device address can be changed by the pin configuration of I2C_ID, as described in the pad description.

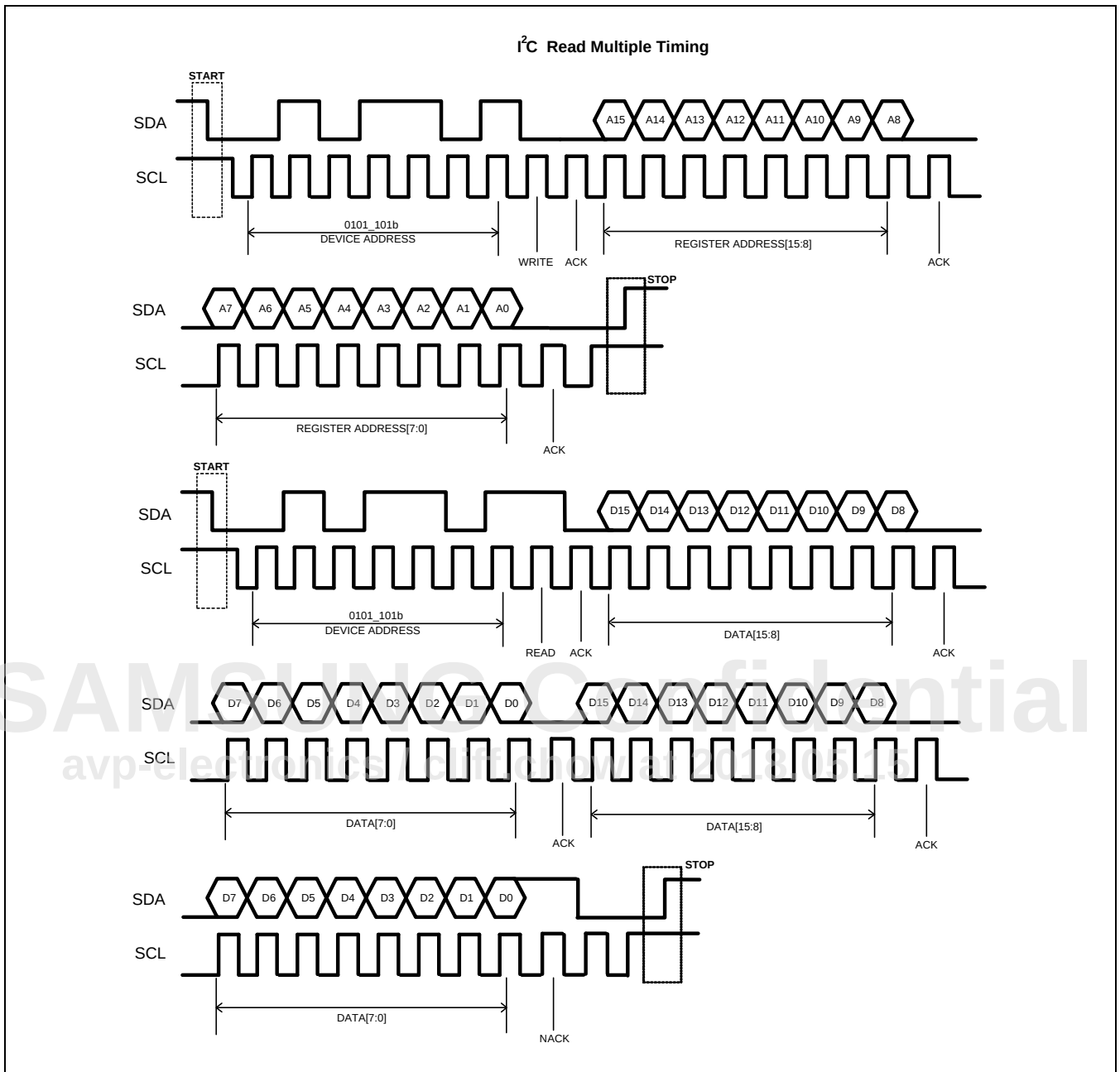


Figure 16 CCI Read Multiple Timing Example

NOTE: The device address can be changed by the pin configuration of I2C_ID, as described in the pad description.

It is possible to configure up to two I2C slave addresses using I2C_ID pins.

Table 12 I2C ID Address (XCE Pad)

XCE	Slave Address (7-bit + Read Mode)	Slave Address (7-bit + Write Mode)	Comment
0	0010_0001b/21h	0010_0000b/20h	Address 1
1	0101_1011b/5Bh	0101_1010b/5Ah	Address 2

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5 Functional Features

5.1 Block Diagram

The S5K3P9SX04 is a progressive-scan sensor that generates a stream of pixel data at a constant frame rate. It uses an on-chip phase-locked loop (PLL) to generate all internal clocks from a single master input clock running between 12MHz and 60MHz. Output interface clocks may be generated by dedicated PLL for maximal flexibility in interface frequency and for EMI avoidance.

The block diagram of the sensor is shown in [Figure 17](#).

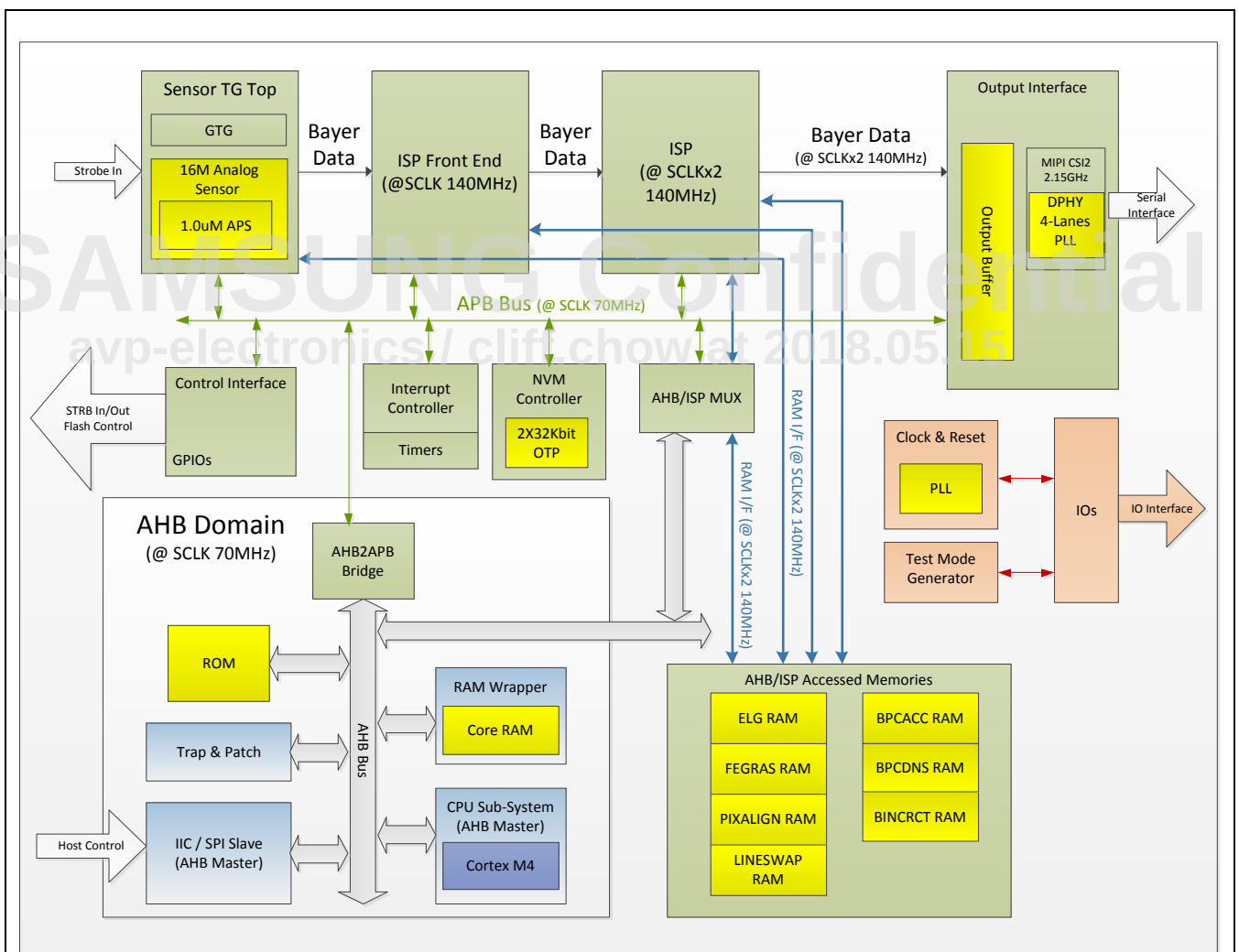


Figure 17 Function Block Diagram

The image sensor has an on-chip ADC. A column parallel ADC scheme is used for low-power analog processing.

The analog output signal of each pixel includes some temporal random noise caused by the pixel reset action and some fixed pattern noise caused by the in-pixel amplifier offset deviation. To eliminate these noise components, a Correlated Double Sampling (CDS) circuit is used before converting to digital.

The output from the ADC is a 10-bit value for each pixel in the array. The ADC output passes through a digital processing signal chain, which provides further data path corrections and applies digital gain.

The digital processing chain performs - row noise reduction, frame black level recovery, compensation for various analog circuit variations and non-uniform pixel levels. These compensations are very flexible allowing different compensation per pixels type & exposure level.

In addition the S5K3P9SX04 perform & activate deterministic pattern generator, a PCM/DPCM compressor and a MIPI CSI-2 frame formatter with embedded line support.

The sensor is interfaced using a set of control and status registers that can be used to control many aspects of the sensor behavior, including frame size, exposure and gain setting. These registers can be accessed through a CCI or SPI interface.

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5.2 Pixel Array Addresses

Addressable pixel array is defined as the pixel address range to be read. The addressable pixel array can be assigned anywhere on the pixel array. The addressed region of the pixel array is controlled by the `x_addr_start`, `y_addr_start`, `x_addr_end`, `y_addr_end`, `x_output_size` and `y_output_size` register.

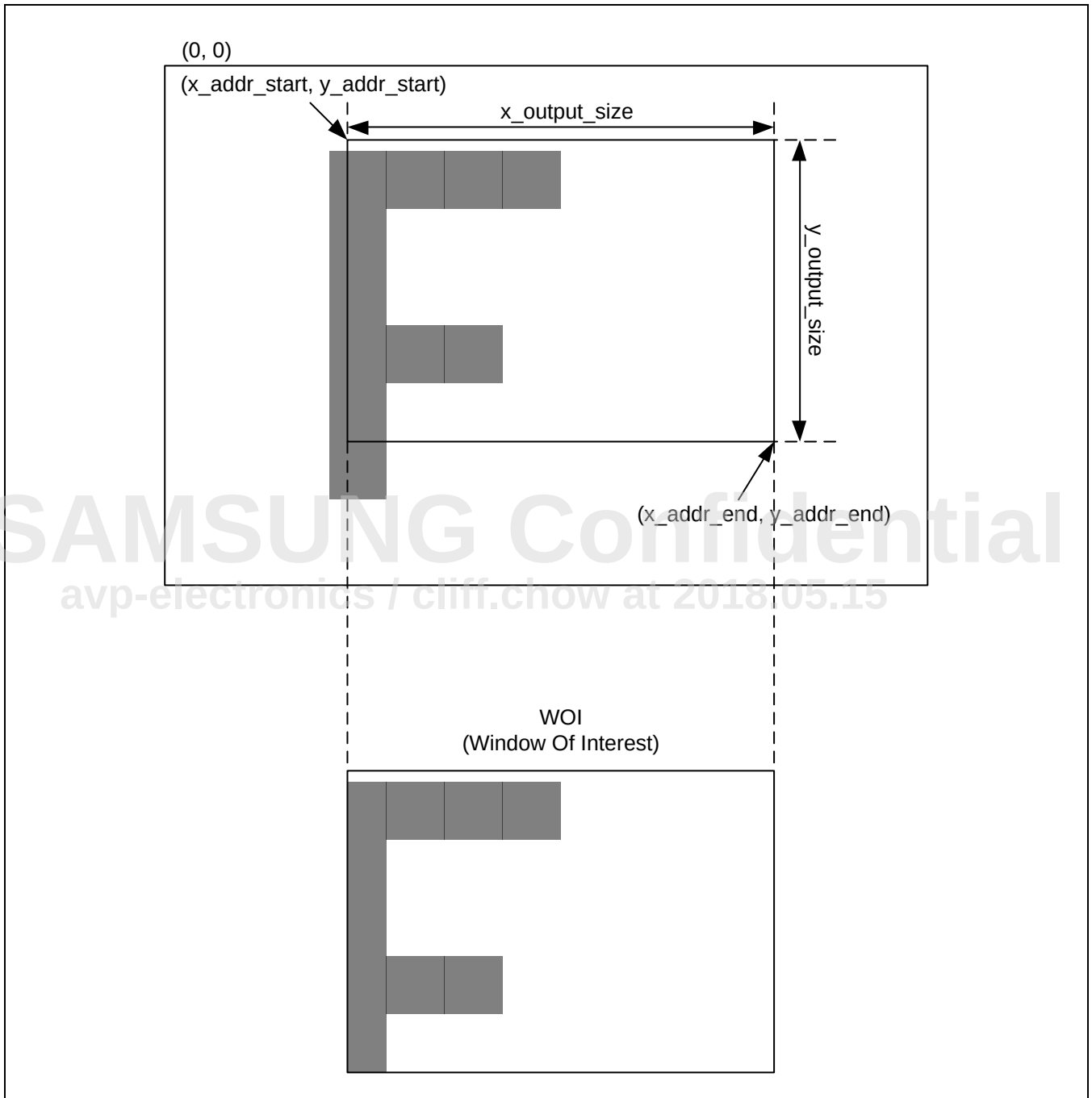


Figure 18 Window of Interest on Pixel Array

5.3 Horizontal Mirror and Vertical Flip

The pixel data is normally read out from left to right in the horizontal direction and from top to bottom in the vertical direction. By changing the mirror/flip mode, the read-out sequence can be reversed, and the resulting image can be flipped like a mirror image. Pixel data is then read out from right to left in horizontal mirror mode and from bottom to top in vertical flip mode. The horizontal mirror and the vertical flip mode can be programmed by the image orientation register.

The sensor module supports four possible pixel readout orders, as described in the sections below:

a) Standard Readout

The addressed region of the horizontal pixel data output is controlled by the `x_addr_start` and `x_output_width` registers. The addressed region of the vertical pixel data output is controlled by the `y_addr_start` and `y_output_depth` registers.

b) Horizontally Mirrored Readout

The addressed region of the horizontal pixel data output is controlled by the `x_addr_end`, and `x_output_width` registers, and the vertical pixel data output is the same as that of the standard readout.

c) Vertical Flipped Readout

The horizontal pixel data output is same as that of the standard readout, and the addressed region of the vertical pixel data output is controlled by the `y_addr_end` and `y_output_depth` registers.

d) Horizontally Mirrored and Vertically Flipped Readout

The addressed region of the horizontal pixel data output is controlled by the `x_addr_end` and `x_output_width` registers. The addressed region of the vertical pixel data output is controlled by the `y_addr_end` and `y_output_depth` registers.

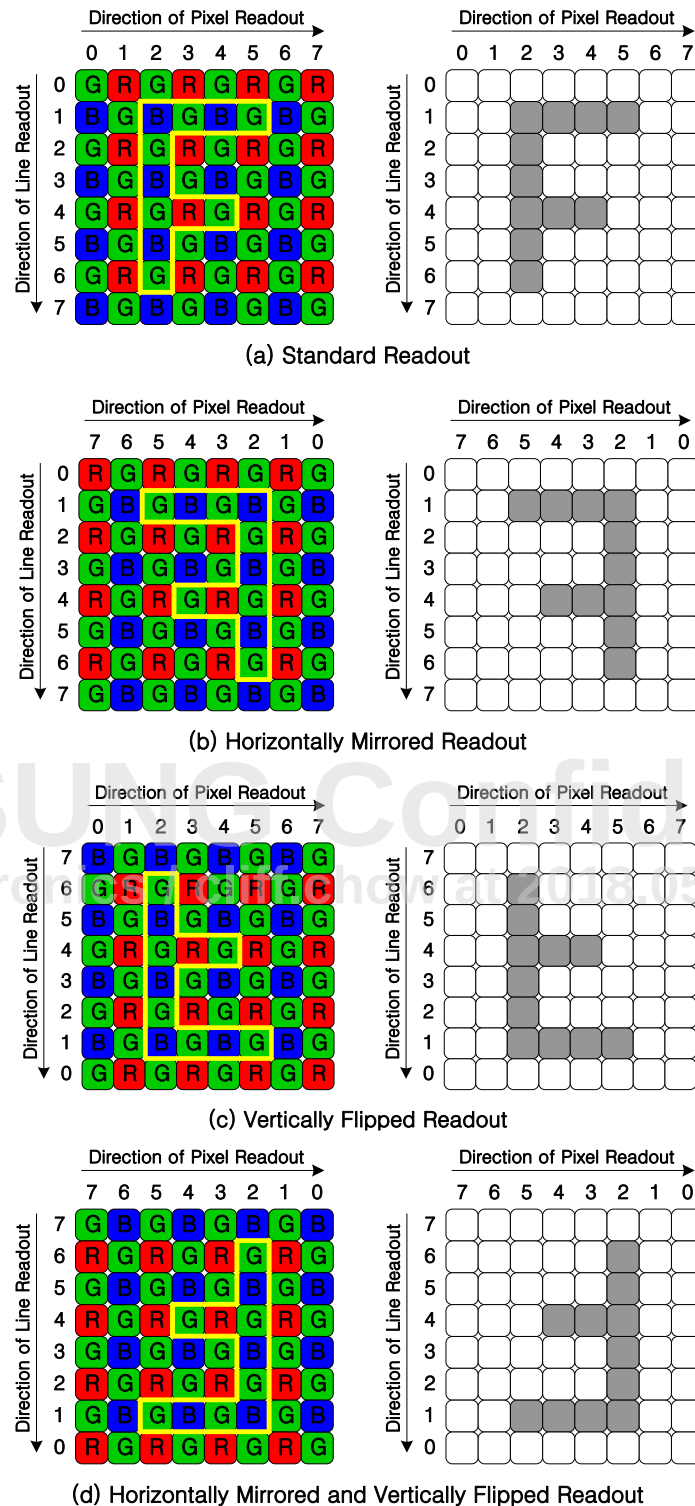


Figure 19 Horizontal Mirror and Vertical Flip @ Normal/Binning Output mode

5.4 Averaged Readout (Binned Readout)

By programming the averaged readout enable register, the sensor can be configured to read out pixel data that has been averaged with adjacent same color pixels.

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5.5.2 Functional Operation Modes

S5K3P9SX04 can support a wide range of operation modes.

Operation mode is function of several mode parameters such as:

- Interface bandwidth
- Image requested quality (including TetraBayer option)
- HW limitations (minimal H/V-blank, blocks' delay)
- Requested sensor output size and sensor operation mode
- CIS output size and number of bits per pixel e.g. RAW10, RAW8+ (8 bit in DPCM/PCM compression)
- Required Vertical blank time
- Power consumption limitations e.g. use single PLL, limited MIPI lanes, system lower power modes.

Typical operation modes and related typical settings are presented in this table.

Table 13 Typical Functional Operation Modes

APS Pattern	Mode	Frame rate	CIS Output		Binning		Image Mode
			H	V	H	V	
Bayer	16MP 4:3 @30fps	30fps	4656	3504	1	1	Full size
	FHD 4:3 @60fps	60fps	2328	1312	2	2	Binning
	HD 16:9 @120fps	120fps	1360	768	3	3	Binning (crop)
	VGA 4:3 @180fps	180fps	656	496	6	6	Binning (crop)

5.6 PLL and Clock Generator

The S5K3P9SX04 clock system uses system phase-locked loop (PLL), system clock dividers, output PLL, and output clock dividers to generate all internal clocks from a single master input clock running between 12 MHz and 60 MHz.

System clocks may be generated by the output PLL. In this case the system PLL will be powered-down and overall system power reduced. The output PLL's maximum effective VCO for MIPI transmission is 2.15GHz. The output PLL's maximum VCO for system requirements is divided by 2 (1075MHz).

Alternatively dedicated System PLL may be used for maximal flexibility in interface frequency and for EMI avoidance. The maximum system PLL VCO frequency is 1000MHz.

Clock dividers are used to generate all system clocks from one or two PLL sources.

The charge pump clock and the ADC clock are used for A/D conversion circuits, pixel clock and pixel clock/2 are used for pixel processing and sensor control. Bit clock and output clock are set according to the required output rate.

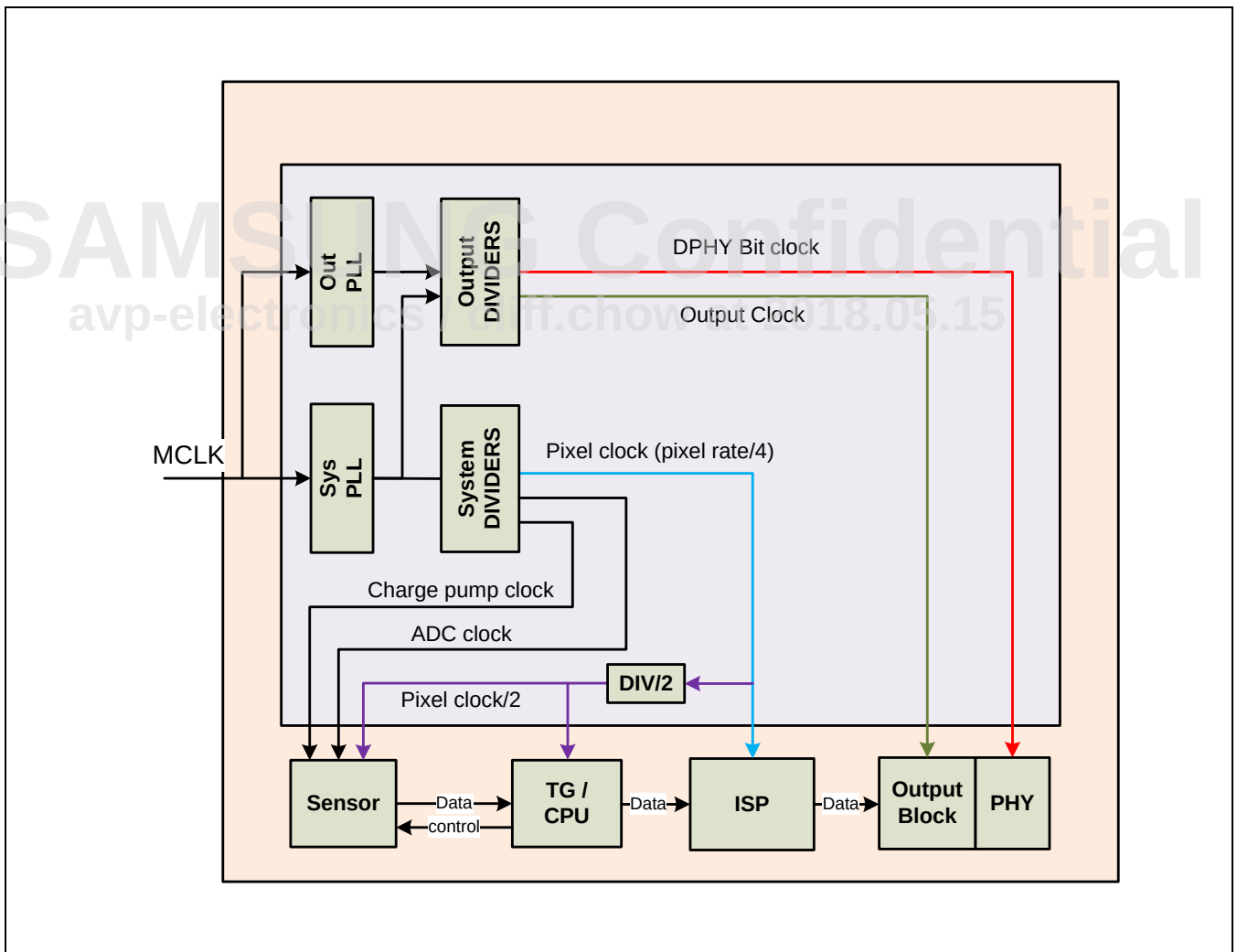


Figure 21 Clock System Block Diagram

5.6.1 Clock Relationships

The host provides the external input clock (with values varying between 12-60MHz) in addition to setting dividers and multipliers in order to get the required video timing and output pixel clocks.

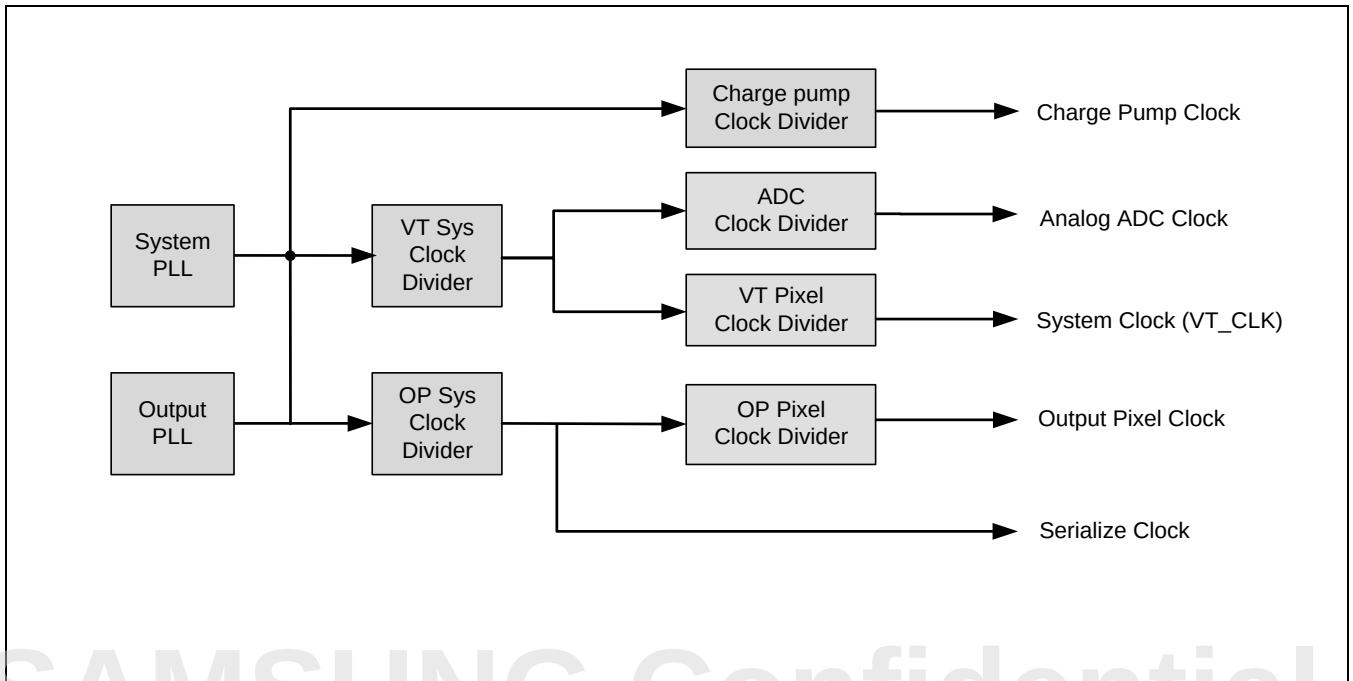


Figure 22 Clock Relationships

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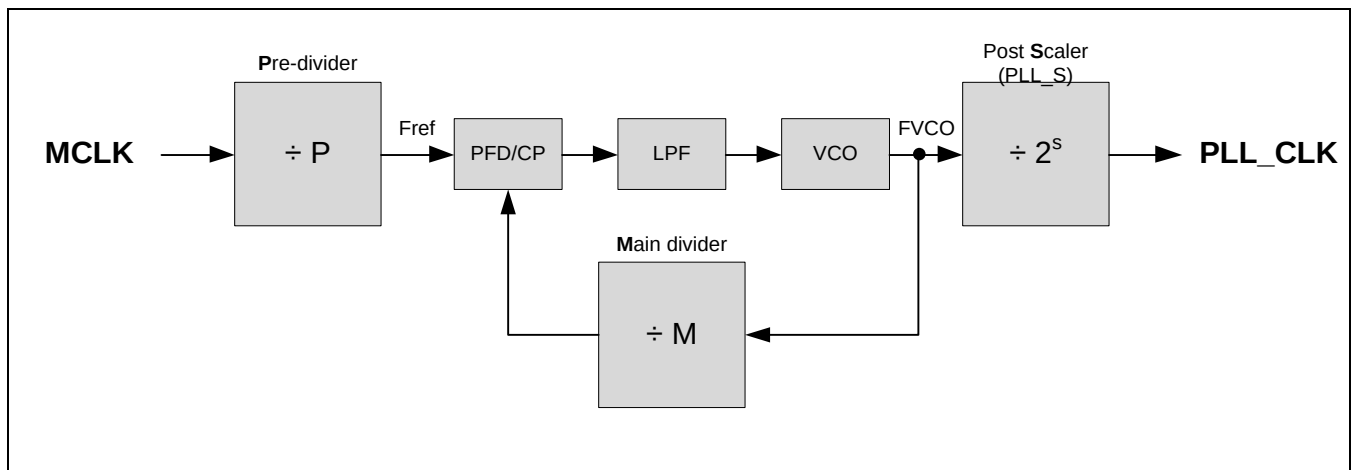


Figure 23 PLL Frequency Synthesis

$$PLL_CLK = MCLK \times \frac{M}{P} \times \frac{1}{2^S}$$

NOTE: Post Scalar Divider (PLL_S) = 0 on S5K3P9SX04 and is not controllable. Post dividers are used for lower frequency synthesis.

Table 14 PLL Component Output Frequency

Parameter	Min.	Typ.	Max.	Unit	Remarks
Input frequency range	12	–	60	MHz	EXTCLK frequency range
System PLL Reference frequency range	2.5	–	5	MHz	Output of system PLL pre-divider (Fref)
System PLL VCO frequency range	500	–	1000	MHz	Output of system PLL multiplier VCO oscillation range (Fvco)
System PLL output frequency range	62.5	–	1000	MHz	Output of system PLL post scaler. Minimum value is just for test. (4 > S ≥ 1)
Output PLL Reference frequency range	6	–	12	MHz	Output of output PLL pre-divider (Fref)
Output PLL VCO frequency range	530	–	1075	MHz	Output of output PLL multiplier VCO oscillation range (Fvco). Effective frequency for MIPI transmission is x2 faster
Output PLL output frequency range	40	–	1075	MHz	Output of output PLL post scaler. Minimum value is just for test. (4 > S ≥ 1). Effective frequency for MIPI transmission is x2 faster

NOTE: For more information about the PLL and clock system control, refer to the application note.

5.6.2 Master Clock Waveform Diagram

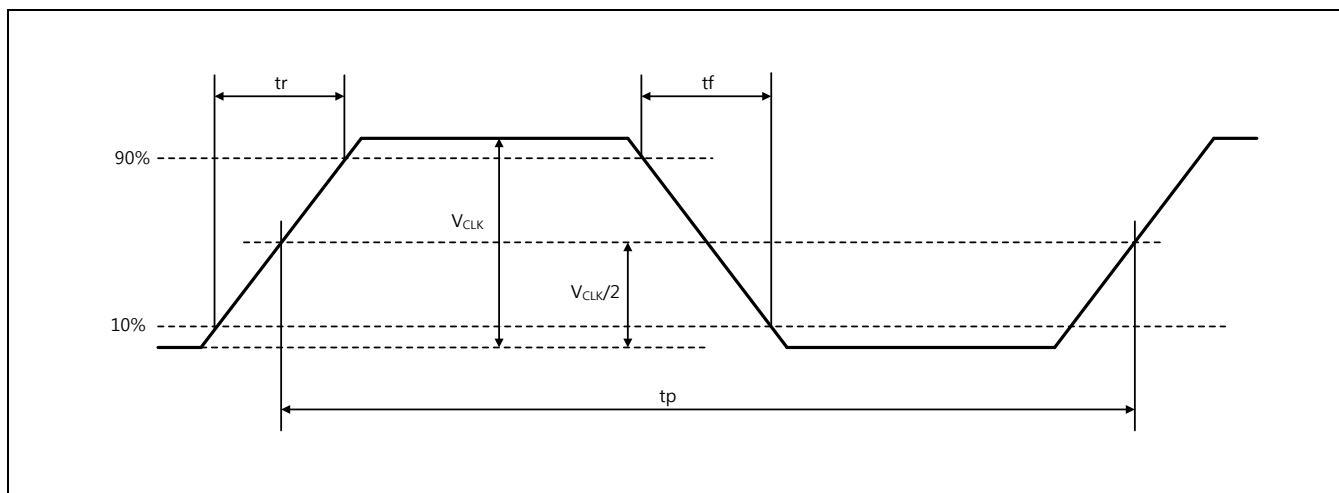


Figure 24 Master Clock Waveform Diagram

MCLK is the input clock to the S5K3P9SX04 sensor, sometimes refer as EXTCLK (external clock).

Table 15 MCLK timing spec

Parameter	Symbol	Min.	Typ.	Max.	Unit	Comment
MCLK clock frequency	MCLK	12	—	100	MHz	—
MCLK period	tp	16.667	—	83.33	ns	—
MCLK rise/fall time	tr/tf	—	—	10		—
MCLK period jitter (peak-to-peak)	Tjitter	—	—	400	ps	—

5.7 Gain Control

5.7.1 Analog Gain Control

Analog gain can be calculated by the following equation:

$$gain = \frac{x}{0x20}$$

NOTE: In S5K3P9SX04, Analog gain is global; there is no per-channel gain. $gain = \frac{x}{32}$
Gain is supported up to X16.

We can also control the divider (default value is 0x20) and multiply it by 2^N .

N is controlled by the following register.

Table 16 Analog Gain Examples

Gain Value	Analog_gain_code_XXX Register Value
X1	0x0020
X2	0x0040
X3	0x0060
X8	0x0100
X10	0x0140
X12	0x0180
X16	0x0200

5.7.2 Digital Gain Control

This block handles digital gains and periodic gain mismatch.

It has two sets of register interfaces:

- SMIA gain registers interface, which is coarse and supports fractional gain of 1/256 scale.
Digital gain of the four Bayer channels is controlled separately using the four parameters shown in the following table. When digital gain is applied, the LSB(s) resulting data shall be padded with zeros.

5.8 Embedded Line

S5K3P9SX04 may be configured to generate an embedded MIPI header with frame information.

GYRO data will be outputted through embedded footer.

5.9 NVM OTP Memory

The OTP memory module is a non-volatile OTP (one-time programming) memory module. This module enables the saving of unique data to each chip at the production stage.

OTP memory is used to store the following information:

- Chip ID data – production history data to be stored during die sorting
- Parameter for Lens shading
- 9Kbit for users

The OTP memory module is a non-volatile OTP (one-time programming) memory module. This module enables the saving of unique data to each chip at the production stage.

OTP memory is used to store the following unique information:

- Chip ID data – production history data to be stored during die sorting
- Parameter for Lens shading

There are two different ways to access each OTP memory:

- Externally, using IO pads by configuring the chip to a dedicated test mode and performing full OTP memory IP write/read protocol.
- Internally, through the internal-bus using the controller.

5.9.1 OTP Read/Write Procedure

Refer to S5K3P9SX04 application note.

5.10 Wide Dynamic Range (WDR) Support

The S5K3P9SX04 Contain a mechanism that enable reaching Wide Dynamic Range (WDR) image capturing at both Video & still modes, allowing high quality image capturing in cases of mixed lightening scenes.

Wide dynamic range acquisition is done by having Multiple Integration Time within the same frame. This is done by interlace pixels of short and long exposure times.

The pixel array of S5K3P9SX04 support interlacing pixels of short and long exposure times within same frame. Amount of exposure difference between short & long exposed pixels is controllable to allow adapting to present scene. ISP functions as Dark Level Correction & Global Offset Correction allows different handling for short & long exposed pixels. Merging short & long exposed pixels is done within external companion ISP (or Application processor).

5.11 Phase Detection Auto Focus (PD AF) Support

TBD

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5.12 Dual camera input Sync

Dual Sync is an attribute that enables the sensor to accept the VSync-in from an external source.

The purpose of such capability is to enable the sensor to sync its transmission of data with other sensors.

Higher level applications will be able to analyze multiple images being transmitted from multiple sensors.

Dual sync will operate correctly only if both sensors are configured to work with the same frame rate, and the sensor can properly fix and adjust only minor drifts which are caused by physical differences between the sensors.

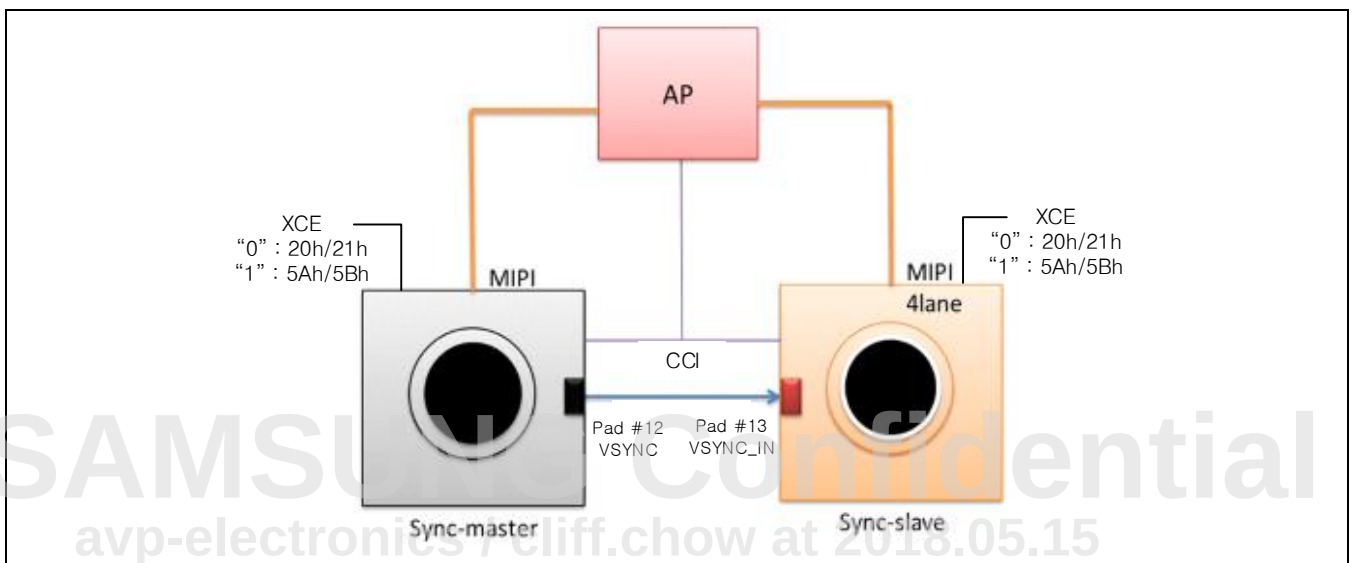


Figure 25 Dual Camera system

The Dual Sync feature monitors 2 intervals on the slave sensor (i.e. the sensor which will be in sync with the external source, i.e. the master sensor) which define the time in lines between:

- VSync-in to VSync-in – The time between 2 consecutive VSync-in received from the master.
- VSync-in to VSync-out – The time between VSync-in from the master and the internal VSync-out generated by the slave.

	Label	Min	Max	Units
Dual Sync frames delay	Tframe_dly	1	2	frames
Vsync (I) to Vsync (O) exposure time < Frame Time - 60 H-times	T0	5	-	H-Times
Vsync (I) to Vsync (O) exposure time >= Frame Time - 60 H-times		60	-	

Dual Sync frames delay in S5K3P9SX04 is 1 to 2 frames as shown below:

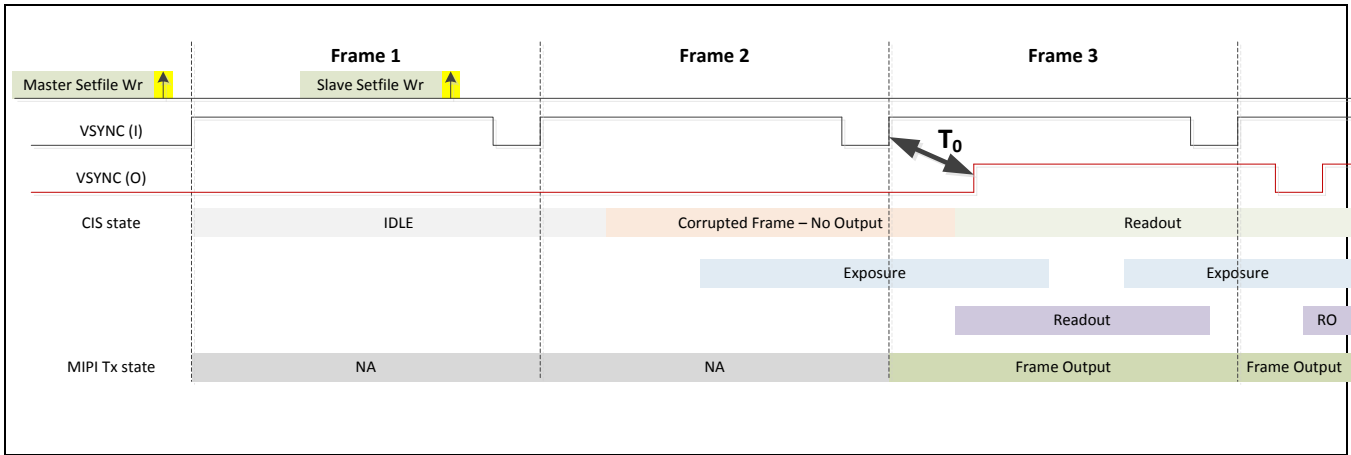


Figure 26 Dual camera sync initialization sequence

5.13 Gyro synchronizer

For image stabilization, CIS provide gyro data synchronized with image capture time.

It delivers gyro data to backend processor through MIPI chain. Backend processor is able to distinguish gyro data with data type (0x30~0x39).

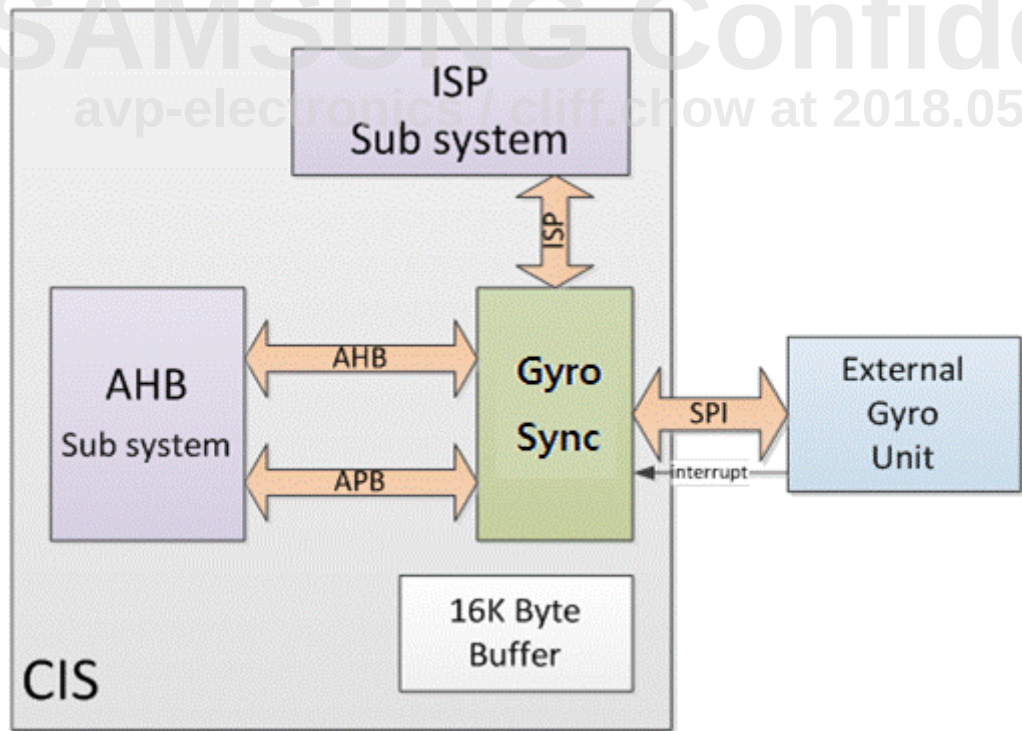


Figure 27 Gyro synchronizer Schematic

The gyro synchronizer block interfaces with an external Gyro Module via SPI protocol (EIS is the master).

The user configures the Gyro module via the gyro synchronizer block. During image capture the gyro synchronizer handles the Gyro samples independently. The external Gyro sends an interrupt once data is ready. Once the EIS receives this interrupt it will read the samples and save them in a local buffer. The gyro synchronizer also receives ISP indications in order to synchronize the Gyro samples with the image stream. The Gyro samples are embedded inside the image footer by the ELG block.

A 16K Byte buffer is used to store the Gyro samples that are received during the frame.

The gyro synchronizer can operate with various Gyro units via SPI protocol and interrupt indication and it supports various axis sample rates combined with image throughputs.

5.14 Test Pattern

S5K3P9SX04 may be configured to generate deterministic test patterns. Refer to the S5K3P9SX04 application note.

5.15 Output Data Interface

S5K3P9SX04 MIPI CSI-2 interface is a four-lane high-speed serial interface that connects the camera sensor to a host processor. Maximum bitrate of MIPI of S5K3P9SX04 is 2.15Gbps per lane.

S5K3P9SX04 supports all mandatory requirements in MIPI CSI-2 version 1.00 and DPHY 1.2 specifications. Please see MIPI DPHY 1.2 specification for details.

- Main output frame rates:
 - Capture: 16MP 30fps
 - FHD video: 4MP 60fps
 - HD video: 120fps
 - VGA video: VGA 180fps
- MIPI CSI-2: Four lanes with a maximum of 2.15Gbps (bit clock rate)
- Supported data types: RAW8, RAW10
- Data output types:
 - Video stream
 - PDAF stream
 - Embedded lines (GYRO output)
- Integrated MIPI DPHY
- MIPI transmit only
- Support up to four data lanes, one clock lane,
- Support swap in MIPI data lanes.

- Only +/- swap (not support between lanes)
- ULPM: Ultra low power mode supported

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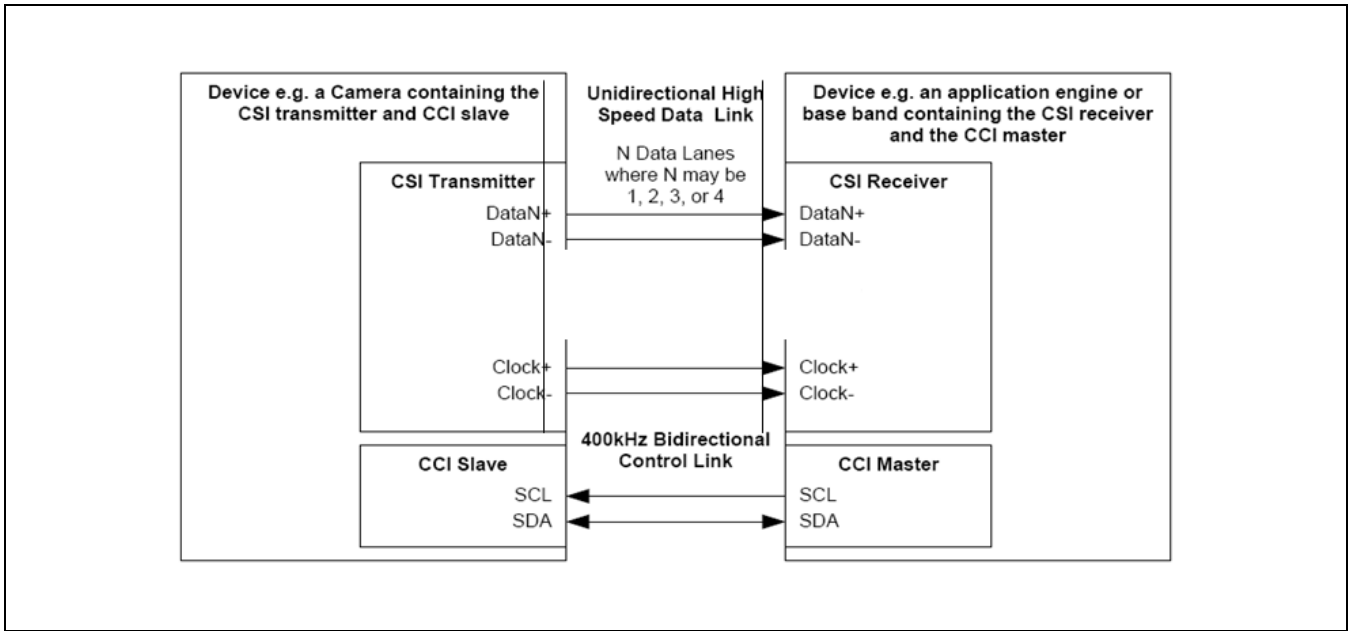


Figure 28 CSI-2 and CCI Transmitter and Receiver Interface

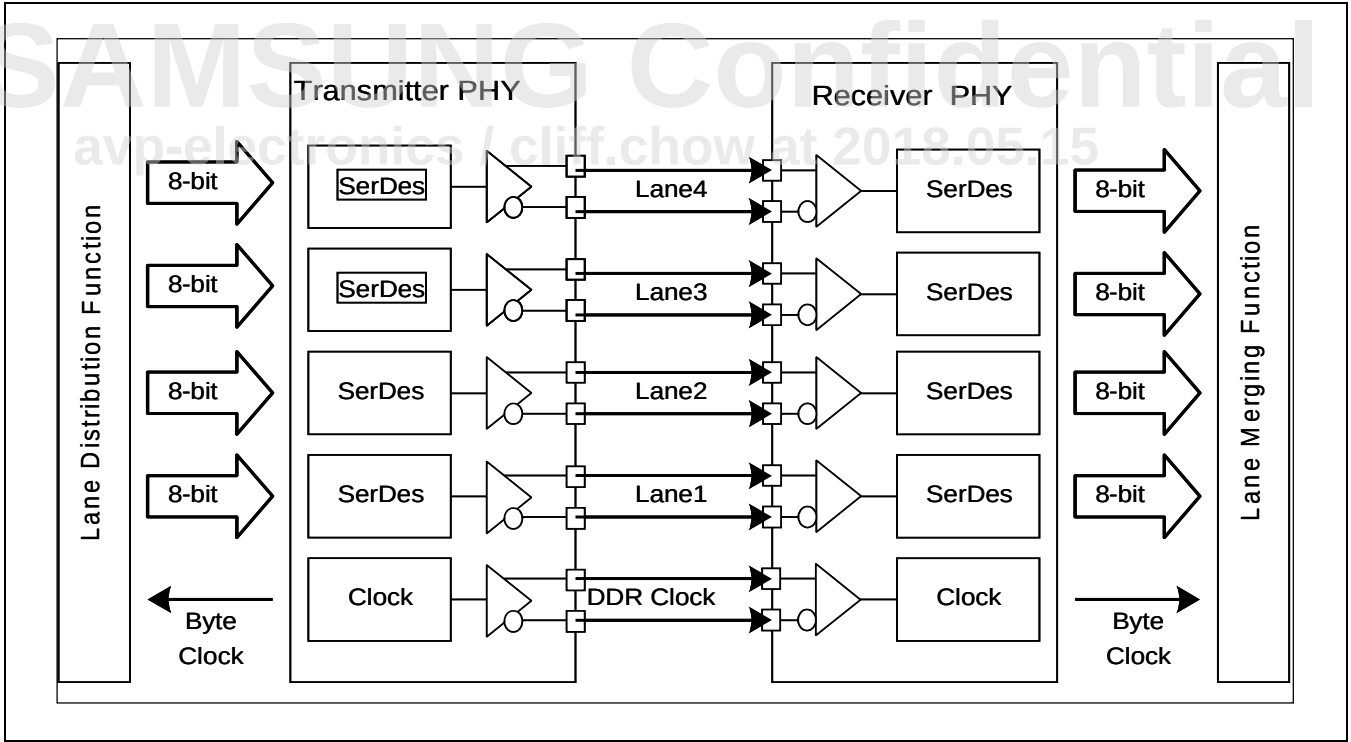


Figure 29 Four Lane Transmitter and Four Lane Receiver Example

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

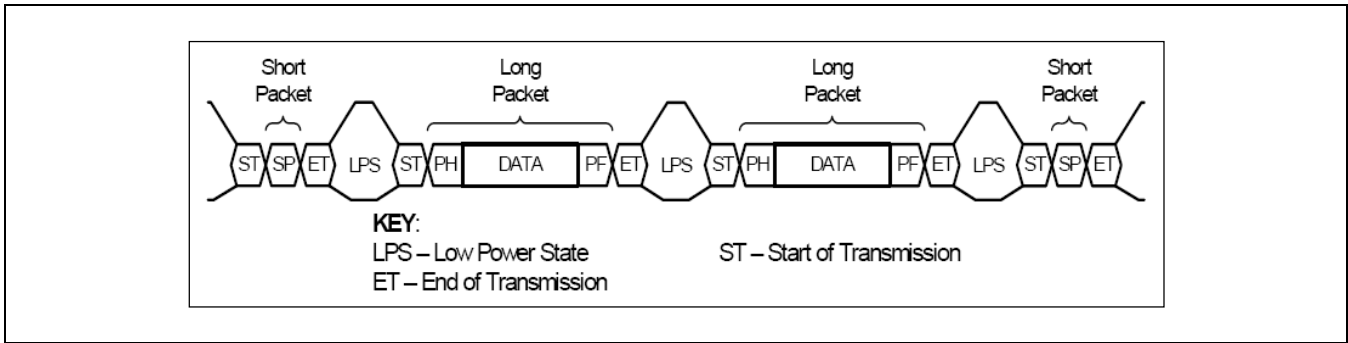


Figure 30 Low Level Protocol Packet Overview

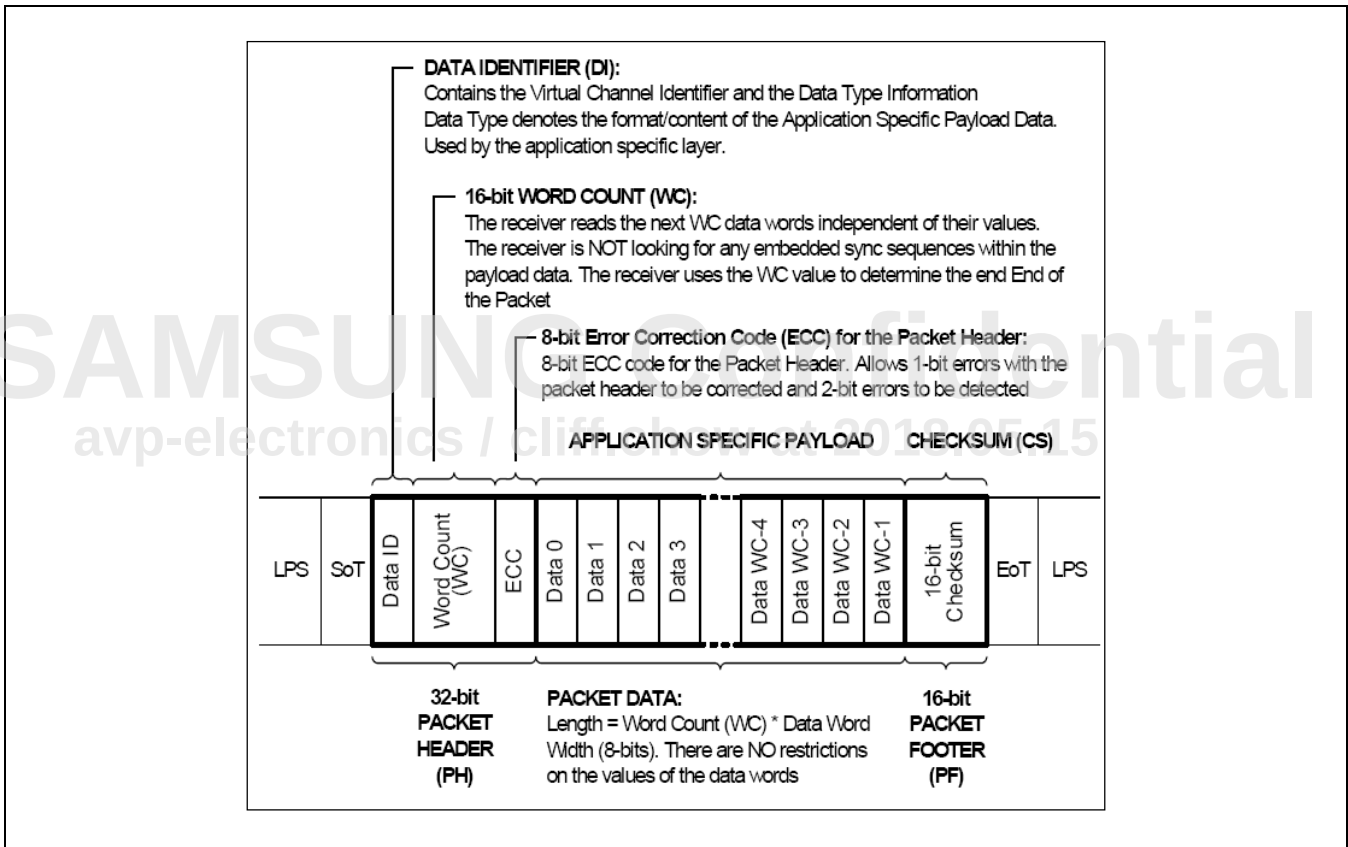


Figure 31 Long Packet Structure

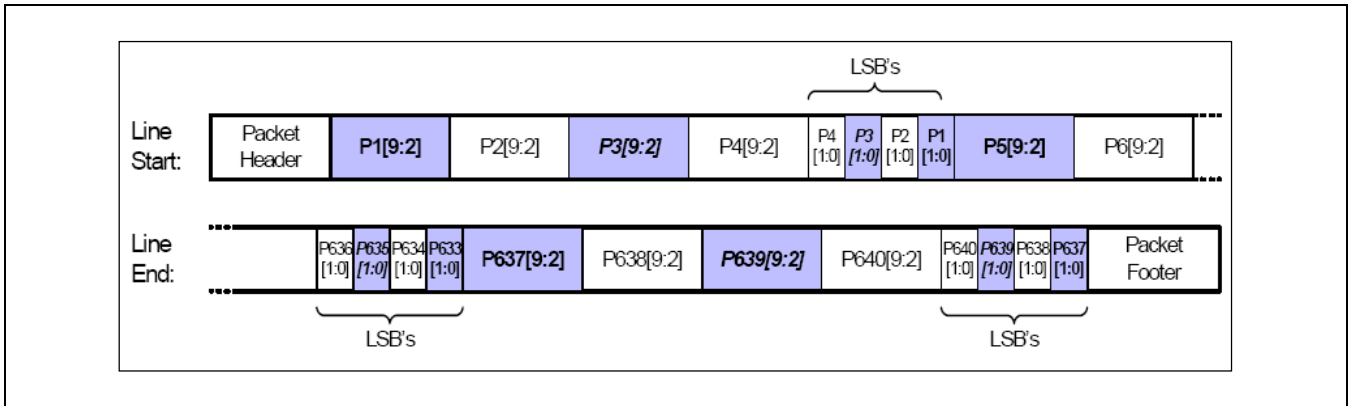


Figure 32 RAW10 Transmission

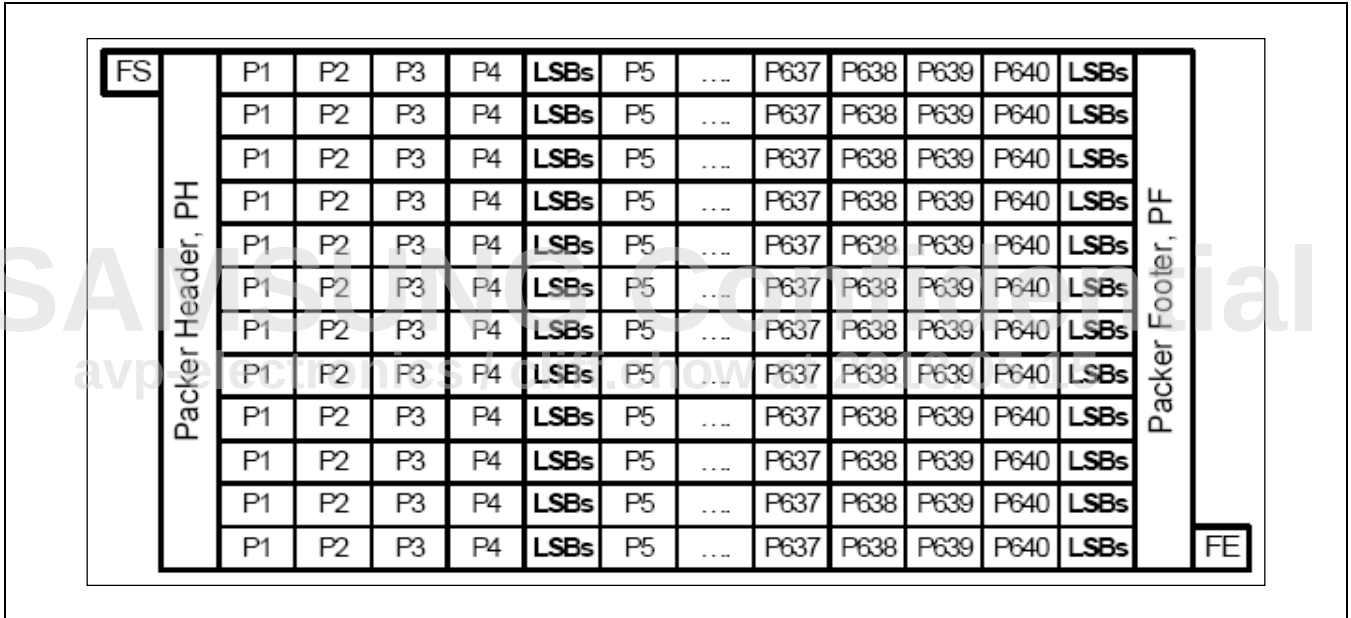


Figure 33 RAW10 Frame Format

6

Electrical Characteristics

6.1 Absolute Maximum Rating

Table 17 Absolute Maximum Rating

Description	Symbol	Min.	Typ.	Max.	Unit
Digital absolute Max.	VDDD (Max.)	-0.5	—	1.5	V
Analog absolute Max.	VDDA (Max.)	-0.3	—	3.6	
I/O absolute Max.	VDDIO (Max.)	-0.3	—	3.6	
Digital input voltages absolute Max ⁽¹⁾	VIP (Max.)	-0.3	—	VDDIO + 0.3	
VCAP analog voltage absolute Max ⁽²⁾	VCAP	-0.3	—	4.2	
Storage temperature	TSTR	-40	—	85	°C

NOTE:

1. Digital Inputs: MCLK, RSTN, I2C_SPI_N_SEL, XCE, SDO, SDI, SCK, GPIO_1/2/3
2. Voltage on external analog capacitors

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6.2 Operating Conditions

Table 18 Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit
Digital power supply ⁽¹⁾	VDDD	0.95	1.05	1.15	V
Analog power supply ⁽²⁾	VDDA	2.7	2.8	2.9	
I/O supply	VDDIO	1.7 (or 2.7)	1.8 (or 2.8)	1.9 (or 2.9)	
Digital input voltages ⁽³⁾	VIP	0	–	VDDIO	
VCAP analog voltage	VCAP	0	–	4.2	
Test temperature(T_a) ⁽⁴⁾	TTEST	21	23	25	°C
Optimum operating temperature(T_j) ⁽⁵⁾	TOPT	0	–	60	
Normal operating temperature(T_j) ⁽⁶⁾	TOPR	–20	–	60	
Functional operating temperature(T_j) ⁽⁷⁾	TFUNC	–20	–	70	

NOTE:

1. Digital Supply Tolerances: Lower limit 1.05V – 100 mV, Upper Limit: 1.05V + 100mV
2. Analog Supply Tolerances: Lower limit 2.8V – 100mV, Upper Limit: 2.8V + 100mV
3. Digital Inputs: MCLK, RSTN (XSHUTDOWN), SCL, SDA
4. Test Temperature – image quality test conditions
5. Optimum Operating Temperature – no visible degradation in image quality
6. Normal Operating Temperature – camera produces acceptable images
7. Functional Operating Temperature – camera fully functional
8. All input power is the standard voltage of the chip input power pad.

6.3 DC Characteristics

Table 19 DC Characteristics

(VDDA = 2.7 V to 2.9 V, VIP = 1.8 V $\pm$ 0.1 V or 2.8 V $\pm$ 0.1 V, T_j = – 20 to + 70 °C, CLOAD = 20 pF)

Characteristics	Symbol	Condition	Min.	Typ.	Max.	Unit
Input voltage	VIH	–	0.7 × VIP	–	VIP + 0.3	V
	VIL	–	–0.3	–	0.3 × VIP	
Input leakage current	IIL	VIN = VIP or VSS	–10	–	10	μA
High level output voltage	VOH	IOH = –100μA	VDDIO – 0.2	–	–	V
Low level output voltage	VOL	IOL = 100μA	–	–	0.2	
High-Z output leakage current	IOZ	VOUT = VSS or VDDD	– 10	–	10	μA
Input capacitance	CIN	–	–	–	5	pF
Supply current	IHWSBA (1)	Hardware standby mode analog	–	N/A	N/A	μA
	IHWSBD (1)	Hardware standby mode digital	–	N/A	N/A	
	ISTRMA (2)	Streaming mode analog 16M@30fps	–	44.5	57.85	mA
	ISTRMD (2)	Streaming mode digital 16M@30fps	–	127.7	166.01	
	ISTRMIO (2)	Streaming mode digital IO 16M@30fps	–	0.5	0.65	

NOTE:

1. Digital IHWSBD - Digital and Analog Power Off is needed.
2. Conditions: VDDA = 2.8 V, VDDD = 1.05 V, VDDIO = 1.8 V, T_j = 60 °C

6.4 AC Characteristics

Table 20 AC Characteristics

Characteristics	Symbol	Condition	Min.	Typ.	Max.	Unit
External clock frequency (NOTE)	fXCLK	–	12.0	–	60	MHz
External clock duty cycle (NOTE)	fXDUTY	–	45	–	55	%
PLL locking time	tLOCK	–	--	–	500	μs

NOTE: Applied to MCLK pin

1. A Fast-mode I2C-bus device can be used the external clock 14 MHz over range

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7

Lens Specification (CRA : Chief Ray Angle Spec.)

7.1 Target CRA

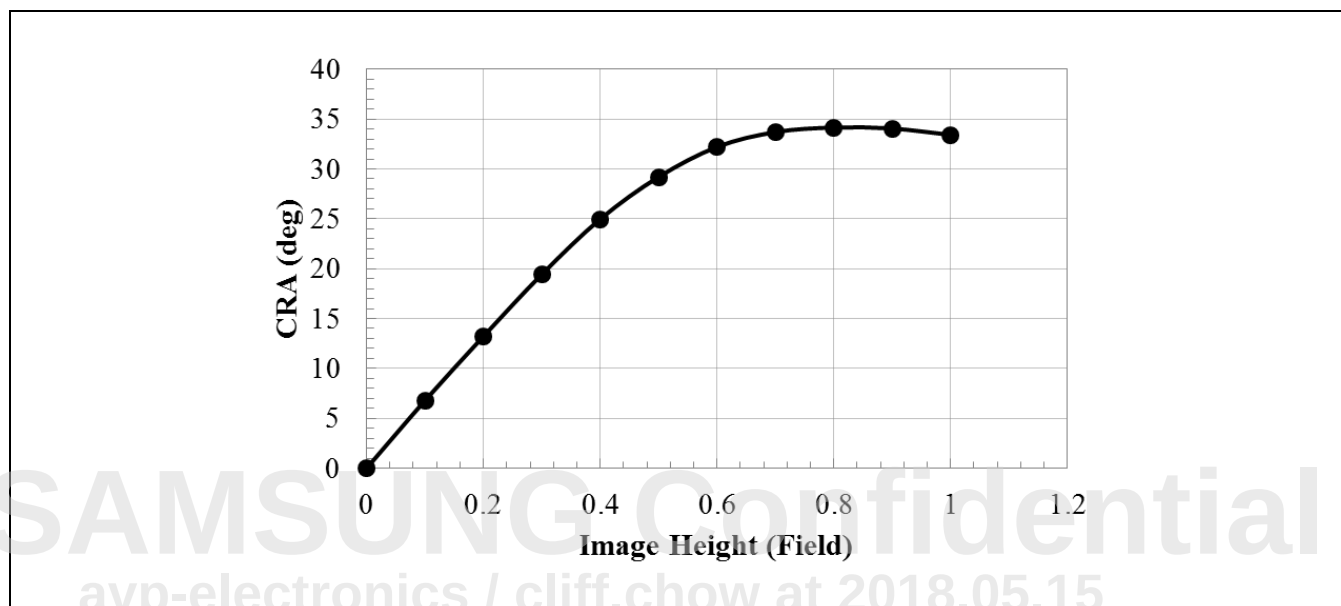


Figure 34 Target CRA

Table 21 Image Height (Field)

Image Height	mm	CRA (deg)
0	0.000	0.00
0.1	0.290	6.79
0.2	0.580	13.22
0.3	0.871	19.43
0.4	1.161	24.98
0.5	1.451	29.20
0.6	1.741	32.21
0.7	2.031	33.70
0.8	2.322	34.16
0.9	2.612	34.06
1	2.902	33.42

7.2 Pixel-Shading characteristics

The following figure illustrates shading characteristics of Green pixel over difference between given lens's CRA and the target CRA by image height.

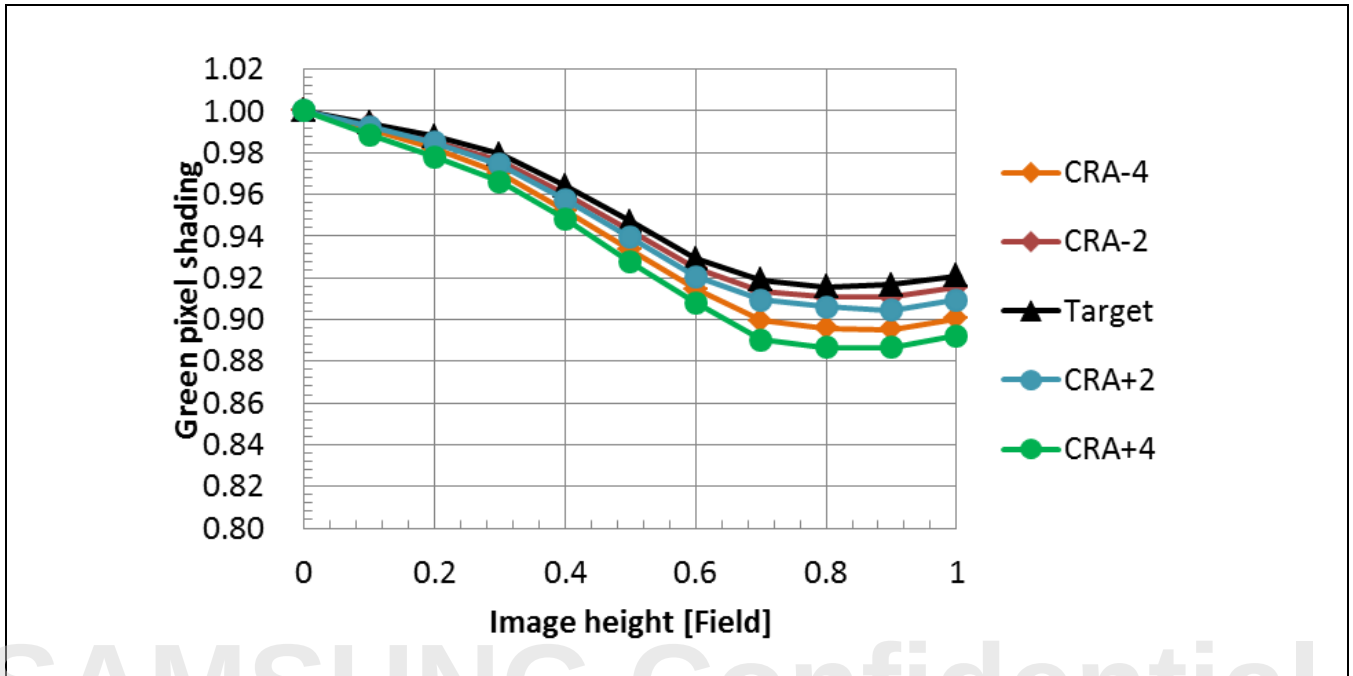


Figure 35 Shading Characteristics of Green

The following figure shows color ratio of B/G and R/G over the target CRA by image height

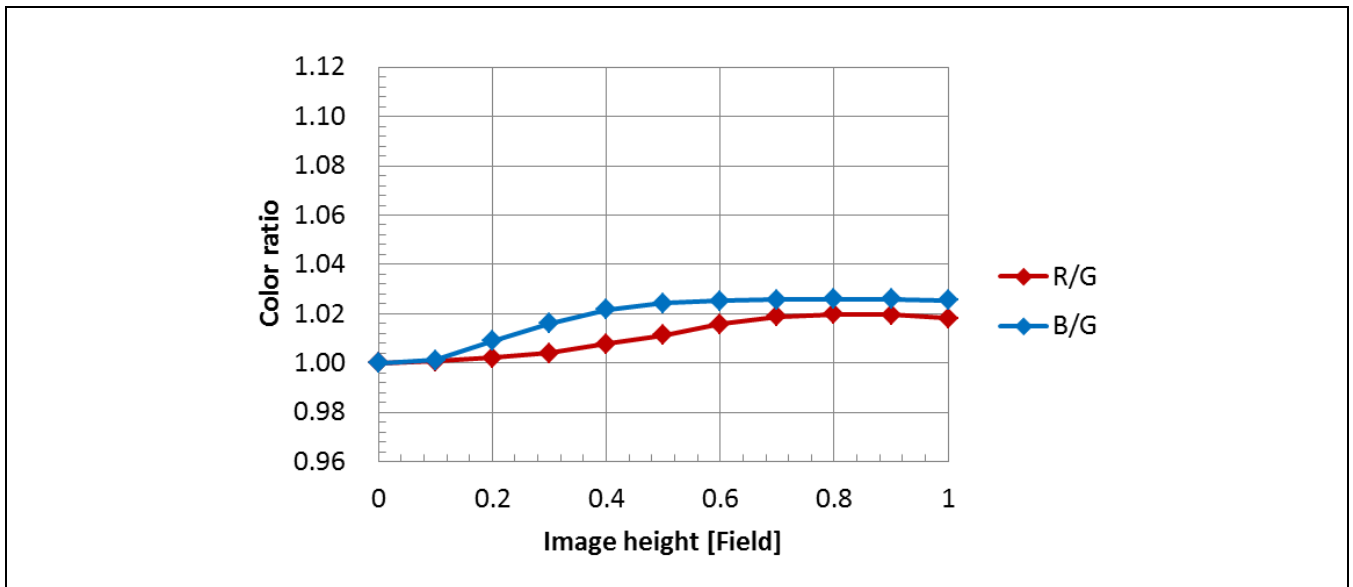


Figure 36 Shading Characteristics of Color ratio

7.3 Spectral Response

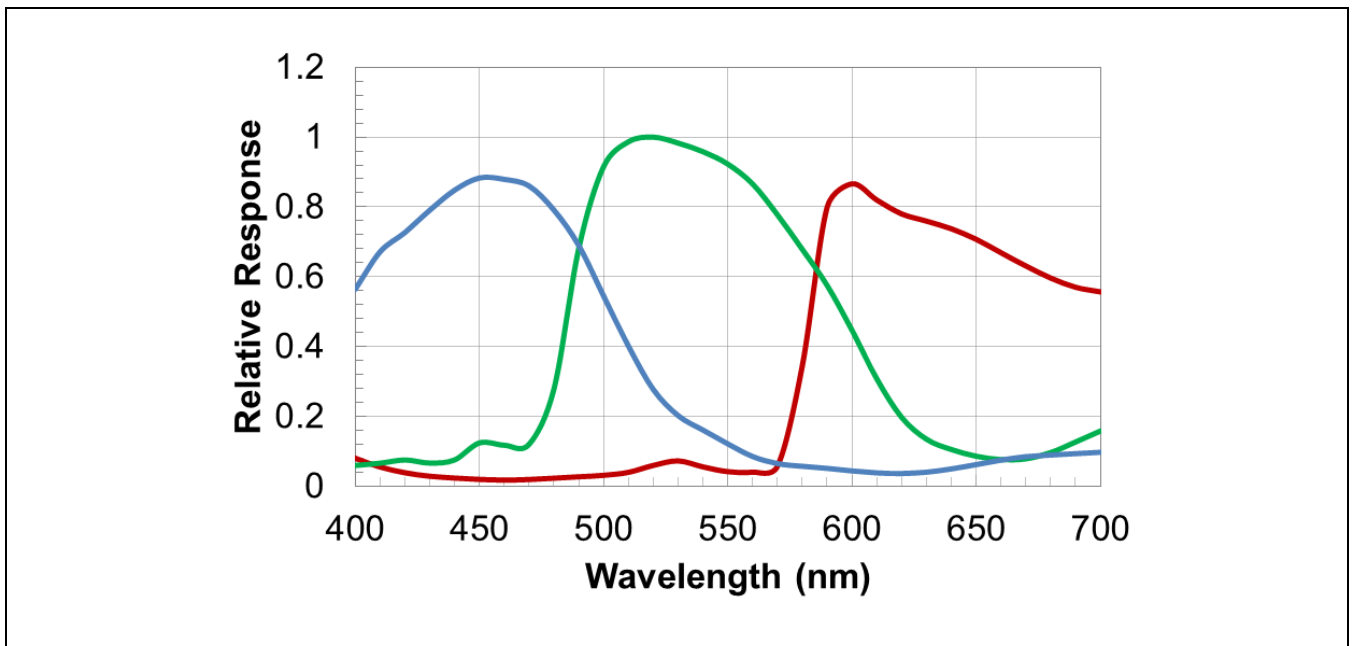


Figure 37 Spectral Response

Table 22 Spectral Response

Wavelength (nm)	R	G	B	Wavelength (nm)	R	G	B
400	0.080	0.060	0.564	560	0.041	0.866	0.085
410	0.055	0.066	0.671	570	0.057	0.777	0.065
420	0.039	0.075	0.727	580	0.346	0.679	0.057
430	0.029	0.067	0.791	590	0.799	0.575	0.051
440	0.024	0.075	0.848	600	0.866	0.447	0.044
450	0.020	0.124	0.883	610	0.820	0.307	0.038
460	0.018	0.118	0.879	620	0.780	0.197	0.036
470	0.019	0.120	0.860	630	0.759	0.135	0.040
480	0.023	0.278	0.791	640	0.737	0.105	0.050
490	0.027	0.681	0.690	650	0.707	0.086	0.062
500	0.032	0.915	0.545	660	0.669	0.076	0.075
510	0.040	0.987	0.401	670	0.631	0.079	0.085
520	0.060	1.000	0.277	680	0.596	0.097	0.089
530	0.073	0.983	0.202	690	0.570	0.127	0.093
540	0.056	0.958	0.161	700	0.557	0.158	0.097
550	0.042	0.923	0.122				

7.4 IR-cut filter selection

The following figure shows IR response of blue channel. The performance of IR-cut filter is very important for image quality at low color temperature. Samsung recommends IR cut filter with cut off from 650 to 1200nm

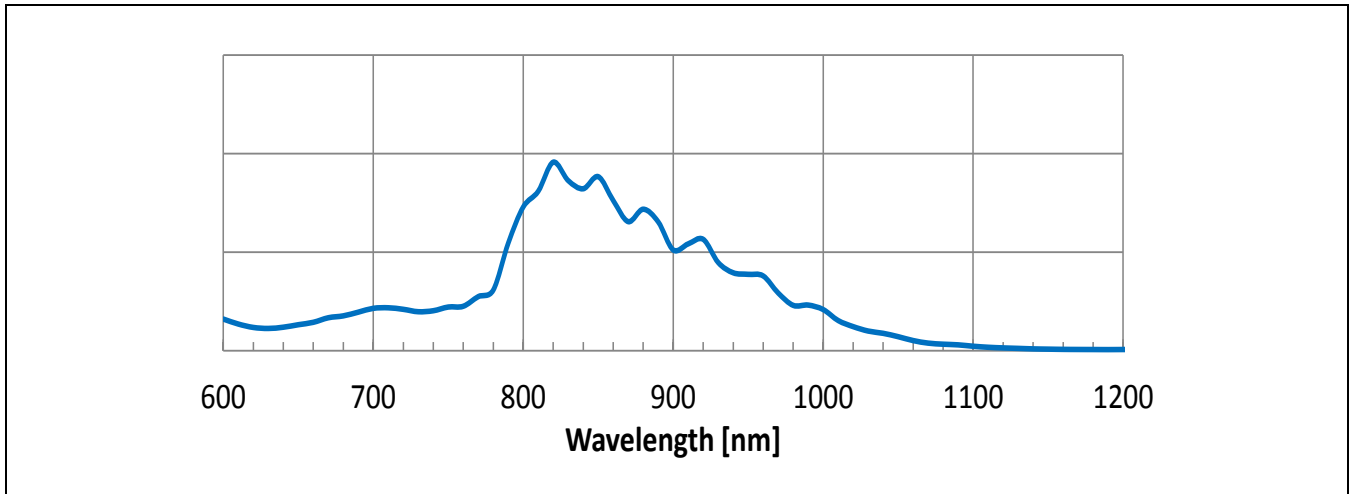


Figure 38 IR Response of Blue channel

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8

Register Description

This section describes the S5K3P9SX04 register map.

Vendor registers are defined for extended functionality control.

8.1 Configuration Register

Table 23 Configuration Register

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rd_general						
api_rd_general_model_id	0x40000000	2	0x3109	RO	EF	Camera module model identification number.
api_rd_general_revision_number_major	0x40000002	1	0xa0	RO	EF	Revision identifier of camera module for tuning/settings change.
api_rd_general_manufacturer_id	0x40000003	1	0x0	RO	EF	Module manufacturers code.
api_rd_general_api_version	0x40000004	1	0x50	RO	EF	API version.
api_rd_general_frame_counter	0x40000005	1	0x0	RO	EF	0 = Wakeup (before first frame). 0xFF = SW stand by. 1-254 = Frame counter value.
api_rd_general_pixel_order	0x40000006	1	0x0	RO	EF	Color pixel order
api_rd_general_config_counter	0x40000007	1	0x0	RO	EF	Configuration change counter(frames aborted or corrupted frames).
api_rd_general_data_pedestal	0x40000008	2	0x40	RO	EF	Data pedestal - typically code 64 for 10-bit systems
api_rd_general_temperature	0x4000000A	2	0x0	RO	EF	16-bit Degrees Celsius in 8:8 format

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rd_general_pixel_depth	0x4000000C	1	0xa	RO	EF	8-bit, 10-bit or 12-bit pixel data
api_rd_general_color_filter_pattern	0x4000000D	1	0x10	RO	EF	Color filter pattern: 0x00 - Bayer GRBG without pdaf 0x10 - Bayer GRBG with pdaf 0x20 - RBW without pdaf 0x30 - RBW with pdaf 0x40 - Black and while CF (all active pixels has the same CF) without pdaf 0x50 - Tetracell pattern (2x2 Quadra CF) without PDAF 0x60 - Tetracell pattern (2x2 Quadra CF) with super PDAF 0x70 - Tetracell pattern (2x2 Quadra CF) with normal PDAF 0x80 - Black and while CF (all active pixels has the same CF) with PDAF
api_rd_general_cra	0x4000000E	1	0x0	RO	EF	CRA - Chief Ray Angle.
api_rd_general_sample_type	0x4000000F	1	0x1	RO	EF	0x01 = Typical sample. 0x00 = Random, Corner samples.
api_rd_general_revision_number_minor	0x40000010	1	0x0	RO	EF	Device revision identifier of the camera module for minor changes.
api_rd_general_additional_specification_version	0x40000011	1	0x6	RO	EF	
api_rd_general_module_date_year	0x40000012	1	0x11	RO	EF	Bits[3:0] - Last digit of manufacturing year.
api_rd_general_module_date_month	0x40000013	1	0x02	RO	EF	Bits[3:0] - Manufacturing month.
api_rd_general_module_date_day	0x40000014	1	0x09	RO	EF	Bits[3:0] - Manufacturing day.
api_rd_general_module_date_phase	0x40000015	1	0x1	RO	EF	Bits[2:0] - Manufacturing phase.
api_rd_general_sensor_model_id	0x40000016	2	0x0	RO	EF	Silicon identification number
api_rd_general_sensor_revision_number	0x40000018	1	0x0	RO	EF	Device Revision Identifier of sensor silicon.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rd_general_sensor_manufacturer_id	0x40000019	1	0x0	RO	EF	Manufacturers Code. Please refer the GIIC Manufacturer Codes Document.
api_rd_general_sensor_firmware_version	0x4000001A	2	0xFA80	RO	EF	Sensor firmware version.
api_rd_general_serial_number	0x4000001C	4	0x0	RO	EF	Running serial number.
api_rd_frame_format						
api_rd_frame_format_model_type	0x40000040	1	0x4	RO	EF	1 = 2 bytes data format.
api_rd_frame_format_model_subtype	0x40000041	1	0x21	RO	EF	Bits[7:4] - number of horizontal descriptors. Bits[3:0] - number of vertical descriptors.
api_rd_frame_format_descriptor_0	0x40000042	2	0x5918	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_1	0x40000044	2	0x5918	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_2	0x40000046	2	0x5db0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_3	0x40000048	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_4	0x4000004A	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_5	0x4000004C	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_6	0x4000004E	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_7	0x40000050	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_8	0x40000052	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_9	0x40000054	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rd_frame_format_descriptor_10	0x40000056	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_11	0x40000058	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_12	0x4000005A	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_13	0x4000005C	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_frame_format_descriptor_14	0x4000005E	2	0x0	RO	EF	[15:12] - Descriptor type. [11:0] - Descriptor Size
api_rd_analog_gain						
api_rd_analog_gain_analog_gain_capability	0x40000080	2	0x2	RO	EF	Describes the sensor analog gain capabilities. 0=Single global analogue gain only 1=Separate channel analogue gains only 2=Analog gain per exposure Short/Long
api_rd_analog_gain_code_min	0x40000084	2	0x20	RO	EF	Minimum recommended analog gain code
api_rd_analog_gain_code_max	0x40000086	2	0x200	RO	EF	Maximum recommended analog gain code
api_rd_analog_gain_code_step	0x40000088	2	0x1	RO	EF	Analog gain code step size
api_rd_analog_gain_type	0x4000008A	2	0x0	RO	EF	Analog gain coding type.
api_rd_analog_gain_m0	0x4000008C	2	0x1	RO	EF	Analog gain m0 constant (16-bit signed integer)
api_rd_analog_gain_c0	0x4000008E	2	0x0	RO	EF	Analog gain c0 constant (16-bit signed integer)
api_rd_analog_gain_m1	0x40000090	2	0x0	RO	EF	Analog gain m1 constant (16-bit signed integer)
api_rd_analog_gain_c1	0x40000092	2	0x20	RO	EF	Analog gain c1 constant (16-bit signed integer)
api_rd_data_format						

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rd_data_format_mode_l_type	0x400000C0	1	0x1	RO	EF	Data Format (0x01 = 2 Byte Data Format)
api_rd_data_format_mode_l_subtype	0x400000C1	1	0x3	RO	EF	Contains the number of data format descriptors used
api_rd_data_format_descriptor_0	0x400000C2	2	0xa0a	RO	EF	Top 10-bits of internal pixel data. Transmitted as RAW10.
api_rd_data_format_descriptor_1	0x400000C4	2	0xa08	RO	EF	Top 10-bits of internal pixel data compressed to 8-bits. Transmitted as RAW8.
api_rd_data_format_descriptor_2	0x400000C6	2	0x808	RO	EF	Top 8-bits of internal pixel data. Transmitted as RAW8.
api_rd_data_format_descriptor_3	0x400000C8	2	0x0	RO	EF	Data type descriptor - not used.
api_rd_data_format_descriptor_4	0x400000CA	2	0x0	RO	EF	Data type descriptor - not used.
api_rd_data_format_descriptor_5	0x400000CC	2	0x0	RO	EF	Data type descriptor - not used.
api_rd_data_format_descriptor_6	0x400000CE	2	0x0	RO	EF	Data type descriptor - not used.
api_rd_current_state						
api_rd_current_state_tg_clk_khz	0x400000D0	4	0x0	RW	ES	Logical TGCLK (on ASIC it's always same as system clock) .
api_rd_current_state_pixel_clk_khz	0x400000D4	4	0x0	RW	ES	the ISP clk * the number of pixels processed in each clk.
api_rd_current_state_mipi_clk_khz	0x400000D8	4	0x0	RW	ES	the MIPI clk.
api_rd_current_state_vendor_error_type	0x400000DC	2	0x0	RW	ES	System level error, caused by wrong user configuration: \n 0 - No error. \n 1 - External clock is not defined. \n 2 - External clock is defined, but out of range. \n 10 - Main pll pre_pll_clk_div is out of range. \n 11 - Main pll pll_ip_freq is out of range. \n 12 - Main pll pll_multiplier is out of range. \n 13 - Main pll pll_s divider is out of

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						range. \n 14 - Main pll VCO freq is out of range. \n 15 - Main pll FOUT freq is out of range. \n 20 - Mipi pll pre_pll_clk_div is out of range. \n 21 - Mipi pll pll_ip_freq is out of range. \n 22 - Mipi pll pll_multiplier is out of range. \n 23 - Mipi pll pll_s divider is out of range. \n 24 - Mipi pll VCO freq is out of range. \n 25 - Mipi pll FOUT freq is out of range. \n 30 - vt_sys_div is out of range. \n 31 - vt_pix_div is out of range. \n 32 - vt_sys_freq is out of range. \n 33 - vt_pix_freq is out of range. \n 34 - op_sys_div is out of range. \n 35 - op_pix_div is out of range. \n 36 - op_sys_freq is out of range. \n 37 - op_pix_freq is out of range. \n 38 - Ddr divider is out of range. \n 39 - DPHY freq divider is out of range. \n 50 - WDR is not allowed on configured binning mode. \n 60 - Two or more functions are assigned to same GPIO. \n 62 - Read from invalid OTP address. \n VENDOR_ERR_NOT_SUPPORTED_COLUMNS_SUBSAMPLING - Configured columns sub-sampling is not supported.
api_rd_current_state_fw_fi rst_error	0x400000DE	2	0x0	RW	ES	FW error - First FW error .
api_rd_current_state_fw_fi rst_error_info	0x400000E0	2	0x0	RW	ES	FW error - FW error information .
api_rd_current_state_isp1 _output_width	0x400000E2	2	0x0	RW	ES	Video output image width include margins .
api_rd_current_state_isp1 _output_height	0x400000E4	2	0x0	RW	ES	Video output image height include margins .
api_rd_current_state_isp2 _output_width	0x400000E6	2	0x0	RW	ES	Virtual channel #2 output image width include margins .
api_rd_current_state_isp2	0x400000E8	2	0x0	RW	ES	Virtual channel #2 output image

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
_output_height						height include margins .
api_rw_general_setup						
api_rw_general_setup_mode_select	0x40000100	1	0x0	RW/S	EF	0 = Software Standby. 1 = Streaming.
api_rw_general_setup_image_orientation	0x40000101	1	0x0	RW/C	EF	Image orientation i.e. horizontal mirror and vertical flip. 0 = No horizontal mirror, no vertical flip. 1 = Horizontal mirror, no vertical flip. 2 = No vertical mirror, vertical flip. 3 = Horizontal mirror and vertical flip.
api_rw_general_setup_software_reset	0x40000103	1	0x0	RW/R	EF	1 = Software reset.
api_rw_general_setup_grouped_parameter_hold	0x40000104	1	0x0	RW/C	EF	The grouped parameter hold register disables the consumption of integration, gain and video timing parameters. 0 = Consume as normal, 1 = Hold.
api_rw_general_setup_mask_corrupted_frames	0x40000105	1	0x1	RW	ES	On Preserve timing mode 0 = Output the corrupted frame. 1 = Don't output the corrupted frame.
api_rw_general_setup_fast_standby_ctrl	0x40000106	1	0x0	RW	ES	0 = Frame completes before mode entry. 1 = Frame may be truncated before mode entry.
api_rw_general_setup_cci_address_control	0x40000107	1	0x20	RW	ES	Control CCI address. Expressed as 8 bit (e.g. 0x20 write, 0x21 read is entered as 0x20).
api_rw_general_setup_context_switching						
api_rw_general_setup_context_switching_select	0x40000108	1	0x0	RW	EF	Select Context. 0=Context A 1=Context B.
api_rw_general_setup_context_switching_frame_duration	0x40000109	1	0x0	RW	EF	Number of frames for Context switching
api_rw_general_setup_context_switching_current	0x4000010A	1	0x0	RO	EF	Selected Context. 0=Context A 1=Context B.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_general_setup_context_switching_count	0x4000010B	1	0x0	RO	EF	Frame counter for new context.
api_rw_general_setup_eis_mode	0x4000010C	1	0x0	RW	ES	CIS EIS solution mode. 0 = Disabled. 1 = Enabled, Read data from external gyro to image footer. 2 = Read data from external gyro and use it to stop transmitting frames until the camera during integration time is stable enough.
api_rw_general_setup_immediate_exposure_enable	0x4000010F	1	0x0	RW	EF	start exposure of next frame immediately (in case exposure hasn't start yet). 0 = Disable immediate update for integration time and analog gain. 1 = Enable immediate update for integration time and analog gain.
api_rw_output						
api_rw_output_channel_identifier	0x40000110	1	0x10	RW	ES	Virtual channel number. [1:0] = VC of isp1 - Main video channel.
api_rw_output_signalling_mode	0x40000111	1	0x2	RW	ES	0 = PVI. 2 = CSI-2 with DPHY (MIPI)
api_rw_output_data_format	0x40000112	2	0xa0a	RW/C	EF	Data Format:[15:8] Max Sensors ADC resolution. [7:0] output data format: 0x08: Top 8 bit of pixel data (RAW8). 0x0A Top 10 bits of pixel data (RAW10).
api_rw_output_lane_mode	0x40000114	1	0x0	RW	EF	Number of CSI-2 lanes to be used. 0 = 1 lane. 1= 2 lanes. 2 = 3 lanes. 3 = 4 lanes.
api_rw_output_isp2_output_mode	0x40000115	1	0x0	RW	ES	ISP2 (second stream) stream output mode. please note that the block that generates the stream must be enabled too. 0 - output disabled 1 - output enabled
api_rw_output_isp2_data_type	0x40000116	1	0x30	RW	ES	Isp2 Data type.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_output_embedded_as_active_enable	0x40000117	1	0x0	RW	EF	for debug footer, use this to allow the embedded to be output as (active) video lines.
api_rw_output_emb_use_header	0x40000118	1	0x0	RW	ES	0 = Disable heading embedded line1 = Enable heading embedded line
api_rw_output_emb_header_lines	0x40000119	1	0x2	RW	ES	Defines the number of embedded lines (1-4).
api_rw_output_emb_footer_lines	0x4000011A	1	0x0	RW	ES	0 = Disable footering embedded line. 1 = Enable footering embedded line
api_rw_output_emb_equal_to_data	0x4000011B	1	0x1	RW	ES	Keep embedded line same length as pixels line: 0 = Embedded line length is shorter then data line length 1 = Embedded line length is equal to data line length. Note:this is only the line length, not the actual byte that used for data (fixed to 256 pixels each line).
api_rw_output_emb_in_bytes	0x4000011C	1	0x1	RW	ES	Keep embedded line pixel bitage same as pixels data bitage. 1 = Each embedded line pixel is byte 0 = Each embedded line pixel is equal to data pixel.
api_rw_output_qual_mode_continues_enable	0x4000011E	1	0x1			OIF_QC_CONT_MODE 0 000 PVI Clock always on - Legacy mode of continuous clock
api_rw_output_standby_power_mode	0x4000011F	1	0x0	RW	EF	MIPI DPHY standby power mode: 0 = ULPM. 1 = LP11 - lane lines stays high. 2 = Disabled (lane lines disconnected)
api_rw_op_cond						
api_rw_op_cond_vana_voltage	0x40000130	2	0x28			
api_rw_op_cond_vdig_voltage	0x40000132	2	0x12			

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_op_cond_vio_voltage	0x40000134	2	0x18			
api_rw_op_cond_extclk_frequency_mhz	0x40000136	2	0x0	RW/S	EF	Nominal External clock frequency (8.8 fixed point number).
api_rw_op_cond_extclk_scale_up	0x4000013D	1	0x0	RW	ES	Allows using input clock higher than 255.99MHz. For example, input clock 260MHz will be set as 0x8200 on op_cond_extclk_frequency_mhz, and as 0x01 on this vendor register (130MHz << 1 = 260MHz). For example, input clock 260MHz will be set as 0x8200 (130MHz) on api register api_rw_op_cond_extclk_frequency_mhz, and as 0x01 on this vendor register (130MHz << 1 = 260MHz).
api_rw_op_cond_extclk_frequency_additional_fraction_khz	0x4000013E	2	0x0	RW/S	ES	additional fraction of external frequency given in khz. external frequency is (extclk_frequency_mhz * 1000/256) << extclk_scale_up + extclk_frequency_additional_fraction_khz
api_rw_integration_time						
api_rw_integration_time_fine_integration_time	0x40000200	2	0x100	RW	EF	Fine integration time (of short exposure in WDR mode) in Pixel clocks.
api_rw_integration_time_coarse_integration_time	0x40000202	2	0x100	RW	EF	Coarse integration time in h-times (of short exposure in WDR mode)
api_rw_analog_gain						
api_rw_analog_gain_code_global	0x40000204	2	0x20	RW	EF	Global analog gain code (on WDR: for short exposure pixels) (8.5 format) example: gain x1 = 32d, gain x1.5 = 48d gain x4 = 128d
api_rw_digital_gain						

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_digital_gain_digital_to_analog_gain_control	0x4000020C	1	0x0	RW	EF	0 - do not change gain control. 1 - move as much gain from digital to analog gain, separate for long and short. for gain product above 16, the rest will be in digital gain. 2 - set both long and short analog gains to the minimum of the 2, and convert the rest to digital gain. NOTE: mode 1, 2 only work with api_rw_digital_gain_mode == 0 / 2. i.e. only 1 digital gain value for short and for long.
api_rw_digital_gain_digital_gain_mode	0x4000020D	1	0x0	RW/F	EF	Use digital gain mode: 0 - use one gain for all short channels, and one gain for all long channels. 1- gain for each channel (Not supported on WDR modes!). 2 - use a single digital gain for short and long digital gains, taken from api_rw_digital_gain_digital_gain_green_red.
api_rw_digital_gain_gains						
api_rw_digital_gain_gains_green_red	0x4000020E	2	0x100	RW	EF	Green (Red Row) channel digital gain value.Gain x1 = digital_gain_step_size
api_rw_digital_gain_gains_red	0x40000210	2	0x100	RW	EF	Red channel digital gain valueGain x1 = digital_gain_step_size
api_rw_digital_gain_gains_blue	0x40000212	2	0x100	RW	EF	Blue channel digital gain valueGain x1 = digital_gain_step_size
api_rw_digital_gain_gains_green_blue	0x40000214	2	0x100	RW	EF	Green (Blue Row) channel digital gain valueGain x1 = digital_gain_step_size
api_rw_wdr						
api_rw_wdr_multiple_exp_mode	0x4000021E	1	0x0	RW	EF	Multiple exposure mode select: 0 = single exposure (short only). 1 = Multiple exposure Mosaic-G. 2

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						= Multiple exposure Line-by-Line.
api_rw_wdr_exposure_control	0x4000021F	1	0x0	RW	EF	Exposure Control [3:0]: 0 - User sets two exposures: short & long. 1 - User sets only short exposure. 2 - User sets only long exposure. long exposure fix [7:4]: 0 keep ratio. 1 - clip long. 2 - convert ratio to digital gain. -
api_rw_wdr_exposure_ratio_numer_to_long	0x40000220	1	0x1	RW	EF	Numerator of Short to long exposure ratio
api_rw_wdr_exposure_ratio_numer_to_short	0x40000221	1	0x1	RW	EF	Numerator of Long to Short exposure ratio
api_rw_wdr_exposure_ratio_denom	0x40000222	1	0x1	RW	EF	Denominator of exposure ratio
api_rw_wdr_exposure_order	0x40000223	1	0x0	RW/C	EF	Exposure order
api_rw_wdr_long_fine_integration_time	0x40000224	2	0x100	RW	EF	Fine long integration time (pixels)
api_rw_wdr_long_coarse_integration_time	0x40000226	2	0x100	RW	EF	Coarse long integration time (lines)
api_rw_wdr_ext						
api_rw_wdr_ext_long_digital_gain						
api_rw_wdr_ext_long_digital_gain_green_red	0x40000230	2	0x100	RW	EF	Green (Red Row) channel digital (for long exposure pixels) gain value. Gain x1 = digital_gain_step_size
api_rw_wdr_ext_long_digital_gain_red	0x40000232	2	0x100	RW	EF	Red channel digital gain (for long exposure pixels) valueGain x1 = digital_gain_step_size
api_rw_wdr_ext_long_digital_gain_blue	0x40000234	2	0x100	RW	EF	Blue channel digital gain (for long exposure pixels) valueGain x1 = digital_gain_step_size
api_rw_wdr_ext_long_digital_gain_green_blue	0x40000236	2	0x100	RW	EF	Green (Blue Row) channel digital gain (for long exposure pixels) valueGain x1 =

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						digital_gain_step_size
api_rw_clocks						
api_rw_clocks_vt_pix_clk_div	0x40000300	2	0x7	RW/S	EF	Video Timing Pixel Clock divider.
api_rw_clocks_vt_sys_clk_div	0x40000302	2	0x1	RW/S	EF	Video Timing System Clock Divider Value
api_rw_clocks_vt_pre_pll_clk_div	0x40000304	2	0x6	RW/S	EF	Pre PLL clock Divider Value
api_rw_clocks_vt_pll_multiplier	0x40000306	2	0xf5	RW/S	EF	PLL multiplier Value
api_rw_clocks_op_pix_clk_div	0x40000308	2	0x8	RW/S	EF	Output Pixel Clock Divider
api_rw_clocks_op_sys_clk_div	0x4000030A	2	0x1	RW/S	EF	Output System Clock Divider Value
api_rw_clocks_vt_pll_post_scalar	0x4000030C	2	0x0	RW/C	EF	2 [^] S divider (PLL Freq = Fvco/2 [^] s) divider.
api_rw_clocks_op_pre_pll_clk_div	0x4000030E	2	0x4	RW/S	EF	2nd PLL Pre PLL clock Divider Value
api_rw_clocks_op_pll_multiplier	0x40000310	2	0xb3	RW/S	EF	2nd PLL multiplier Value
api_rw_clocks_op_pll_post_scalar	0x40000312	2	0x0	RW/C	EF	MIPI PLL 2 [^] S divider.
api_rw_frame_timing						
api_rw_frame_timing_dynamic_frame_rate_enable	0x40000335	1	0x0	RW	ES	Enable dynamic frame rate. frame rate change according to integration time, but not smaller than frame_length_lines
api_rw_frame_timing_frame_length_lines	0x40000340	2	0xe24	RW	EF	Frame Length in timing lines. (Can be increase by "long_exposure" registers).
api_rw_frame_timing_line_length_pck	0x40000342	2	0x0	RW/C	EF	Line length (H-time) in pixel clocks (VT clock domain) (Can be increase by "long_exposure" registers). (0 = FW select

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						shortest time possible)
api_rw_image_size						
api_rw_image_size_x_addr_start	0x40000344	2	0x0	RW/C	EF	X-address of the top left corner of the visible pixel data (offset from full image size).
api_rw_image_size_y_addr_start	0x40000346	2	0x0	RW/C	EF	Y-address of the top left corner of the visible pixel data (offset from full image size).
api_rw_image_size_x_addr_end	0x40000348	2	0x122f	RW/C	EF	X-address of the bottom right corner of the visible pixel data (offset from full image size).
api_rw_image_size_y_addr_end	0x4000034A	2	0xdaf	RW/C	EF	Y-address of the bottom right corner of the visible pixel data (offset from full image size).
api_rw_image_size_x_output_size	0x4000034C	2	0x1230	RW	EF	Width of image data output from the sensor module.
api_rw_image_size_y_output_size	0x4000034E	2	0xdb0	RW	EF	Height of image data output from the sensor module.
api_rw_image_size_digital_crop_x_offset	0x40000350	2	0x0	RW	EF	Allow to cut off left side of output image (Crop&Pad)
api_rw_image_size_digital_crop_y_offset	0x40000352	2	0x0	RW	EF	Allow to cut off top side of output image (Crop&Pad)
api_rw_sub_sample						
api_rw_sub_sample_x_even_inc	0x40000380	2	0x1	RW/C	EF	Visible columns readout sub-sampling - even increment.
api_rw_sub_sample_x_odd_inc	0x40000382	2	0x1	RW/C	EF	Visible columns readout sub-sampling - odd increment.
api_rw_sub_sample_y_even_inc	0x40000384	2	0x1	RW/C	EF	Visible rows readout sub-sampling - even increment.
api_rw_sub_sample_y_odd_inc	0x40000386	2	0x1	RW/C	EF	Visible rows readout sub-sampling - odd increment.
api_rw_scaling						

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_scaling_tetracell_digital_binning_factor	0x40000400	2	0x1010	RW/C	EF	tetracell Down scale factor: Range: 1 / 2 upwards Should determine only [7:0] - Row (Vertical) scaling factor, 4 fraction bits. [15:8] - Column (Horizontal) scaling factor, 4 fraction bits.
api_rw_scaling_digital_scaling	0x40000402	2	0x1010	RW/C	EF	Down scale factor: Range: 1 to 16 upwards Should determine only [7:0] - Row (Vertical) scaling factor, 4 fraction bits. [15:8] - Column (Horizontal) scaling factor, 4 fraction bits. the factors of 1.5;2;3;4 (i.e. 24;32;48;64)
api_rw_scaling_hbin_digital_binning_factor	0x40000404	1	0x10	RW/C	EF	enable using digital horizontal binning NOTE: this digital binning is added onto the analog binning. 0 - do not use HBIN in digital. 0x10 - do not use HBIN in digital. 0x20 - use FE-ISP horizontal binning 2. and so on up to 0x80. also see t_isp_hbin... fields.
api_rw_compression						
api_rw_compression_compression_mode	0x40000500	2	0x0	RW	EF	0 = Compression disabled.1 = Compression enabled.
api_rw_test_pattern						
api_rw_test_pattern_test_pattern_mode	0x40000600	2	0x0	RW	EF	Controls the output of the test pattern module 0=no pattern (default) 1= solid colour. 2= 100% colour bars 3= fade to grey colour bars 4=PN9.
api_rw_test_pattern_test_data_red	0x40000602	2	0x0	RW	EF	The test data used to replace red pixel data.
api_rw_test_pattern_test_data_green_red	0x40000604	2	0x0	RW	EF	The test data used to replace green pixel data on rows that also have red pixels
api_rw_test_pattern_test_data_blue	0x40000606	2	0x0	RW	EF	The test data used to replace blue pixel data
api_rw_test_pattern_test_data_green	0x40000608	2	0x0	RW	EF	The test data used to replace green pixel data on rows that

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
data_green_blue						also have blue pixels.
api_rw_test_pattern_horizontal_cursor_width	0x4000060A	2	0x0	RW	EF	Defines the width of the horizontal cursor (in pixels)
api_rw_test_pattern_horizontal_cursor_position	0x4000060C	2	0x0	RW	EF	Defines the top edge of the horizontal cursor
api_rw_test_pattern_vertical_cursor_width	0x4000060E	2	0x0	RW	EF	Defines the width of the vertical cursor (in pixels)
api_rw_test_pattern_vertical_cursor_position	0x40000610	2	0x0	RW	EF	Defines the left hand edge of the vertical cursor.
api_rw_test_pattern_crc	0x40000612	1	0x0	RW	EF	Enable CRC calculation. Every last line, the data will be replaced with the 2 Bytes of CRC output.
api_rw_test_pattern_fade_aux	0x40000613	1	0x0	RW	EF	Quantization level of right side of each bar.
api_rw_test_pattern_fade_factor	0x40000614	1	0x0	RW	EF	Smooth fade to gray gradient.
api_rw_test_pattern_fedtp_gradient_enable	0x40000615	1	0x0	RW	EF	bit 1: gradient enable Y. bit 0: gradient enable X.
api_rw_test_pattern_fedtp_gradient_invert	0x40000616	1	0x0	RW	EF	bit 1: Invert address Y. bit 0: Invert address X.
api_rw_test_pattern_fedtp_gradient_hor_shift	0x40000617	1	0x0	RW	EF	Scaling factor for X dependent gradient.
api_rw_test_pattern_fedtp_gradient_ver_shift	0x40000618	1	0x0	RW	EF	Scaling factor for Y dependent gradient.
api_rw_test_pattern_fedtp_address_noise	0x40000619	1	0x0	RW	EF	Enable horizontal address dependent 6bits noise.
api_rw_test_pattern_fedtp_hor_addr_noise_shift	0x4000061A	1	0x0	RW	EF	Right shift for Horizontal Noise generated values.
api_rw_test_pattern_fedtp_ver_addr_noise_shift	0x4000061B	1	0x0	RW	EF	Right shift for Vertical Noise generated values.
api_rw_test_pattern_fedtp_random_noise_right	0x4000061C	1	0x0	RW	EF	Right shift the random LFSR14 values.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_test_pattern_fedtp_mode56_shift_right	0x4000061D	1	0x0	RW	EF	Right shift for modes 5 and 6 generated values.
api_rw_long_frame_timing						
api_rw_long_frame_timing_frame_length_lines_shifter	0x40000702	1	0x0	RW/C	EF	Left shifter for frame length in lines (max value is 11).
api_rw_long_frame_timing_line_length_pck_shifter	0x40000703	1	0x0	RW/C	EF	shift-up value for line length pck.
api_rw_long_frame_timing_coarse_integration_time_sh	0x40000704	1	0x0	RW/C	EF	Left shifter for coarse integration time (max value is 11) Max exposure shouldn't be more than 1200 seconds.
api_rw_long_frame_timing_long_coarse_integration_time_sh	0x40000705	1	0x0	RW/C	EF	Left shifter of coarse integration time (for long exposure on wdr mode) (max value is 11) Max exposure shouldn't be more than 1200 seconds.
api_rw_phy						
api_rw_phy_dphy						
api_rw_phy_dphy_request_ed_link_rate	0x40000800	4	0x0	RW	ES	Target bitrate for CSI-2 transmission: with D-PHY, this is bitrate (Mbit/s) in 16 fraction bits.
api_rw_phy_dphy_ctrl	0x40000804	1	0x0	RW	ES	0 = Use automatic control. 1 = Use UI control. 2 = Use register control.
api_rw_phy_dphy_skew						
api_rw_phy_dphy_skew_calibration_min_freq_mhz	0x40000806	2	0x5dc	RW	ES	Minimal frequency for invocation of SKEW synchronization (Added by samsung).
api_rw_phy_dphy_skew_periodic_calibration_control	0x40000808	1	0x0	RW	ES	0 = do not send calibration sequence. 2 = send calibration sequence during frame blanking.
api_rw_phy_dphy_skew_periodic_calibration_interval	0x40000809	1	0x0	RW	ES	Output frame interval of period skew calibration. 0=no output. 1=output with all frames. otherwise, output once by X

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						frames. (for example, number 2 means every second frame.
api_rw_phy_dphy_skew_init_calibration_control	0x4000080A	1	0x1			Value 0: do not send calibration sequence Value 1: send calibration sequence during start of streaming
api_rw_phy_dphy_timing						
api_rw_phy_dphy_timing_t_clk_post	0x4000080E	1	0x0	RW	ES	Time that the transmitter shall continue sending HS clock after the last associated Data Lane has transitioned to LP mode.
api_rw_phy_dphy_timing_t_hs_prepare	0x4000080F	1	0x0	RW	ES	THS-PREPARE, is the time to drive LP-00 before starting the HS transmission on a Data Lane.
api_rw_phy_dphy_timing_t_hs_zero_min	0x40000810	1	0x0	RW	ES	THS-ZERO, MIN is the time to send HS-0, i.e. turn on the line termination and drive the interconnect with the HS driver, prior to sending the SoT Sync sequence.
api_rw_phy_dphy_timing_t_hs_trail	0x40000811	1	0x0	RW	ES	THS-TRAIL is the time the transmitter must drive the flipped last data bit after sending the last payload data bit of a HS transmission burst. This time is required by the receiver to determine EoT.
api_rw_phy_dphy_timing_t_clk_trail_min	0x40000812	1	0x0	RW	ES	Time to drive HS differential state after last payload clock bit of a HS transmission burst.
api_rw_phy_dphy_timing_t_clk_prepare	0x40000813	1	0x0	RW	ES	Time to drive LP-00 to prepare for HS clock transmission.
api_rw_phy_dphy_timing_t_clk_zero	0x40000814	1	0x0	RW	ES	Time for lead HS-0 drive period before starting Clock.
api_rw_phy_dphy_timing_t_lpx	0x40000815	1	0x0	RW	ES	Length of any Low-Power state period.
api_rw_phy_dphy_timing_t_hs_exit	0x40000816	1	0x0	RW	ES	ths_exit timing.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_binning						
api_rw_binning_mode	0x40000900	1	0x1	RW/C	EF	0 = Analog binning disabled. 1 = Analog binning enabled. 2 = Analog summation enabled, only in 2PD projects.
api_rw_binning_type	0x40000901	1	0x11	RW/C	EF	[3:0] - Row (Vertical) Binning factor. [7:4] - Column (Horizontal) Binning factor. for analog summation set row binning factor to 2 (in 2PD project only).
api_rw_binning_weighting	0x40000902	1	0x0	RW/C	EF	0=averaged weighting. 1 = summed weighting. 2 = Bayer weighting 3 = module-specific weighting.
api_rw_binning_binning_correction_force_bypass	0x40000903	1	0x0			Bypass Binning correction block. block is bypassed when theres no analog binning, EC)
api_rw_binning_use_dark_lines_bin	0x40000904	1	0x1	RW	EC	Analog vertical binning for dark lines
api_rw_data_transfer						
api_rw_data_transfer_ctrl	0x40000A00	1	0x0	RW	EF	Bit 0: 1 = enable, 0 = disable. Bit 1: 1 = write enable, 0 = read enable. Bit 2: 1 = pending errors, read errors register.(See Protocol documentation)
api_rw_data_transfer_errors	0x40000A01	1	0x0	RW	EF	Error flags: Bit 0 : data corrupted error (failed to read from OTP) Bit 1 : error bit wasn't cleared before operation. Bit 2 : trying to write to bit that is already set in OTP Bit 3: request set (ctrl bit 0 set) while processing request. Bit 4: wrong page number.
api_rw_data_transfer_page_select	0x40000A02	2	0x0	RW	EF	Page each page is 64 bytes. Do not exceed OTP size!
api_rw_data_transfer_data	0x40000A04	64	0x0	RW	EF	64 x 8-bit register for read or write accesses

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_data_transfer_data_0_	0x40000A04	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_1_	0x40000A05	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_2_	0x40000A06	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_3_	0x40000A07	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_4_	0x40000A08	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_5_	0x40000A09	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_6_	0x40000A0A	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_7_	0x40000A0B	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_8_	0x40000A0C	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_9_	0x40000A0D	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_10_	0x40000A0E	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_11_	0x40000A0F	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_12_	0x40000A10	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_13_	0x40000A11	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_14_	0x40000A12	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_15_	0x40000A13	1	0x0	EF	EF	64 x 8-bit register for read or write accesses

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_data_transfer_data_16_	0x40000A14	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_17_	0x40000A15	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_18_	0x40000A16	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_19_	0x40000A17	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_20_	0x40000A18	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_21_	0x40000A19	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_22_	0x40000A1A	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_23_	0x40000A1B	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_24_	0x40000A1C	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_25_	0x40000A1D	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_26_	0x40000A1E	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_27_	0x40000A1F	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_28_	0x40000A20	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_29_	0x40000A21	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_30_	0x40000A22	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_31_	0x40000A23	1	0x0	EF	EF	64 x 8-bit register for read or write accesses

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_data_transfer_data_32_	0x40000A24	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_33_	0x40000A25	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_34_	0x40000A26	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_35_	0x40000A27	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_36_	0x40000A28	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_37_	0x40000A29	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_38_	0x40000A2A	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_39_	0x40000A2B	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_40_	0x40000A2C	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_41_	0x40000A2D	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_42_	0x40000A2E	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_43_	0x40000A2F	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_44_	0x40000A30	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_45_	0x40000A31	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_46_	0x40000A32	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_47_	0x40000A33	1	0x0	EF	EF	64 x 8-bit register for read or write accesses

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_data_transfer_data_48_	0x40000A34	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_49_	0x40000A35	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_50_	0x40000A36	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_51_	0x40000A37	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_52_	0x40000A38	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_53_	0x40000A39	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_54_	0x40000A3A	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_55_	0x40000A3B	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_56_	0x40000A3C	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_57_	0x40000A3D	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_58_	0x40000A3E	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_59_	0x40000A3F	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_60_	0x40000A40	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_61_	0x40000A41	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_62_	0x40000A42	1	0x0	EF	EF	64 x 8-bit register for read or write accesses
api_rw_data_transfer_data_63_	0x40000A43	1	0x0	EF	EF	64 x 8-bit register for read or write accesses

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_isp						
api_rw_isp_shading_correction_mode	0x40000B00	1	0x0	RW	EF	Enable Shading correction (GRAS) 0 - disable shading correction. 1 - load shading correction grid and enable shading correction. 2 - only load shading correction grid.
api_rw_isp_luminance_correction_level	0x40000B01	1	0x80	RW	EF	Level of luminance correction can be used to adjust level of correction depending on use cases or wanted tuning settings.
api_rw_isp_bitreduct_qe_method	0x40000B02	1	0x1	RW	EF	Bit Reduction method of quantization error (QE) 0 - floor. 1 - round. 2 - Error diffusion.
api_rw_isp_black_level_correction_enable	0x40000B05	1	0x1	RW	EF	0 = Black level correction disabled. 1 = Black level correction enabled (enable fadlc)
api_rw_isp_mapped_couplet_correct_enable	0x40000B06	1	0x0	RW	EF	0 = Disabled static bad pixels correction. 1 = Enable static bad pixels correction.
api_rw_isp_dithering_before_gains_enable	0x40000B07	1	0x1			bypass the dithering before mixer gains.
api_rw_isp_dynamic_couplet_correct_enable	0x40000B09	1	0x0	RW	EF	0 = Disabled Dynamic correction. 1 = Enable Dynamic Correction
api_rw_isp_msm_offsets_enable	0x40000B0A	1	0x1	RW	EF	MSM Offsets enable disable flag.
api_rw_isp_msm_gains_enable	0x40000B0B	1	0x1	RW	EF	MSM Gains enable disable flag. disabling this flag will result in MSM Periodic Gains to be set to 1.
api_rw_isp_msm_channel_gains_enable	0x40000B0C	1	0x1	RW	EF	MSM channel Gains enable disable flag. disabling this flag will result in MSM Channel Gains to be set to 1.
api_rw_isp_denoise_enable	0x40000B0D	1	0x0	RW	EF	0 = Disabled denoise. 1 = Enable denoise [0:0] Disparity

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						Enable [4:4] Denoise Enable.
api_rw_flash						
api_rw_flash_duration_pixels	0x40000B10	4	0x0	RW	EF	Flash duration in pixel clocks
api_rw_flash_enable	0x40000B14	1	0x0	RW	EF	0 = Flash disable 1= Flash enable
api_rw_flash_active_low_polarity	0x40000B15	1	0x0	RW	EF	controls polarity of flash signal
api_rw_flash_gpio_index	0x40000B16	1	0x0	RW	ES	Index of flash shutter GPIO (1-3), 0 for None
api_rw_dual_sensor_sync						
api_rw_dual_sensor_sync_dual_sensor_sync_mode	0x40000B30	1	0x0	RO	ES	Sync frames with external sensor. 0: disable the dual sensor sync. 1: enable the dual sensor sync and sync immediately. 2: enable the dual sensor sync, but do not sync. this allows the sensor to function normally, with the dual sync functionality enabled, change the configuration of the sensor on-the-fly when ever required. also see skl_dual_sensor_sync for other configs 8 enable dual sensor without resync. assumes that the HOST has already synchronized between the 2 sensors and just keep the current configuration of VSYNC-IN to VSYNC-OUT.
api_rw_dual_sensor_sync_ratio_mode	0x40000B31	1	0x0	RW	EF	this field enables using a ratio between master and slave ratio. 0 - 1:1 clock ratio between master and slave 1 - 1:N clock ratio between master and slave (faster master) 2 - N:1 clock ratio between master and slave (faster slave)
api_rw_dual_sensor_sync	0x40000B32	1	0x2	RW	EF	the ratio between master and

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
_master_to_slave_ratio						slave
api_rw_dual_sensor_sync_vsync_in_to_vsync_out_time	0x40000B34	2	0x1	RW	EC	the time which should pass between the VSync-in to VSync-out signals. used for calculation of drift. configured in timing lines.
api_rw_sensor						
api_rw_sensor_out_pedestal	0x40000BC0	2	0x40	RW	ES	Output pedestal.
api_rw_sensor_offset_x	0x40000BC2	2	0x0	RW/C	EF	Offset from X-address of the top left corner of the visible pixel data after analog crop, bin and subsample (Resizer)
api_rw_sensor_offset_y	0x40000BC4	2	0x0	RW/C	EF	Offset from Y-address of the top left corner of the visible pixel data after analog crop, bin and subsample (Resizer)
api_rw_sensor_out_type	0x40000BC6	1	0x0	RW	ES	0 = Visible only. 1 = Visible + Dark. 2 = Visible + TMC. 3 = Visible + Dark + TMC.
api_rw_sensor_physical_pixel_order	0x40000BC7	1	0x0	RW	ES	Defines color of physical pixel 0, 0:0: Gr Rg / Bg Gb1: Rg Gr / Gb Bg2: Bg Gb / Gr Rg3: Gb Bg / Rg Gr
api_rw_sensor_tetracell_full_out_pattern_mode	0x40000BC8	1	0x0	RW/C	EC	0 = Bayer pattern. 1 = Tetracell pattern.
api_rw_sensor_static_bpc_calc_mode	0x40000BC9	1	0x0			this field is used to select when the static BPC calculations are done. 0 - offline. can cause the first frame to be without the BPC correction. 1 - during fast frame. adds time to the fast frame, if the exposure is too short. 2 - as part of FW configuration change.
api_rw_sensor_b_abort_timing_on_ch_cfg	0x40000BCA	1	0x1	RW	ES	0 = Preserve timing (Keep frame rate)1 = Abort timing if next frame is corrupted (On configuration change)

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_sensor_abort_timing_method_on_rolling_sh	0x40000BCB	1	0x1	RW	EF	This register describe when to Abort when using rolling shutter: 0 = Abort timing Immediately, even during read out. 1 = Abort timing after read out ends. 2 = Abort timing on end of frame.
api_rw_sensor_abort_timing_on_sw_stby	0x40000BCC	1	0x1	RW	EF	This register describe when to Abort when entering software standby: 1 = Abort timing after read out ends. 2 = Abort timing on end of frame. Note: when general_setup.fast_standby_ctrl = 1, this register is not relevant and the mode is "Abort timing Immediately".
api_rw_pdaf						
api_rw_pdaf_pdaf_exist	0x40000D00	1	0x1	RW	EC	PDAF readout (in case APS have PDAF it's possible only on binning modes. PDAF pixels wont be averaged with normal pixels. No special treatment on ISP chain, BPC, H-Bin. 1 = enable PDAF readout. Enable PDAF readout .
api_rw_pdaf_pdaf_correct	0x40000D01	1	0x1	RW	EC	Enable PDAF defect correction on for visible pixel area.
api_rw_pdaf_pdcollect_enable_collect	0x40000D02	1	0x0	RW	EC	Enable PD collect (Tail mode) .
api_rw_pdaf_virtual_channel	0x40000D03	1	0x1	RW	EC	Register to control PDAF virtual channel when PDAF uses interleaved readout mode.
api_rw_pdaf_interleaved_enable	0x40000D04	1	0x1	RW	EC	1 = output pdaf interleaved (Tail mode). 0 = interlaced (according to bit 2, 3 on control register)
api_rw_pdaf_collect_from	0x40000D05	1	0x2	RW	EC	PD collect output. 0 - before FE cluster binning. 1 - after H-bin. 3 - After Bit reduct.
api_rw_pdaf_pdcollect_output_width	0x40000D06	2	0x0	RW	EC	total PDAF tail output width (with padding). If set to 0 FW calculate the minimal width needed for

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						PDAF in output image the width might be rounded up by 4. Monitor: m_oif_vc2_output_width .
api_rw_pdaf_pdcollect_out_height	0x40000D08	2	0x0	RW	EC	total PDAF tail output height. If set to 0 FW calculate the minimal height needed for PDAF in output image Monitor: m_oif_vc2_output_height .
api_rw_gras						
api_rw_gras_colour_temperature	0x40000D80	2	0xc80			Relevant only if b_set_explicit_color_temperature is set
api_rw_gras_rgb_gains	0x40000D82	6	0x0			Gains per color.
api_rw_gras_rgb_gains_0_	0x40000D82	2	0x100			
api_rw_gras_rgb_gains_1_	0x40000D84	2	0x100			
api_rw_gras_rgb_gains_2_	0x40000D86	2	0x100			
api_rw_gras_b_set_explicit_color_temperature	0x40000D88	1	0x1			Whether color temperature is set directly by colour_temperature field or derived from RGB gains
api_rw_bracketing						
api_rw_bracketing_lut_control	0x40000E0B	1	0x0	RW	EF	0 = disabled 1...n = bracketing over n frames. Maximum value of n is bracketing_LUT_size. If n is smaller than bracketing_LUT_size then camera will use only those n values from the bracketing LUT. E.g. if bracketing_LUT_size is 5, host can use bracketing only for 3 frames by setting bracketing_LUT_control to value 3.
api_rw_bracketing_lut_mo	0x40000E0	1	0x0	RW	EF	Bit 0: 0 = return to SW Standby after bracketing, 1 = continue in

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
de	C					streaming after bracketing Bit 1: 0 = single bracketing, 1 = loop mode. Bit 2-7: reserved for future use.
api_rw_bracketing_lut_entry_control	0x40000E0D	1	0x0			Reserved for future use.
api_rw_bracketing_selected_fields	0x40000E0E	2	0x0	RW	EF	Bit 0: 1 = coarse integration time LUT entry supported Bit 1: 1 = global analog gain LUT entry supported Bit 2: reserved Bit 3: reserved Bit 4: 1 = flash LUT entry supported Bit 5: 1 = global digital gain LUT entry supported Bit 6-7: reserved for future use All bits: 0 = select all.
api_rw_bracketing_lut	0x40000E10	240	0x0	RW	EF	bracketing look up table.
api_rw_bracketing_lut_0_	0x40000E10	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_1_	0x40000E12	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_2_	0x40000E14	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_3_	0x40000E16	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_4_	0x40000E18	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_5_	0x40000E1A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_6_	0x40000E1C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_7_	0x40000E1E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_8_	0x40000E20	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_9_	0x40000E22	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_10_	0x40000E24	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_	0x40000E26	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_12 –	0x40000E28	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_13 –	0x40000E2A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_14 –	0x40000E2C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_15 –	0x40000E2E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_16 –	0x40000E30	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_17 –	0x40000E32	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_18 –	0x40000E34	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_19 –	0x40000E36	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_20 –	0x40000E38	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_21 –	0x40000E3A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_22 –	0x40000E3C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_23 –	0x40000E3E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_24 –	0x40000E40	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_25 –	0x40000E42	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_26 –	0x40000E44	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_27 –	0x40000E46	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_28 –	0x40000E48	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_29 –	0x40000E4A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_30 –	0x40000E4C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_31 –	0x40000E4E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_32 –	0x40000E50	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_33 –	0x40000E52	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_34 –	0x40000E54	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_35 –	0x40000E56	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_36 –	0x40000E58	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_37 –	0x40000E5A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_38 –	0x40000E5C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_39 –	0x40000E5E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_40 –	0x40000E60	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_41 –	0x40000E62	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_42 –	0x40000E64	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_43 –	0x40000E66	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_44 –	0x40000E68	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_45 –	0x40000E6A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_46 –	0x40000E6C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_47 –	0x40000E6E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_48 –	0x40000E70	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_49 –	0x40000E72	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_50 –	0x40000E74	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_51 –	0x40000E76	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_52 –	0x40000E78	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_53 –	0x40000E7A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_54 –	0x40000E7C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_55 –	0x40000E7E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_56 –	0x40000E80	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_57 –	0x40000E82	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_58 –	0x40000E84	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_59 –	0x40000E86	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_60 –	0x40000E88	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_61 –	0x40000E8A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_62 –	0x40000E8C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_63 –	0x40000E8E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_64 –	0x40000E90	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_65 –	0x40000E92	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_66 –	0x40000E94	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_67 –	0x40000E96	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_68 –	0x40000E98	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_69 –	0x40000E9A	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_70 –	0x40000E9C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_71 –	0x40000E9E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_72 –	0x40000EA0	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_73 –	0x40000EA2	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_74 –	0x40000EA4	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_75 –	0x40000EA6	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_76 –	0x40000EA8	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_77 –	0x40000EAA	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_78 –	0x40000EAC	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_79 –	0x40000EAE	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_80 –	0x40000EB0	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_81 –	0x40000EB2	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_82 –	0x40000EB4	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_83 –	0x40000EB6	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_84 –	0x40000EB8	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_85 –	0x40000EBA	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_86 –	0x40000EBB	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_87 –	0x40000EBE	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_88 –	0x40000EC0	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_89 –	0x40000EC2	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_90 –	0x40000EC4	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_91 –	0x40000EC6	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_92_	0x40000EC8	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_93_	0x40000ECA	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_94_	0x40000EC C	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_95_	0x40000EC E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_96_	0x40000ED0	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_97_	0x40000ED2	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_98_	0x40000ED4	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_99_	0x40000ED6	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_100_	0x40000ED8	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_101_	0x40000EDA	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_102_	0x40000EDC	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_103_	0x40000EDE	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_104_	0x40000EE0	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_105_	0x40000EE2	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_106_	0x40000EE4	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_107_	0x40000EE6	2	0x0	EF	EF	bracketing look up table.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_rw_bracketing_lut_10_8_	0x40000EE8	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_10_9_	0x40000EEA	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_0_	0x40000EEC	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_1_	0x40000EE E	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_2_	0x40000EF0	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_3_	0x40000EF2	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_4_	0x40000EF4	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_5_	0x40000EF6	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_6_	0x40000EF8	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_7_	0x40000EFA	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_8_	0x40000EFC	2	0x0	EF	EF	bracketing look up table.
api_rw_bracketing_lut_11_9_	0x40000EFE	2	0x0	EF	EF	bracketing look up table.
api_rw_general_setup_ext						
api_rw_general_setup_ext_debug_out	0x40000FE0	2	0x0	RW/C	EF	Debug select mask for ISP debug regs: Normal=0, ATOP_DTP(on FPGA)=1, FEDTP=2, DCDS=3, HBIN=4, MIXER=5, BPC=6, BINCR=7, HVBIN=8, BDS=9, CROP=10, SMIADTP=11, COMPRESS=12
api_rw_general_setup_ext_otp_address_gras	0x40000FE8	2	0x0	RW	ES	If set to 0 - assuming the NVM header is written directly by the

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						HOST (not read from NVM)
api_rw_general_setup_ext_otp_address_bpc	0x40000FEA	2	0x0	RW	ES	Address of Static BPC table in OTP.
api_rw_general_setup_ext_otp_address_gos	0x40000FEC	2	0x0	RW	ES	Address of GOS data in OTP. 0 = Don't load table from OTP.
api_ro_integration_time						
api_ro_integration_time_capability	0x40001000	2	0x1	RO	EF	0=coarse integration but no fine integration. 1=course and smooth (1 pixel) fine integration.
api_ro_integration_time_coarse_integration_time_min	0x40001004	2	0x2	RO	EF	Minimum coarse integration in timing lines
api_ro_integration_time_coarse_integration_time_max_margin	0x40001006	2	0x2	RO	EF	Current frame length - current max coarse exp
api_ro_integration_time_fine_integration_time_min	0x40001008	2	0x0	RO	EF	Minimum fine integration in pixels
api_ro_integration_time_fine_integration_time_max_margin	0x4000100A	2	0x0	RO	EF	Current line length - current max fine exp.
api_ro_digital_gain						
api_ro_digital_gain_capability	0x40001080	2	0x2	RO	EF	0 = None. 1 = Per channel digital gain. 2 = Per channel or exposure digital gain (based on vendor_sensor_digital_gain_mode)
api_ro_digital_gain_min	0x40001084	2	0x100	RO	EF	Minimum recommended digital gain value
api_ro_digital_gain_max	0x40001086	2	0x8000	RO	EF	Maximum recommended digital gain value
api_ro_digital_gain_step_size	0x40001088	2	0x100	RO	EF	Defines the resolution of the digital gain control parameters.
api_ro_pll_clock_limits						

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_pll_clock_limits_clock_tree_pll_capability	0x400010F0	1	0x3	RO	EF	Bit 0: 0 = 2-PLL clock tree not supported, 1 = 2-PLL clock tree supported. Bit 1: 0 = 1-PLL clock tree not supported, 1 = 1-PLL clock tree supported.
api_ro_pll_clock_limits_min_ext_clk_freq_mhz	0x400010F4	4	0x40c00000	RO	EF	Minimum external clock frequency (IEEE754 format)
api_ro_pll_clock_limits_max_ext_clk_freq_mhz	0x400010F8	4	0x42c80000	RO	EF	Maximum external clock frequency (IEEE754 format)
api_ro_pll_clock_limits_main						
api_ro_pll_clock_limits_main_min_vt_pre_pll_clk_div	0x400010FC	2	0x1	RO	EF	Minimum Pre PLL divider value
api_ro_pll_clock_limits_main_max_vt_pre_pll_clk_div	0x400010FE	2	0x3f	RO	EF	Maximum Pre PLL divider value
api_ro_pll_clock_limits_main_min_vt_pll_ip_freq_mhz	0x40001100	4	0x40200000	RO	EF	Minimum PLL input clock frequency (IEEE754 format)
api_ro_pll_clock_limits_main_max_vt_pll_ip_freq_mhz	0x40001104	4	0x40a00000	RO	EF	Maximum PLL input clock frequency (IEEE754 format)
api_ro_pll_clock_limits_main_min_vt_pll_multiplier	0x40001108	2	0x40	RO	EF	Minimum PLL multiplier
api_ro_pll_clock_limits_main_max_vt_pll_multiplier	0x4000110A	2	0x3ff	RO	EF	Maximum PLL multiplier
api_ro_pll_clock_limits_main_min_vt_pll_vco_freq_mhz	0x4000110C	4	0x43fa0000	RO	EF	Minimum VT PLL VCO frequency (IEEE754 format)
api_ro_pll_clock_limits_main_max_vt_pll_vco_freq_mhz	0x40001110	4	0x447a0000	RO	EF	Maximum VT PLL VCO frequency (IEEE754 format)
api_ro_pll_clock_limits_main_max_vt_pll_post_scaler	0x40001114	2	0x5	RO	EF	Maximum Post Scaler S (2^S divider) value .

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_pll_clock_limits_min_min_op_pll_freq_mhz	0x40001118	4	0x427c0000	RO	EF	Minimum PLL output clock frequency (IEEE754 format)
api_ro_pll_clock_limits_min_max_op_pll_freq_mhz	0x4000111C	4	0x447a0000	RO	EF	Maximum PLL output clock frequency (IEEE754 format)
api_ro_pll_clock_limits_min_min_vt_sys_clk_div	0x40001120	2	0x1	RO	EF	Minimum video timing system clock divider value
api_ro_pll_clock_limits_min_max_vt_sys_clk_div	0x40001122	2	0xc	RO	EF	Maximum video timing system clock divider value
api_ro_pll_clock_limits_min_min_vt_sys_clk_freq_mhz	0x40001124	4	0x427c0000	RO	EF	We actually force here the vt_sys divider to be 1 since $DDR (ramp\ clock) = VCO / (vt_sys_div * DDR_div)$ Minimum video timing system clock frequency (IEEE754 format)
api_ro_pll_clock_limits_min_max_vt_sys_clk_freq_mhz	0x40001128	4	0x447a0000	RO	EF	Maximum video timing system clock frequency (IEEE754 format)
api_ro_pll_clock_limits_min_min_vt_pix_clk_div	0x4000112C	2	0x1	RO	EF	Minimum video timing pixel clock divider value
api_ro_pll_clock_limits_min_max_vt_pix_clk_div	0x4000112E	2	0xc	RO	EF	Maximum video timing pixel clock divider value
api_ro_pll_clock_limits_min_min_vt_pix_clk_freq_mhz	0x40001130	4	0x42c00000	RO	EF	Minimum video timing pixel clock frequency (IEEE754 format)
api_ro_pll_clock_limits_min_max_vt_pix_clk_freq_mhz	0x40001134	4	0x440c0000	RO	EF	Maximum video timing pixel clock frequency (IEEE754 format)
api_ro_pll_clock_limits_output						
api_ro_pll_clock_limits_output_min_op_pre_pll_clk_div	0x40001138	2	0x1	RO	EF	Minimum Pre OP PLL clock Divider Value
api_ro_pll_clock_limits_output_max_op_pre_pll_clk_div	0x4000113A	2	0x21	RO	EF	Maximum Pre OP PLL clock Divider Value

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_pll_clock_limits_output_min_op_pll_ip_freq_mhz	0x4000113C	4	0x40c00000	RO	EF	Minimum OP PLL input clock frequency (IEEE754 format)
api_ro_pll_clock_limits_output_max_op_pll_ip_freq_mhz	0x40001140	4	0x41400000	RO	EF	Maximum OP PLL input clock frequency (IEEE754 format)
api_ro_pll_clock_limits_output_min_op_pll_multiplier	0x40001144	2	0x19	RO	EF	Minimum OP PLL Multiplier
api_ro_pll_clock_limits_output_max_op_pll_multiplier	0x40001146	2	0xc8	RO	EF	Maximum OP PLL Multiplier
api_ro_pll_clock_limits_output_min_op_pll_vco_freq_mhz	0x40001148	4	0x44048000	RO	EF	Minimum OP PLL VCO frequency (IEEE754 format)
api_ro_pll_clock_limits_output_max_op_pll_vco_freq_mhz	0x4000114C	4	0x44866000	RO	EF	Maximum OP PLL VCO frequency (IEEE754 format)
api_ro_pll_clock_limits_output_max_op_pll_post_scaler	0x40001150	2	0x4	RO	EF	Maximum Post Scaler S (2^S divider) value .
api_ro_pll_clock_limits_output_min_op_pll_op_freq_mhz	0x40001154	4	0x42080000	RO	EF	Minimum OP PLL output clock frequency (IEEE754 format)
api_ro_pll_clock_limits_output_max_op_pll_op_freq_mhz	0x40001158	4	0x44866000	RO	EF	Maximum OP PLL output clock frequency (IEEE754 format)
api_ro_pll_clock_limits_output_min_op_sys_clk_div	0x4000115C	2	0x1	RO	EF	Minimum output system clock divider value
api_ro_pll_clock_limits_output_max_op_sys_clk_div	0x4000115E	2	0xc	RO	EF	Maximum output system clock divider value
api_ro_pll_clock_limits_output_min_op_sys_clk_freq_mhz	0x40001160	4	0x42080000	RO	EF	Minimum output system clock frequency
api_ro_pll_clock_limits_output_max_op_sys_clk_freq_mhz	0x40001164	4	0x44866000	RO	EF	Maximum output system clock frequency

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_pll_clock_limits_output_min_op_pix_clk_div	0x40001168	2	0x1	RO	EF	Minimum output pixel clock divider value
api_ro_pll_clock_limits_output_max_op_pix_clk_div	0x4000116A	2	0xc	RO	EF	Maximum output pixel clock divider value
api_ro_pll_clock_limits_output_min_op_pix_clk_freq_mhz	0x4000116C	4	0x41000000	RO	EF	Minimum output pixel clock frequency
api_ro_pll_clock_limits_output_max_op_pix_clk_freq_mhz	0x40001170	4	0x43480000	RO	EF	Maximum output pixel clock frequency
api_ro_pll_clock_limits_output_min_dphy_freq_mhz	0x40001174	4	0x42a00000	RO	EF	Minimum DPHY input clock frequency (IEEE754 format)
api_ro_pll_clock_limits_output_max_dphy_freq_mhz	0x40001178	4	0x45066000	RO	EF	Maximum DPHY input clock frequency (IEEE754 format)
api_ro_frame_timing						
api_ro_frame_timing_min_frame_length_lines	0x400011A0	2	0x64	RO	EF	Minimum Frame Length allowed.
api_ro_frame_timing_max_frame_length_lines	0x400011A2	2	0xffff	RO	EF	Maximum possible number of lines per Frame.
api_ro_frame_timing_min_line_length_pck	0x400011A4	2	0x0	RO	EF	Minimum Line Length allowed. (0 = Auto select)
api_ro_frame_timing_max_line_length_pck	0x400011A6	2	0xffff	RO	EF	Maximum possible number of pixel clocks per line.
api_ro_frame_timing_min_line_blanking_pck	0x400011A8	2	0xb8	RO	EF	Minimum line blanking time in pixel clocks
api_ro_frame_timing_min_frame_blanking_lines	0x400011AA	2	0x36	RO	EF	Minimum frame blanking in video timing lines.
api_ro_frame_timing_min_line_length_pck_step_size	0x400011AC	1	0x8	RO	EF	Minimum step size of line length in pixel clocks
api_ro_size_limits						
api_ro_size_limits_x_addr_min	0x400011B0	2	0x0	RO	EF	Minimum X-address of the addressable pixel array

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_size_limits_y_addr_min	0x400011B2	2	0x0	RO	EF	Minimum Y-address of the addressable pixel array
api_ro_size_limits_x_addr_max	0x400011B4	2	0x1230	RO	EF	Maximum X-address of the addressable pixel array
api_ro_size_limits_y_addr_max	0x400011B6	2	0xdb0	RO	EF	Maximum Y-address of the addressable pixel array
api_ro_size_limits_min_x_output_size	0x400011B8	2	0x100	RO	EF	Minimum x output size in pixels
api_ro_size_limits_min_y_output_size	0x400011BA	2	0x90	RO	EF	Minimum y output size in pixels
api_ro_size_limits_max_x_output_size	0x400011BC	2	0x1230	RO	EF	Maximum x output size in pixels
api_ro_size_limits_max_y_output_size	0x400011BE	2	0xdb0	RO	EF	Maximum y output size in pixels
api_ro_sub_sample_cap						
api_ro_sub_sample_cap_min_even_inc	0x400011C0	2	0x1	RO	EF	Minimum Increment for even pixels
api_ro_sub_sample_cap_max_even_inc	0x400011C2	2	0x1	RO	EF	Maximum increment for even pixels
api_ro_sub_sample_cap_min_odd_inc	0x400011C4	2	0x1	RO	EF	Minimum Increment for odd pixels
api_ro_sub_sample_cap_max_odd_inc	0x400011C6	2	0xf	RO	EF	Maximum Increment for odd pixels
api_ro_scaling_cap						
api_ro_scaling_cap_scaling_capability	0x40001200	2	0x2	RO	EF	0 = None. 1 = Horizontal. 2 = Full (Horizontal & Vertical)
api_ro_scaling_cap_scaler_m_min	0x40001204	2	0x10	RO	EF	Down scale factor: Minimum M value (x1 - No scale)
api_ro_scaling_cap_scaler_m_max	0x40001206	2	0x80	RO	EF	Down scale factor: Maximum M value (x4 scale)
api_ro_scaling_cap_scaler_n_min	0x40001208	2	0x10			

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_scaling_cap_scaler_n_max	0x4000120A	2	0x10			
api_ro_scaling_cap_spatial_sampling_capability	0x4000120C	2	0x1			
api_ro_scaling_cap_digital_crop_capability	0x4000120E	2	0x1	RO	EF	0 = None. 1 = input digital crop supported.
api_ro_compression_cap						
api_ro_compression_cap_compression_capability	0x40001300	2	0x1	RO	EF	Image Compression Capability Register - [0x1300-0x1301] 0 = None. 1 = DPCM/PCM.
api_ro_color_matrix						
api_ro_fifo						
api_ro_fifo_size_pixels	0x40001430	2	0x0			
api_ro_fifo_support_capability	0x40001432	1	0x2	RO	EF	0 = not supported. 1 = supports derating. 2 = supports VT/OP domain decoupling.
api_ro_dphy						
api_ro_dphy_ctrl_capability	0x40001440	1	0x1f	RO	EF	Bit 0: 1 = automatic DPHY control supported. Bit 1: 1 = UI based DPHY control supported. Bit 2: 1 = manual register control based DPHY control supported for non-extended Time_and_UI_1 register. Bit 3: 1 = manual register control based DPHY control supported for non-extended Time_and_UI_2 register. Bit 4: 1 = manual register control based DPHY control supported for non-extended Time register.
api_ro_dphy_lane_mode_capability	0x40001441	1	0xf	RO	EF	Bit 0: 1 = 1 lane supported, 0 = not supported. Bit 1: 1 = 2 lane supported, 0 = not supported. Bit 2: 1 = 3 lane supported, 0 = not supported. Bit 3: 1 = 4 lane supported, 0 = not supported. Bit 4: 1 = 5 lane supported, 0 = not supported.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						supported. Bit 5: 1 = 6 lane supported, 0 = not supported. Bit 6: 1 = 7 lane supported, 0 = not supported. Bit 7: 1 = 8 lane supported, 0 = not supported.
api_ro_dphy_signalling_mode_capability	0x40001442	1	0x4	RO	EF	Bit 0: PVI. Bit 1: reserved. 2 = CSI-2 with D-PHY supported. 0 = not supported. Bit 3: 1 = CSI-2 with C-PHY supported. 0 = not supported.
api_ro_dphy_fast_standby_capability	0x40001443	1	0x1	RO	EF	0 = Frame truncation not supported for rolling shutter. 1 = Frame truncation supported for rolling shutter.
api_ro_dphy_cci_address_control_capability	0x40001444	1	0x0	RO	EF	Bit 0: 1 = SW changeable CCI address supported. 0 = not supported.
api_ro_binning_capability						
api_ro_binning_capability_min_frame_length_lines_bin	0x40001700	2	0x64	RO	EF	Minimum frame length in timing lines on binning mode.
api_ro_binning_capability_max_frame_length_lines_bin	0x40001702	2	0xffff	RO	EF	Maximum frame length in timing lines on binning mode.
api_ro_binning_capability_min_line_length_pck_bin	0x40001704	2	0x0	RO	EF	0 = FW choose shortest length lines possible. Otherwise, this is the minimum value.
api_ro_binning_capability_max_line_length_pck_bin	0x40001706	2	0xffff	RO	EF	Maximum frame length in timing lines on binning mode.
api_ro_binning_capability_min_line_blanking_pck_bin	0x40001708	2	0xb8	RO	EF	Minimum line blank time in Pixel clocks.
api_ro_binning_capability_fine_integration_time_min_bin	0x4000170A	2	0x0	RO	EF	Minimum fine integration time allowed in binning mode.
api_ro_binning_capability_fine_integration_time_max_mar_bin	0x4000170C	2	0x0	RO	EF	Margin used to determine the maximum fine integration time allowed in binning mode.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
api_ro_binning_capability_binning_capability	0x40001710	1	0x1	RO	EF	Bit 0 = No. 1 = Yes.
api_ro_binning_capability_binning_weighting_capability	0x40001711	1	0x1	RO	EF	Bit 0: 1 = Averaged Bit 1: 1 = Summed (for low light) Bit 2: 1 = Bayer-corrected Bit 3: 1 = Module specific weighting.
api_ro_binning_capability_binning_sub_types	0x40001712	1	0x5	RO	EF	Number of Binning subtypes available.
api_ro_binning_capability_binning_type_1	0x40001713	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_2	0x40001714	1	0x12	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_3	0x40001715	1	0x13	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_4	0x40001716	1	0x14	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_5	0x40001717	1	0x16	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_6	0x40001718	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_7	0x40001719	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_8	0x4000171A	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						factor.
api_ro_binning_capability_binning_type_9	0x4000171B	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_10	0x4000171C	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_11	0x4000171D	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_12	0x4000171E	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_binning_capability_binning_type_13	0x4000171F	1	0x11	RO	EF	Supported analog binning bits[7:4] = columns binning factor. Bits[3:0] = rows binning factor.
api_ro_data_transfer						
api_ro_data_transfer_if_capability	0x40001800	1	0x1	RO	EF	Bit 0: 1 = interface 1 supported Bit 1: reserved, report value 0. Bit 2: 0 = polling not needed in reading. 1 = polling needed in reading. Bit 3: 0 = polling not needed in writing. 1 = polling needed in writing. Bit 4: 0 = supports single DTI functionality. 1- supports multi DTI functionality Bits 0-3 are valid in single DTI functionality only
api_ro_isp						
api_ro_isp_shading_correction_capability	0x40001810	1	0x3	RO	EF	bit 0: 1 = On-module shading correction. bit 1: 1 = Luminance correction adjust available.
api_ro_isp_green_imbalan	0x40001811	1	0x0	RO	EF	bit 0: 1 = on-module green imbalance filtering. bit 1: 1 =

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
ce_capability						Green imbalance filter weight adjust available.
api_ro_isp_black_level_capability	0x40001812	1	0x1	RO	EF	Bit 0: 1= On module black level correction.
api_ro_isp_module_specific_correction_capability	0x40001813	1	0x0			Bit 0: 0 = not supported. 1 = Manual supported. Bit 1: 0 = not supported. 1 = Limited auto supported.
api_ro_isp_defect_correction_capability	0x40001814	2	0x40fd	RO	EF	Bit 0=1: internal module mapped defect correction available. Bit 1=1: internal module mapped defect correction weight available. Bit 2=1: internal module dynamic couplet correction available. Bit 3=1: manual dynamic couplet correction weight adjust available. Bit 4=1: limited auto dynamic couplet correction weight adjust available. Bit 5=1: internal module dynamic single pixel defect correction available. Bit 6=1: manual dynamic single pixel defect correction weight adjust available. Bit 7=1: limited auto dynamic single pixel defect correction weight adjust available. Bit 8=1: combined internal module dynamic couplet & single pixel defect correction available. Bit 9=1: manual dynamic couplet & single pixel defect correction weight adjust available. Bit 10=1: limited auto dynamic couplet & single pixel defect correction weight adjust available. Bits 11-B13: reserved Bit 14=1: mapped defect limited auto weight adjust available Bit 15: reserved for future extensions
api_ro_isp_defect_correction_capability_2	0x40001816	2	0x0	RO	EF	B0=1: internal module mapped triplet correction available. B1=1: manual mapped triplet correction weight adjust available. B2=1: limited auto mapped triplet

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						correction weight adjust available. B3=1: internal module dynamic triplet correction available. B4=1: manual dynamic triplet correction weight adjust available. B5=1: limited auto dynamic triplet correction weight adjust available. B6=1: internal module dynamic line correction available. B7=1: manual dynamic line correction weight adjust available. B8=1: limited auto dynamic line correction weight adjust available. Other bits: reserved for future extensions.
api_ro_edof_cap						
api_ro_edof_cap2						
api_ro_timer						
api_ro_ms_af_cap						
api_ro_static_af_camera						
api_ro_static_af_camera_actuator_type	0x40001B10	2	0x0	RO	EF	Reserved for host usage for the mentioned purpose.
api_ro_static_af_camera_af_device_address	0x40001B12	1	0x0	RO	EF	I2C address of AF device.
api_ro_pdaf						
api_ro_pdaf_capability	0x40001B20	1	0xf5	RO	EF	Bit 0: 1 = PDAF supported. Bit 1: 1 = processed PDAF data readout in bottom embedded data. Bit 2: 1 = processed PDAF data readout in interleaved manner. Bit 3: 1 = RAW PDAF data readout in bottom embedded data. Bit 4: 1 = RAW PDAF data readout in interleaved manner. Bit 5: 1 = visible pixel area PDAF pixel specific defect correction supported. Bit 6: 1 = virtual channel interleaving supported. Bit 7: 1 = data type

Name	Address	Size	Reset Value	Access Type	Timing Type	Description
						interleaving supported.
api_ro_bracketing						
api_ro_bracketing_lut_capability_1	0x40001B30	1	0x33	RO	EF	Bit 0: 1 = coarse integration time LUT entry supported Bit 1: 1 = global analog gain LUT entry supported Bit 2: reserved. Bit 3: reserved. Bit 4: 1 = flash LUT entry supported. Bit 5: 1 = global digital gain LUT entry supported. Bit 6-7: reserved for future use. All bits: 0 = not supported
api_ro_bracketing_lut_capability_2	0x40001B31	1	0x3	RO	EF	Bit 0: 1 = supports single bracketing mode. Bit 1: 1 = supports looped bracketing mode.
api_ro_bracketing_lut_size	0x40001B32	1	0x1e	RO	EF	Max number of frames supported by LUT, e.g. 5 = LUT can contain settings for 5 frames. 240 bytes, 4 field types, each field type is 2 bytes,

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